Provisional: descriptive bidding patterns and regulatory context

MTU15 thesis companion

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This file collects descriptive facts that any identification design must rest on: how firms bid across hour-classes (critical vs flat), how pivotal and non-pivotal firms differ, how technologies differ, and the regulatory / RT2 / reforzada context in which the reforms operate. Nothing here is an identification claim. The regression outputs from the current reform-window design are collected in a companion sibling document.

Contents

Glossary of acronyms

Acronym	Meaning
CCGT	Combined-cycle gas turbine: flexible, dispatchable thermal generator.
DA	Day-ahead market (Mercado Diario), cleared once per day.
IDA	Intraday auctions (3 European SIDC sessions post-2024-06-14).
MCP	Market clearing price.
MTU	Market time unit (MTU60 = hourly bidding; MTU15 = 15-minute bidding).
ISP15	Imbalance Settlement Period at 15 minutes (settlement granularity, started 2024-12-01).
REE	Red Eléctrica de España, the Spanish transmission-system operator.
OMIE	Operador del Mercado Ibérico de Energía, the wholesale market operator.
PDBF	Programa Diario Base de Funcionamiento — first post-DA viable schedule (REE-built).
PDBC	Programa Diario Base de Casación — pre-PDBF DA-cleared schedule (OMIE-built).
PHF	Programa Horario Final — post-IDA + post-RT viable schedule.
RES	Renewable Energy Sources (wind, solar PV, solar thermal, RoR hydro).
TR	Tiempo Real — real-time restrictions phase (PO 3.4–3.5), pay-as-bid.
Fase I/II	Restricciones técnicas Fase 1 (post-PDBF, pre-IDA) and Fase 2 (post-IDA), pay-as-bid.
aFRR	Automatic Frequency Restoration Reserve (Regulación secundaria), PICASSO platform.
mFRR	Manual Frequency Restoration Reserve (Regulación terciaria), MARI platform.
RR	Replacement Reserves (Reserva de Sustitución), TERRE platform.
BSP	Balance Service Provider (firm enrolled to supply balancing energy).
EUPHEMIA	Day-ahead pan-European auction algorithm (clears DA market).

Reform-window timeline

The five reform windows used throughout this document are defined as follows. The “pre-IDA window” (pre 2024-06-14) is the baseline regime, never directly compared in the main tables but referenced for context.

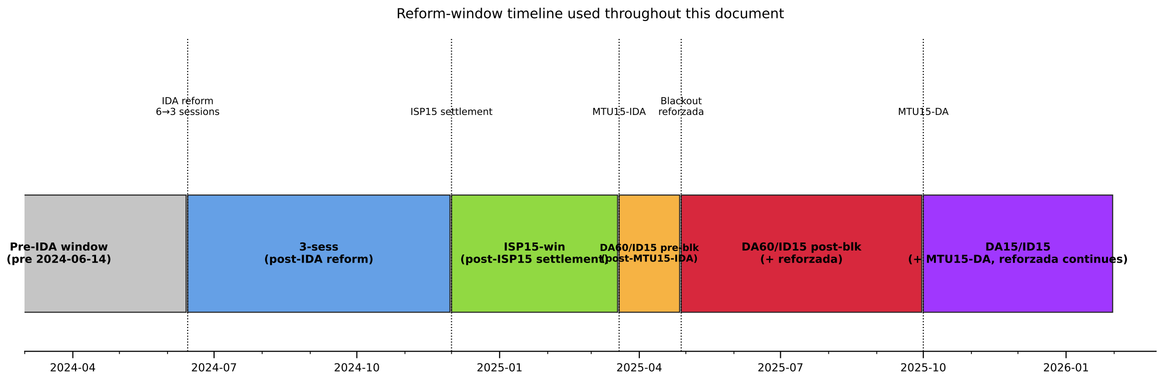


Figure 1: Reform-window timeline. Five named windows partition the post-2024-06-14 sample; the pre-IDA window is the baseline regime. Dotted vertical lines mark the five reform dates that produce the window boundaries.

Calendar coverage and same-calendar baselines

Why this matters. Spanish electricity is strongly seasonal: heating-driven demand peaks in winter, cooling in summer, hydro inflows peak in spring, solar production peaks at the summer solstice, wind blows hardest in winter. The five reform windows above do *not* span the calendar uniformly — each window covers a specific subset of calendar months, and these subsets differ across windows. Any cross-regime comparison that does not hold the calendar fixed therefore confounds the regime change with the seasonal pattern. **This applies to every variable in this document**: bid prices, D_w, Δp / Δq, redispatch volumes, imbalance MWh, system-cost levels, RES capture rates. A reader who picks any table in this document should first read Table 1 below to know which calendar months that regime covers and what comparator is available.

Table 1: Calendar coverage of each reform window, and same-calendar baseline availability. "Calendar months covered" is the set of clock months that fall inside the window. "Days" is the window length. "Same-calendar pre-IDA placebo" indicates whether a clean same-calendar baseline exists in the pre-IDA period (2018-06-14 through 2024-06-13), where no granularity reform was active. "Reforzada overlay" flags windows in which the post-blackout reinforced-operation policy (active 2025-04-28 onward) is also in effect.

Regime	Calendar dates	Months covered	Days	Reforzada?	Same-calendar pre-IDA placebo
pre-IDA	$\leq$ 2024-06-13	all 12, multi-year	–	no	N/A (is the baseline)
3-sess	2024-06-14 – 2024-11-30	Jun–Nov	170	no	Clean (Jun–Nov 2018–2023)
ISP15-win	2024-12-01 – 2025-03-18	Dec–Mar	108	no	Clean (Dec–Mar 2018–2024)
MTU15-IDA pre-blk	2025-03-19 – 2025-04-27	late Mar – Apr	40	no	Clean (same 40 days 2018–2024)
MTU15-IDA post-blk	2025-04-28 – 2025-09-30	May – Sep	156	**yes**	Clean for granularity (May–Sep 2018–2023, pre-IDA); none for reforzada (no pre-2025-04-28 reforzada in data)
DA15/ID15	2025-10-01 – present	Oct–May (partial, ongoing)	220+	**yes**	None clean. 2024–25 same-calendar comparators are themselves a mix of post-MTU15-IDA + ISP15-win regimes

Three categories. Each regime falls into one of three buckets for the purpose of cross-regime reading:

1. **Clean same-calendar placebo available.** *3-sess, ISP15-win, MTU15-IDA pre-blk.* For each of these, we can restrict the pre-IDA baseline to the same calendar months in 2018–2024 (and average across years), and the difference (regime − pre-IDA same-cal) is interpretable as the regime effect with seasonality differenced out. Any cross-regime reading that uses these windows defaults to this same-calendar comparison unless explicitly stated otherwise.

2. **Reforzada-confounded, partial calendar-match available.** *MTU15-IDA post-blk.* The window spans May–Sep 2025, which has clean same-calendar comparators in pre-IDA (May–Sep 2018–2023). But these comparators are pre-reforzada. So the difference (MTU15-IDA post-blk − pre-IDA same-cal) confounds (a) the granularity reform, (b) operación reforzada, and (c) the post-blackout system state. There is no calendar period in the data in which reforzada is active but MTU15 is not, so reforzada cannot be differenced out via a same-calendar pre-reform window. For descriptive purposes, May–Sep 2025 is best read as "post-MTU15-IDA + reforzada + post-blackout, jointly"; see §6 on disentangling these channels.

3. **No clean calendar match.** *DA15/ID15.* The window covers Oct 2025 onward. The same calendar months one year earlier (Oct 2024 – May 2025) are themselves a mix of pre-MTU15-IDA, ISP15-win, MTU15-IDA pre-blk, and MTU15-IDA post-blk regimes — so calendar-matching to "one year ago" does not produce a single-regime comparator. The cleanest available calendar comparators for DA15/ID15 are the same months in pre-IDA years (Oct 2023 – May 2024 was largely pre-IDA, with the IDA-3 reform starting only mid-Jun 2024; Oct 2022 – May 2023 and earlier are all pre-IDA). But even with pre-IDA same-calendar baselines, the reforzada overlay is unresolved by this comparator: DA15/ID15 is the union of MTU15-DA, MTU15-IDA, and reforzada in the same calendar window, and our data does not contain a period where any one of these three is active without the others. Therefore any DA15/ID15 reading should be flagged as descriptive only, with the three confounds explicitly named.

Fourier-deseasonalization: formal specification.

For an outcome y_t observed on daily series across regimes

$$\mathcal{R} \;=\; \{\,\text{pre-IDA, 3-sess, ISP15-win,} \\ \text{MTU15-IDA pre-blk, MTU15-IDA post-blk, DA15/ID15}\,\},$$

we fit a Frisch-Waugh-Lovell pooled regression on $t \in [2022\text{-}01\text{-}01, 2026\text{-}05\text{-}15]$:

$$g(y_t) \;=\; \alpha + \sum_{r\in\mathcal{R}\setminus\{0\}} \beta_r\,\mathbf{1}\{\text{regime}(t)=r\} + \underbrace{\sum_{k=1}^{K}\left[a_k\,c_k(t) + b_k\,s_k(t)\right]}_{\text{Fourier basis on day-of-year}} + \underbrace{\sum_{j=1}^{6}\delta_j\,\mathbf{1}\{\text{dow}(t)=j\}}_{\text{day-of-week dummies}} + \varepsilon_t,$$

where the omitted regime is pre-IDA, the omitted day-of-week is Monday, the link $g(\cdot)$ is $\log(\cdot)$ for positive levels (prices, costs, MWh aggregates) and $\text{logit}(\cdot) = \log(p/(1-p))$ for bounded shares (in-band MW share, capture rate, channel-share fractions), and the day-of-year sinusoids are

$$c_k(t) \;=\; \cos\!\left(\frac{2\pi k\cdot \text{doy}(t)}{365}\right), \qquad s_k(t) \;=\; \sin\!\left(\frac{2\pi k\cdot \text{doy}(t)}{365}\right), \qquad k = 1,2,3,4.$$

We use $K = 4$ harmonics throughout: the annual fundamental ($k = 1$, $\sim$ 365-day period), the semi-annual harmonic ($k = 2$, $\sim$ 183-day period), the four-month harmonic ($k = 3$), and the quarterly harmonic ($k = 4$, $\sim$ 91-day period); higher harmonics ($k \geq 5$) are not identified separately from intra-month noise on daily data and do not improve out-of-sample fit on the year-fold CV diagnostics we ran for bid-shape (§3.4). The six day-of-week dummies absorb weekly periodicity in prices and MWh aggregates; with the shortest regime window (MTU15-IDA pre-blk) only 40 days long, DOW imbalance can otherwise leak into $\widehat{\beta}_r$. We do not report standard errors on the $\widehat{\beta}_r$: the spec is used purely as a point-estimation device for the SA values defined below, not for inference on the regime contrast itself.

Pooled seasonality across regimes is a constraint, not a modelling choice. With only 40 days in MTU15-IDA pre-blk, regime-specific Fourier coefficients (a_k interacted with regime dummies) are not estimable — there is not enough within-regime calendar variation to separate them from the regime fixed effect.

Seasonality-adjusted (SA) regime value. We define $\hat{y}_r^{\mathrm{SA}}$ as the predicted level of y in regime r at *annual-mean Fourier* (which is exactly zero on the integer doy grid, by orthogonality of c_k, s_k to a constant) and *within-week DOW mean* (one-seventh weight on each day-of-week dummy). For an identity-link outcome:

$$\hat{y}_r^{\mathrm{SA,\ identity}} = \widehat{\alpha} + \widehat{\beta}_r + \tfrac{1}{7}\sum_{j=1}^{6}\widehat{\delta}_j.$$

For $g \in \{\log, \mathrm{logit}\}$ the naive plug-in $g^{-1}(\widehat{\alpha} + \widehat{\beta}_r + \frac{1}{7}\sum_j \widehat{\delta}_j)$ is biased by Jensen's inequality; we correct it non-parametrically via Duan smearing across the in-sample residuals $\{\widehat{e}_i\}_{i=1}^n$:

$$\hat{y}_r^{\mathrm{SA}} = \frac{1}{n}\sum_{i=1}^{n} g^{-1}\left(\widehat{\alpha} + \widehat{\beta}_r + \tfrac{1}{7}\sum_{j=1}^{6}\widehat{\delta}_j + \widehat{e}_i\right).$$

Duan smearing is robust to non-normal residuals and handles the regime-heterogeneous variance case automatically (winter residual variance is larger than summer; the smear factor adjusts for the pooled scale). For log outcomes Duan typically multiplies the naive plug-in by $\exp(\sigma^2/2) \approx$ 1.10–1.37 in our data — a material correction that the original (uncorrected) spec was missing. Where SA values appear in tables, they sit in coloured brackets next to the raw regime mean; where the value is calendar-matched by construction (Dec-vs-Dec, etc.), no SA is computed because seasonality is already differenced out.

Why Fourier+DOW rather than month or year fixed effects: with our reform windows (Table 1), some regimes do not span all 12 calendar months (e.g., MTU15-IDA pre-blk is 40 days in late March–April only). A month-of-year fixed effect interacted with regime would be partially absorbed into the regime dummy itself — not separately identified. A smooth Fourier basis extrapolates the seasonal cycle to any calendar date, allowing the same parametric form to apply across regimes regardless of which calendar months they happen to cover.

Discipline applied throughout this document.

- Per-regime descriptive tables report the regime's raw level alongside bracketed SA: the latter is $\hat{y}_r^{\mathrm{SA}}$ from the spec above, applied to that outcome's daily series. Bucket-1 regimes (§§above) additionally have same-calendar pre-IDA means available for cross-validation against the SA reading; bucket-2 and bucket-3 regimes do not.
- For bucket-2 (MTU15-IDA post-blk), the SA value differences out seasonality but the granularity and reforzada confounds remain (the SA value is the joint effect of granularity + reforzada + post-blackout at average doy).
- For bucket-3 (DA15/ID15), same: the SA value combines all three reforms with reforzada at average doy. Wherever a DA15/ID15 SA number is reported, we name the confounds.

- The functional-PCA exercise in §3.4 uses pairwise reform-window comparisons with *calendar-shifted placebo years* (2022, 2023) to explicitly difference out year-on-year seasonality within each placebo. That is a tighter seasonality-control implementation than the Fourier basis used elsewhere; we treat the fPCA placebo as the methodological benchmark, the Fourier-FWL spec above as the workhorse.
- Where the raw and SA values move in the same direction across regimes, the documented pattern survives seasonality control. Where they diverge (large bracketed-vs-raw gaps), the raw cross-regime reading is mostly a calendar-mix artifact and only the SA value is meaningfully comparable across regimes.

Reading guide. Part A is about bidding and trading patterns. Part B is about the regulatory/post-clearing channels (reforzada, geographic dominance, CNMC sanctions, etc.) that are independent of the granularity reform but constrain how the identification can be set up.

Part I
Part A — Bidding and trading patterns

Reading guide for Part A. Three setup sections come first. §1 defines the **hour-class taxonomy** (critical / flat / midday) used to slice every table and figure that follows. §2 characterises **who price-sets** in the day-ahead market (§2.1) and, methodologically, in the intraday markets (§2.2). §3 then defines the **bid-shape metrics** (D_w, Δp, Δq), the choice of band centre c and half-width h, and the cell-level aggregator. Visual results (per tech × firm × quarter bid curves with mean clearing prices overlaid) come in §3.1; quantitative tables and histograms ($D_w > 0$ rates, $\Delta p/\Delta q$ channel shares, within-hour shaping by firm × tech) come in §3.2.

1 Hour-class taxonomy: critical, flat, midday

Empirical classification: within-hour standard deviation of residual demand. The crit/flat/midday split is **not assumed** on demand-level grounds ("critical hours are the ones with the highest demand"). It is derived from the within-hour *variability* of residual demand. The intuition is mechanical: *MTU15 granularity has economic content only where the within-hour profile actually moves.* If residual demand is essentially flat within a clock-hour, the hourly clearing price is already a good summary; granular per-quarter pricing has no signal to add. If residual demand changes sharply within the hour (morning ramp, evening peak, solar fade), the hourly clearing price was a compromise across rapidly-changing within-hour conditions; granular pricing unlocks the ability to discriminate across quarters and is therefore where strategic bidding can have a quarter-specific footprint.

Concretely, for clock-hour h on date d, let $D_{h,q,d}$ be the residual demand (= load − wind − solar) at quarter $q \in \{1, 2, 3, 4\}$ of that hour. The hour's quarter-mean is $\bar{D}_{h,d} = \frac{1}{4}\sum_q D_{h,q,d}$. We compute, per (date, hour), the within-hour standard deviation

$$\sigma_{\text{within}}(h, d) \;=\; \sqrt{\tfrac{1}{4}\sum_{q=1}^{4}\big(D_{h,q,d} - \bar{D}_{h,d}\big)^2},$$

and take the expectation over all 2025 dates to get a clock-hour metric

$$\sigma_{\text{within}}(h) \;=\; \mathbb{E}_d\big[\sigma_{\text{within}}(h, d)\big] \quad \text{[MW]}.$$

The 24-hour ranking of $\sigma_{\text{within}}(h)$ is the actual driver of the classification. Hours with high σ_{within} are critical; hours with low σ_{within} are flat; midday hours are excluded for a separate reason (below). In economic terms, high σ_{within} means the within-hour residual-demand curve moves enough that equilibrium quarter prices and quantities should differ; hour-level pricing was a binding pooling constraint, and granular pricing relaxes it. Strategic firms with within-hour market power therefore have scope to exploit it in critical hours and essentially no scope in flat hours — which is the heterogeneity Part A's CCGT bid-shape analysis is set up to detect.

Data, window, and frequency. $D_{h,q,d}$ is computed from two ENTSO-E Transparency Platform sources, both at 15-minute ISP resolution:

- Spanish system load: ENTSO-E A65 (ENTSO-E A65 load series), filtered to MTU = 15 minutes.
- Spanish wind + solar generation: ENTSO-E A75 (ENTSO-E A75 wind+solar actual generation series), PSR types B16 (solar) + B18 (wind offshore) + B19 (wind onshore), filtered to MTU = 15 minutes.

Residual demand is the per-ISP arithmetic difference $D = \text{load} - \text{solar} - \text{wind}$.

Window. The classifier runs on **2023-01-01 through 2026-05-14 (≈ 3.4 years)**, which is the largest window for which both A65 and A75 are uniformly available at 15-min granularity. ENTSO-E reports Spanish A65/A75 at 60-min before 2022 and at mixed 60/15-min during 2022 (transition to ISP15 was 2024-11-13 in the regulatory calendar but ENTSO-E's data-side switch for Spain happened earlier in 2022), so pre-2023 the within-hour variance of residual demand is undefined — there is one observation per clock-hour rather than four. We use the longest clean 15-min window because the classifier is a structural feature of the within-day pattern, not a regime-specific statistic; averaging over more years gives a more stable hour-of-day ranking.

Why ENTSO-E and not ESIOS, given ESIOS has 5-min for demand. ESIOS (REE) does publish demand at 5-minute native resolution (`demand_real` id 1293, available back to 2018) and our ingestion catalog stores it downsampled to 15-min ISP. However, ESIOS' curated wind/solar generation indicators are forecasts (`wind_forecast_pen` id 541, `solar_pv_forecast` id 542, `solar_forecast_combined` id 10034), not actuals, and they are 15-min only from 2022 onward (60-min before). For actual wind/solar generation, ENTSO-E A75 is the only consistent source and binds the window at 2023-01-01. Even if 5-min demand were used on the demand side, the market clears at 15-min ISP and bids are quoted at 15-min ISP, so within-hour variability at 5-min resolution would pick up sub-ISP fluctuations that the market cannot price-discriminate against by construction — the economic content of granularity, which is what σ_{within} is meant to proxy for, lives at 15-min.

Day-type pool. All dates 2023–2026 are pooled (weekday + weekend), consistent with the bid-side analysis in Part A. The weekday-only partition produces the same class assignments (no rank swaps); weekend σ_{within} is uniformly smaller in absolute terms but the morning-ramp / evening-peak pattern survives.

The 24-hour ranking. Table 2 is reproduced from the main paper's Appendix A.7. It shows the per-hour mean load, solar, wind, residual demand, σ_{within}, and the signed within-hour load change $\Delta\text{load} = \text{load}_{q_4} - \text{load}_{q_1}$, averaged over all dates 2023-01-01 through 2026-05-14. The critical-flat-midday classification falls out of σ_{within} directly.

Table 2: Hour-of-day classification metrics, Spain 2023–2026 (ENTSO-E A65 + A75, 15-min ISP, $n \approx 3.4$ years of daily observations per hour). The classifier is σ_{within} (column 6, MW). Critical hours: $\sigma_{\text{within}} \in [522, 1303]$. Flat hours: $\sigma_{\text{within}} \in [336, 357]$. Midday: $\sigma_{\text{within}} \in [468, 502]$ — intermediate, but excluded for the separate reason discussed below. Dropped (transition) hours sit at the regime boundaries. *Year-stability:* across the 3 full years 2023, 2024, 2025 (computed separately), the critical-set min σ_{within} across years is 530 MW, exceeding the flat-set max (387 MW) by 1.4×; no within-class swaps across years.

Clock-hour	Demand (GW)	Solar (GW)	Wind (GW)	Residual demand (GW)	σ_{within} (MW)	Δ demand (MW)	Class
00:00–01:00	22.8	0.1	7.3	15.4	412	−834	*dropped*
01:00–02:00	22.0	0.1	7.1	14.8	347	−470	*flat*
02:00–03:00	21.6	0.1	6.9	14.6	336	−190	*flat*
03:00–04:00	21.5	0.1	6.7	14.7	357	176	*flat*
04:00–05:00	22.4	0.1	6.7	15.7	587	1114	*dropped*
05:00–06:00	24.4	0.2	6.7	17.5	721	1743	*critical*
06:00–07:00	26.5	1.5	6.6	18.4	915	1423	*critical*
07:00–08:00	27.7	4.9	6.2	16.6	1078	627	*critical*
08:00–09:00	28.3	9.1	5.8	13.4	1006	253	*critical*
09:00–10:00	28.3	12.0	5.5	10.9	753	−104	*dropped*
10:00–11:00	28.2	13.3	5.4	9.5	499	−66	*dropped*
11:00–12:00	28.2	13.8	5.5	8.9	468	94	*dropped*
12:00–13:00	28.3	13.8	5.7	8.8	502	−104	*dropped*
13:00–14:00	27.9	13.5	5.8	8.5	485	−331	*dropped*
14:00–15:00	27.5	12.9	6.1	8.5	469	−122	*dropped*
15:00–16:00	27.6	11.7	6.4	9.6	677	291	*dropped*
16:00–17:00	28.2	9.0	6.8	12.3	1109	553	*critical*
17:00–18:00	29.2	5.4	7.4	16.4	1303	891	*critical*
18:00–19:00	30.5	2.2	7.7	20.5	1065	884	*critical*
19:00–20:00	31.3	0.6	7.9	22.8	522	176	*critical*
20:00–21:00	30.4	0.3	7.9	22.2	622	−1444	*critical*
21:00–22:00	28.0	0.2	7.9	20.0	781	−1851	*critical*
22:00–23:00	25.8	0.2	7.7	18.0	664	−1407	*critical*
23:00–00:00	24.2	0.1	7.5	16.5	510	−1062	*dropped*

Classification rule. The partition is read off the table by contiguous-block structure: **critical** = the morning-ramp block (5h–8h) + the evening-peak block (16h–22h), 11 hours all with $\sigma_{\text{within}} \in [522, 1303]$ MW; **flat** = the overnight block (1h–3h), 3 hours all with $\sigma_{\text{within}} \in [336, 357]$ MW; **midday** = the solar-peak block (11h–14h), 4 hours with $\sigma_{\text{within}} \in [468, 502]$ MW *and* residual demand < 9 GW. The 6 remaining hours (0, 4, 9, 10, 15, 23, $\sigma_{\text{within}} \in [412, 753]$ MW) straddle the thresholds and are dropped. No clustering or supervised threshold tuning is involved — the partition can be reconstructed from the sorted table by visual inspection.

Why we distinguish critical from flat. Critical-hour σ_{within} is roughly 1.5–4× that of flat hours (522–1,303 MW vs 336–357 MW), reflecting fast within-hour shifts in residual demand: morning demand ramp + solar rise at 5h–9h, solar fade + evening peak at 16h–22h. Granular per-quarter pricing in critical hours can in principle discriminate prices across quarters that face very different supply/demand conditions, so it is where strategic bidding has scope. Flat hours have nearly-flat residual demand within the clock-hour and granular pricing has no economic content to extract — the four quarter prices, absent friction, should be essentially the same. This is why flat is the within-day control: the (crit − flat) differential strips out any day-level shock (gas, weather, regulation) that affects both hour-classes proportionally, while flat itself carries no granularity-exploitation signal.

Falsifier already documented in the main paper. Main paper §2.4: CCGT price-setting share is 9.1% critical vs 9.0% flat (almost identical), so TTF/gas-cost shocks cannot propagate asymmetrically across the two hour-classes. The within-day differencing therefore identifies behaviour, not cost shocks.

Outcome-side validation: post-MTU15-DA actually price-discriminates more in critical hours. The classifier is a structural proxy: high σ_{within} should predict where granular pricing delivers measurable price dispersion. Post-MTU15-DA (2025-10-01 to 2026-05-14, the only window with 15-min DA quarter prices), the within-hour DA quarter clearing-price standard deviation averages **6.58 EUR/MWh in critical hours vs 2.13 in flat hours (3.1× ratio)** — midday is even tighter (1.51 EUR/MWh, solar-marginal regime). The classifier correctly identifies the hours where post-reform quarter prices actually diverge.

Why we exclude midday from the control. Midday hours (11–14) have $\sigma_{\text{within}} \in [468, 502]$ MW, intermediate between critical and flat. By the variability criterion alone, midday is neither clearly "treated" (granularity has some scope) nor clearly "control" (granularity has zero scope). What pushes midday into a third bucket is the *level* of residual demand and the implied marginal-pricing technology: midday solar peak drives residual demand down to 8.5–8.9 GW (vs $\sim$ 15–23 GW outside midday), and the marginal price-setting technology shifts from CCGT/hydro to solar PV + storage. Pooling midday into flat as a control would absorb both (a) the strategic-bidding signal we want and (b) the operational solar-vs-thermal differential (a separate channel that has its own dynamics). We exclude midday from the control set entirely. It appears in some descriptive overlays (channel-level histograms of Δp, Δq show crit/midday/flat side by side for comparison), but it is never used as control in the DDD.

Why a 6-hour transition gap. Hours 0, 4, 9, 10, 15, 23 sit at the boundaries between regimes. Their σ_{within} values are 412, 587, 753, 499, 677, 510 MW — straddling the flat/critical thresholds. Rather than force a binary classification, we drop them. The 14-hour analytic sample (11 crit + 3 flat) is the analytic universe for every table in Part A; the 4 midday hours are reserved for descriptive overlay only.

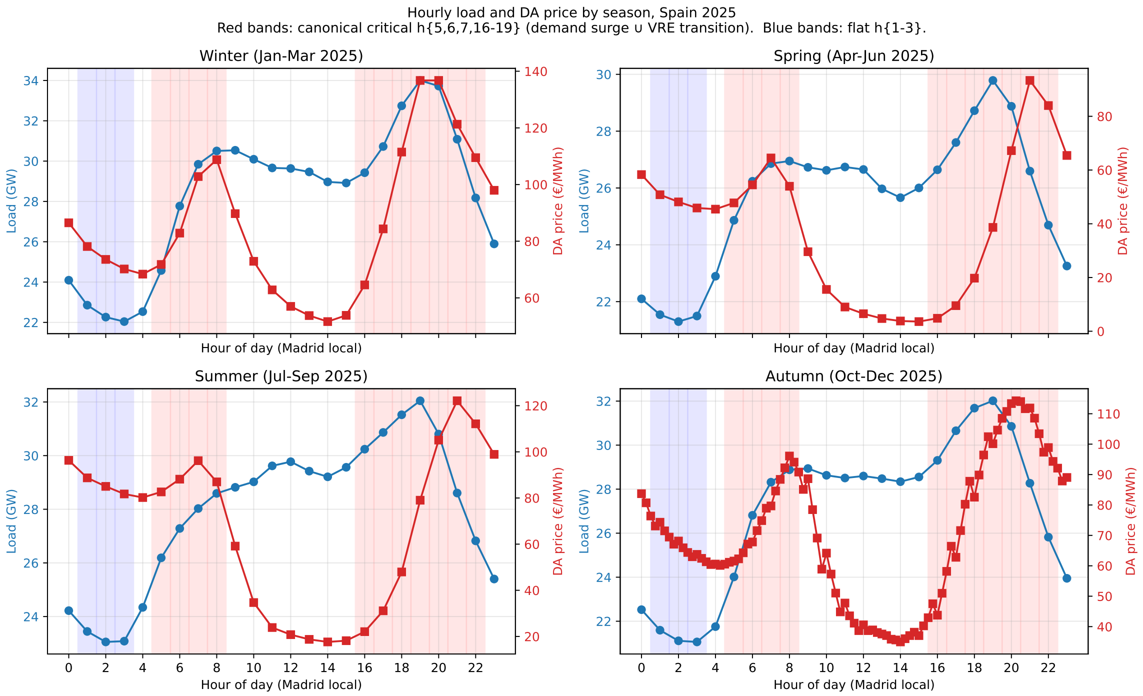


Figure 2: Hourly load and DA price by season, Spain 2025. Blue: demand; red: DA clearing price. Red bands: critical hours (05:00–09:00, 16:00–23:00); blue bands: flat hours (01:00–04:00). *Note:* demand level is shown here for descriptive context only — the formal classifier is σ_{within} of residual demand (see Table 2), not demand level. The seasonal-stability of the within-day pattern is the relevant point. Reproduced from main paper §2.1.

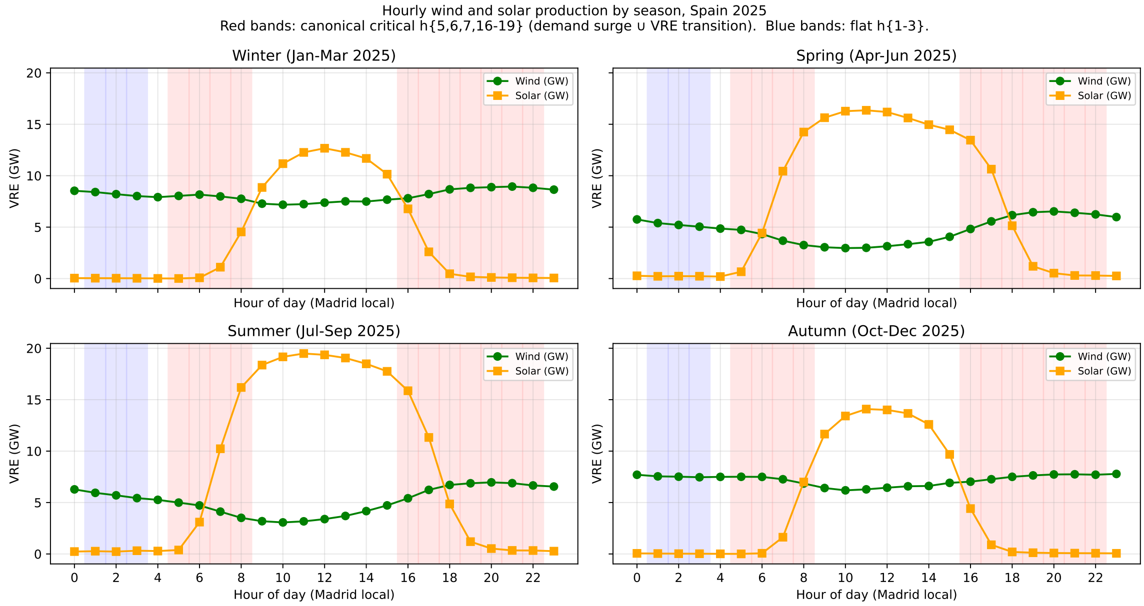


Figure 3: Hourly wind and solar production by season, Spain 2025. Red bands: critical hours; blue bands: flat hours. Solar production concentrates in midday hours (11–14, white band between the two red bands) — precisely the hours we exclude from the control. The high σ_{within} in critical hours coincides with the morning solar rise and the evening solar fade (see Table 2 for the per-hour values); the within-hour residual-demand swings in those windows are driven jointly by demand ramps and solar transitions. Reproduced from main paper §2.1.

2 Who price-sets

2.1 For the day-ahead market

Method (compressed). OMIE submits stepwise orders to EUPHEMIA: in-the-money fully accepted, out-of-the-money fully rejected, **at-the-money can be curtailed**. The price-setter is the unit whose at-MCP step is partially accepted. Operationally, with q_{below} = sum of unit's bids strictly below MCP, q_{at} = sum at MCP (±0.01 EUR/MWh), and q_{assigned} = unit's PDBC cleared MWh:

$$q_{\text{below}} + \varepsilon \; < \; q_{\text{assigned}} \; < \; q_{\text{below}} + q_{\text{at}} - \varepsilon$$

Excluded: full acceptance ($q_{\text{assigned}} \geq q_{\text{below}} + q_{\text{at}}$, inframarginal at MCP) and curtailed-to-zero ($q_{\text{assigned}} \leq q_{\text{below}}$, often MIC deactivation). Sources: clearing price from OMIE day-ahead clearing-price series; bid stack from OMIE day-ahead bid-detail and bid-header tables; cleared MWh from OMIE day-ahead programs. **Not** PDBF (bilaterals) or PHF (REE redispatch). Aggregations below MW-weighted by q_{at} unless stated. Paradoxical rejection only applies to block orders (not in DET), so no separate filter is needed.

Naive vs corrected. A naive rule that flags every unit with p_{bid} = MCP (this is the rule CNMC uses in its annual market-supervision report, IS/DE/013/24; see Table 5 below) would pool all three cases; the direction of bias is tech-specific. For CCGT 2025Q4 critical, the CNMC-style 9.1% becomes 2.7% under partial-acceptance (most CCGT at-MCP steps are MIC-deactivated to zero); for 2024Q4 critical, 13.8% becomes 18.1%.

Side composition: sell-side dominates critical, buy-side dominates flat/midday in MTU15-DA. Periods where at least one sell unit / at least one buy unit has an at-MCP step.

Regime, hour-class	sell-only	buy-only	both	neither	n periods
DA15/ID15 Critical	53%	25%	11%	11%	4,048
DA15/ID15 **Flat**	36%	**42%**	10%	12%	1,104
DA15/ID15 Midday	28%	**38%**	18%	17%	1,472
DA60/ID15 post-blk Critical	53%	19%	14%	14%	1,716
DA60/ID15 post-blk Flat	60%	17%	9%	13%	468
3-sess Critical	52%	21%	11%	15%	1,870

Table 3: At-MCP side presence. "neither" = EUPHEMIA price-indeterminacy zone (MCP between bid steps, mid-point rule). In DA15/ID15 flat and midday, more periods are buy-side-only price-setting than sell-side-only.

Table 4: Price-setter shares by technology and hour-class, across 5 reform-window regimes (sell-side partial-acceptance, MW-weighted by q_{at}). C = Critical, F = Flat. Caveat: DA60/ID15 pre-blackout is 40 days only — directional.

	3-sess		**ISP15-win**		**DA60/ID15 pre**		**DA60/ID15 post**		**DA15/ID15**	
Tech	C	F	C	F	C	F	C	F	C	F
Wind	28.3	17.9	37.8	63.9	47.5	70.9	29.6	17.3	29.3	53.0
Solar PV	17.7	0.0	1.4	0.0	20.6	0.0	29.9	0.0	7.7	0.0
Hydro pump	10.4	13.3	17.0	2.7	2.2	10.8	8.0	19.3	28.4	13.0
Hydro	11.9	36.1	19.6	10.9	2.7	1.7	9.4	21.5	23.7	20.0
Nuclear	12.5	20.2	4.2	6.5	15.6	13.5	7.9	5.5	4.4	3.3
CCGT	8.0	8.9	17.2	11.1	0.8	0.0	6.2	33.1	2.7	2.0
Cogen	3.0	0.3	0.1	0.1	2.5	0.1	2.1	0.1	0.5	0.9
Other RES	3.9	1.5	0.6	0.2	3.4	0.8	3.1	1.2	1.2	2.2
Import	0.8	0.8	0.5	3.1	0.0	0.0	0.0	0.0	0.5	3.3

Sell-side price-setter shares by technology and regime (headline table).

Cross-check: CNMC's broader marginal-tech metric. The Spanish regulator CNMC publishes an analogous metric in its annual market-supervision report (*Informe de supervisión del mercado peninsular mayorista al contado de electricidad. Año 2023*, IS/DE/013/24, 28 Jan 2025), where it states for example that *"los ciclos combinados se convierten en la tecnología marginal en un porcentaje elevado de las horas (21,7% cuota ciclos, 43,3% cuota renovable en septiembre"* of 2023 (§2.5.3, p. 51). CNMC's metric counts a unit as marginal whenever its highest-accepted sell tranche lands at MCP, *regardless of whether the at-MCP step is curtailed or fully accepted.* Our partial-acceptance metric above is strictly tighter (it requires the at-MCP step to be *strictly partially* accepted). Table 5 reports the CNMC-style version of the CCGT row of Table 4 for our 5 reform windows; the two metrics give the same qualitative ranking but the CNMC version is mechanically larger.

Table 5: CCGT marginal-tech share, CNMC-style vs partial-acceptance. CNMC-style = % of MW-weighted votes where the rank-1 accepted sell tranche (highest p_{bid} with $p_{\text{bid}} \leq$ MCP) belongs to a CCGT unit; ties at rank 1 split the vote equally across units. This matches the share concept used in CNMC IS/DE/013/24 Cuadro 11 (p. 81). Partial-acceptance = the strict EUPHEMIA-aware rule of Table 4. C = Critical, F = Flat, M = Midday. CNMC's reported September 2023 figure for CCGT (21.7%, peninsular monthly average) is roughly the order of magnitude of our 3-sess critical-hour CCGT share under the CNMC-style metric (13.1%) and the ISP15-win critical-hour share (14.1%).

	CNMC-style (rank-1 at MCP)			**Partial-acceptance (ours)**		
Regime	C	F	M	C	F	M
3-sess (Jun–Nov 2024)	13.1	5.9	6.9	8.0	8.9	—
ISP15-win (Dec24–Mar25)	14.1	13.0	14.7	17.2	11.1	—
DA60/ID15 pre-blk (Mar19–Apr27)	2.2	0.0	0.7	0.8	0.0	—
DA60/ID15 post-blk (Apr28–Sep)	11.1	20.5	0.4	6.2	33.1	—
DA15/ID15 (Oct–Dec 2025)	9.1	9.3	1.9	2.7	2.0	—

Why the two metrics diverge. The largest gap is DA15/ID15 critical (9.1 CNMC vs 2.7 ours): under MTU15-DA, most CCGT at-MCP steps are *MIC-deactivated to zero* ($q_{\text{assigned}} \leq q_{\text{below}}$), which the CNMC rule still counts as "marginal" but the partial-acceptance rule does not. The DA60/ID15 post-blackout flat row goes the other way (20.5 CNMC vs 33.1 ours), because in that regime a couple of Endesa CCGT units (SROQ2, PGR5) submit single-block full-nameplate tranches at MCP that get *partially* accepted (driving up the partial-acceptance share) but are not the rank-1 marginal step in many of those periods. Both metrics agree that DA60/ID15 pre-blackout has near-zero CCGT marginal presence and that DA15/ID15 critical is much lower than ISP15-win or 3-sess.

CCGT firm-level: GN never price-sets, IB and GE swap dominance. Within the CCGT column above, who owns the at-MCP step?

Regime	Hour-class	IB	GE	GN	HC	OTHER
3-sess	Critical	**60**	31	0	0	8
3-sess	Flat	16	**50**	0	0	25
ISP15-win	Critical	35	**57**	0	0	5
ISP15-win	Flat	2	**74**	0	3	20
DA60/ID15 post-blk	Critical	31	**69**	0	0	0
DA60/ID15 post-blk	Flat	19	**80**	0	0	0
DA15/ID15	Critical	**65**	29	0	6	0
DA15/ID15	Flat	**80**	0	6	14	0

Table 6: Per-firm shares within the CCGT price-setter pool (% of CCGT q_{at}). **GN's CCGT bid stack is so heavily concentrated at scarcity prices (≥ 500 EUR/MWh) that GN essentially never lands at MCP**. IB and GE alternate as the dominant CCGT price-setter. The 33.1% CCGT flat-hour figure in DA60/ID15 post-blackout is 80% GE.

Top price-setting units in DA15/ID15 (for context). MUEL (Iberdrola's Muela del Cortes pump-storage) is the workhorse: 496 events / ∼5.5 events per day. Other Iberdrola hydro (DUER 201, TAJO 171, SIL 130 events) and the IBEVD11 wind aggregator (153 events, but ∼ 1,613 MW per event = 247 GW total q_{at}) dominate the top 7. Endesa's MLTG pump-storage (164 events) is the only non-Iberdrola unit in the top 5.

Reading across regimes (synthesis).

- **Wind is the largest sell-side price-setter** in almost every cell (18–71% MW share). RE-Mercado aggregator portfolios systematically land at-the-money; 70–99% of wind/solar/cogen at-MCP events involve a single block at MCP with no in-the-money bids (the deep-dive companion data). CCGT, hydro, and pump-storage instead use true stacks (24–34% single-block).

- **Iberdrola dominates the marginal MW in DA15/ID15** (top-1 firm-share ≈ 65% critical, 80% flat; firm HHI > 4,000), through Hydro pump (MUEL), Hydro (DUER, TAJO, SIL), and the IBEVD11 wind aggregator.

- **CCGT's share fluctuates strongly across regimes**: 8% (3-sess) → 17% (ISP15-win) → 0.8% (DA60/ID15 pre-blk, tiny window) → 6% / 33% (flat) (DA60/ID15 post-blk) → 3% (DA15/ID15) in critical hours. The 33% flat-hour spike is 80% GE-owned, driven by ∼ 2 units (SROQ2, PGR5 — both Endesa) submitting single full-nameplate (∼360 MW) tranches at MCP (MIC-reservation style); cell-count-weighted alternative is 17%.

- **GN does not partial-accept at MCP, but is strategically active within-hour.** GN's CCGT bid mass sits at scarcity prices (≥ 500 EUR/MWh) and rarely lands at MCP for partial acceptance, but the bid *shape* across the 4 within-hour quarters is the largest of any CCGT firm in terms of median price shift (see Table 13 below). Direct inspection of BES4 (GN) confirms: on 2025-10-15 hour 18, Q1 is binary (1 EUR / 1000 EUR), Q2 the same, Q3 adds a small near-clearing tier, Q4 has a full 10-step ladder at 81–97 EUR/MWh.

- **Solar PV** dominates midday post-MTU15-DA (35.1%, see CSV) and the negative/zero-MCP scarcity bin (40–65%). CCGT never sets the price at MCP ≤ 0.
- **Buy-side price-setters** are dominated by pump-load demand bids (46–84%) and retailer demand bids (14–53%). In DA15/ID15 flat hours, buy-side sets the price about 42% of periods (vs 36% sell-side).
- **Cross-regime CCGT symmetry** (simple TTF/gas-shock falsifier) holds in 3-sess (8.0 vs 8.9) and DA15/ID15 (2.7 vs 2.0), but breaks in ISP15-win (17.2 vs 11.1) and DA60/ID15 post-blk (6.2 vs 33.1).

2.2 For the intraday markets

Method. Identical partial-acceptance rule as §2.1, with three data substitutions: clearing price from OMIE intraday-auction clearing-price series (per session × period), bid stack from OMIE intraday-auction bid-detail data + OMIE intraday-auction bid-header data (sell-side, latest version, pools across the 3 IDA sessions per day), cleared MWh per unit from OMIE intraday-auction programs per unit. Matched-period share is 26–32% across regimes (similar to DA).

Table 7: IDA price-setter shares by technology, hour-class, and regime (at-the-money partial-acceptance, sell side), MW-weighted by q_{at}. C = Critical, F = Flat. Shares sum to 100 per (regime, hour-class).

	3-sess		**ISP15-win**		**DA60/ID15 pre**		**DA60/ID15 post**		**DA15/ID15**	
Tech	C	F	C	F	C	F	C	F	C	F
Wind	35.5	15.7	24.8	40.2	52.6	53.8	37.8	18.8	31.9	42.4
CCGT	14.1	29.2	16.1	15.3	3.1	2.8	9.7	23.1	**24.2**	**32.6**
Hydro pump	12.9	23.4	17.2	2.2	2.5	13.0	6.2	26.1	14.0	3.8
Solar PV	10.5	0.0	1.7	0.1	20.3	0.0	22.8	0.0	7.5	0.1
Hydro	6.4	15.9	18.4	23.4	4.4	3.4	5.6	15.6	10.0	10.6
Retailer	10.2	6.2	10.4	12.1	5.3	7.5	7.3	11.8	5.7	5.2
Nuclear	2.7	3.2	0.0	0.0	1.0	13.8	2.5	1.3	2.5	0.0
Other RES	1.0	1.1	1.7	0.0	3.1	0.7	2.4	0.2	0.6	0.4

Reading the IDA table.

- **CCGT is much more active as an IDA price-setter than as a DA price-setter in DA15/ID15.** Critical-hour CCGT IDA share is **24.2%** (vs **2.7%** in DA); flat-hour CCGT IDA share is **32.6%** (vs **2.0%** in DA). A factor of $\sim$ 9–16. The natural reading: CCGTs sit out of DA (bidding above the clearing price; see also the q_1-drop story in Part B, §5) and re-enter via IDA, where they now systematically clear at the marginal step.
- **Wind dominates IDA too**, but less than in DA: 32% IDA critical share in DA15/ID15 vs 29% DA critical share. Wind aggregators play in both markets.
- **CCGT IDA share collapsed in the DA60/ID15 pre-blackout window** (3.1% critical) but recovered post-blackout (9.7%) and grew to 24.2% under DA15/ID15. The post-MTU15-DA jump is striking and consistent with a DA-to-IDA strategic substitution.
- **Retailer (buy-side, IDA)** sets the price 5–12% of IDA critical-hour mass — comparable order to DA. Buy-side matters in IDA too.

3 Setup: bid-shape metrics (D_w, Δp, Δq)

D_w, in full. Let cell (f, u, d, h) index a firm f, unit u, date d, clock-hour h. The hour partitions into 4 quarters $q \in \{1, 2, 3, 4\}$. In quarter q the unit submits a multiset of price-quantity sell tranches with $p \in [\underline{p}, \overline{p}] \subset \mathbb{R}$ (Spanish SDAC: $\underline{p} = -500$, $\overline{p} = 4{,}000$ EUR/MWh; negative bids are admissible) and $x > 0$ (MWh, energy for the quarter-slot). Let $c_{d,h}$ be the hour-mean clearing price (justified below) and fix a strategic half-band of half-width $h = 50$ EUR/MWh (justified below) around it. **We work throughout with band-restricted supply curves**: N_q is the number of *distinct* prices submitted in quarter q *within* $[c - h, c + h]$, and x_{qk} is the sum of quantities at p_{qk}. The band-local quarter-q supply curve is the right-continuous step function

$$S_q(p) \;=\; \sum_{k=1}^{N_q} x_{qk}\,\mathbf{1}\{p_{qk} \le p\}, \qquad p \in [c - h, c + h],$$

with $S_q(c - h) = 0$ and $S_q(c + h) = M_q$ (the band-mass of quarter q — the total MWh offered *anywhere inside* the band $[c - h, c + h]$, not the offer at the maximum-price threshold). Infra-marginal mass below $c - h$ is excluded by construction: D_w measures the within-band *shape* of supply across quarters, not infra-marginal level offsets.

For two quarters i, j, the 1-D Wasserstein-1 distance is the L^1 area between cumulative curves, equivalently the L^1 distance between (extended) quantile functions:

$$W_1^{\text{band}}(S_i, S_j) \;=\; \int_{c-h}^{c+h} \big|S_i(p) - S_j(p)\big|\,dp \;=\; \int_0^{\max(M_i, M_j)} \big|S_i^{-1}(Q) - S_j^{-1}(Q)\big|\,dQ,$$

where $S_q^{-1}(Q) = c + h$ for $Q > M_q$ pads the smaller curve to common total mass (Vallender, 1973). To isolate variation near the clearing price, weight by an (unnormalized) Epanechnikov-shape kernel:

$$K_h(p - c) \;=\; \max\left\{0,\; 1 - \left(\frac{p - c}{h}\right)^2\right\}, \quad K_h \in [0, 1], \quad K_h(c) = 1,$$

$$D_w^{(i,j)} \;=\; \int_{c-h}^{c+h} K_h(p - c)\,\big|S_i(p) - S_j(p)\big|\,dp \;\le\; W_1^{\text{band}}.$$

D_w has units EUR (quantity in MWh times price in EUR/MWh).

Why c = hour-mean clearing price. The band $[c - h, c + h]$ is meant to isolate the strategic pricing region — the price range where the unit's bids actually had a non-trivial probability of clearing in this hour. We choose $c = c_{d,h}$ = the hour-mean of the four within-hour quarter clearing prices. Three reasons:

- **Symmetric across quarters.** The metric is comparing the 4 quarter bid curves; if we used each quarter's *own* clearing price as its own reference, the band would shift across quarters and the within-quarter dissimilarity statistic would no longer be comparable across quarters of the same hour — we would be measuring shifts of the reference, not shifts of the curve.

- **In the relevant pricing region.** The hour-mean sits, by construction, in the middle of the within-hour clearing-price range, so $[c - h, c + h]$ contains all quarter clearings (typically) plus the surrounding band where firms are competing on margin.

- **One reference per (cell, hour).** A single c per (date, clock-hour) cell keeps the statistic $D_w^{(i,j)}$ defined on a common interval for all 6 pairs in the cell.

Alternative for robustness: use the hourly DA clearing price (which is exogenous to within-hour quarter shaping by construction — the DA clears at the hour level even post-MTU15-IDA). This would be more exogenous to the firm's own shaping but would lose connection to actually realised within-hour prices. Flagged for sensitivity.

Why $h = 50$ EUR/MWh. The half-width h is the tuning parameter that controls how much of the bid curve enters the dissimilarity statistic. We pick $h = 50$ EUR/MWh on three grounds:

- **Order-of-magnitude of clearing prices.** Spanish DA clearing prices in 2024–25 typically sit in the 50–150 EUR/MWh range; a ±50 band captures bid mass within roughly one "clearing-price width" of clearing, which is the region where bid placement matters for being on the margin.
- **CCGT marginal-cost scale.** CCGT marginal costs in our sample range roughly 80–200 EUR/MWh; $h = 50$ is well below this range, so the band excludes far-from-clearing bids that effectively never compete.
- **SDAC range scale.** On a $[-500, 4000]$ price domain $h = 50$ is a small slice ($\sim 1\%$ of the upper bound) — it does *not* include strategic high-cap bids (which sit at 800–3000 EUR/MWh and play a different role).

This is a choice that should be tested empirically, not asserted. The check below sweeps $h \in \{20, 50, 100, 200\}$ and confirms the qualitative ranking (GN dominant on Δq, GE leaning on Δp, IB/HC negligible, flat hours ≈ 0) holds at all four values. A more principled empirical anchor for a future pass would be to pick h as the price interval that captures, say, 90% of clearing-price-setter bids for the median CCGT unit, computed from the bid-side data; we use $h = 50$ as the working choice and treat it as a tuning parameter, not a structural one.

Table 8: h-sensitivity of the channel-active percentages, CCGT DA Oct–Dec 2025. Each cell: % of (firm, unit, date, hour) cells with the channel active ($\Delta q > 0$ or $\Delta p > 0$) under band half-width h EUR/MWh. The qualitative ranking (GN dominant on Δq, GE leaning on Δp, IB/HC negligible, flat hours ≈ 0) holds across all four h values. The working choice $h = 50$ is bolded.

		Δq-active (%)				Δp-active (%)			
Firm	Hour	h=20	**h=50**	h=100	h=200	h=20	**h=50**	h=100	h=200
GN	critical	12.51	18.37	24.07	24.54	0.01	0.01	0.01	0.01
GN	flat	0.00	0.00	0.05	0.05	0.00	0.00	0.00	0.00
GE	critical	0.46	0.33	0.27	0.27	0.69	1.56	1.64	1.64
GE	flat	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
IB	critical	0.32	1.94	4.25	5.67	0.01	0.10	0.37	2.52
IB	flat	0.28	1.03	2.20	2.48	0.14	0.89	1.31	1.40
HC	critical	0.05	0.05	0.05	0.05	0.00	0.00	0.00	0.00
HC	flat	0.00	0.00	0.00	0.00	0.00	0.00	0.18	0.55

Reading the sensitivity. Across all four h:

- **GN's Δq dominance is robust.** GN critical-hour Δq-active rises monotonically with h ($12.5\% \to 18.4\% \to 24.1\% \to 24.5\%$) because a wider band captures more pairs that differ in band-mass. The *ranking* (GN $\gg$ everyone else on Δq critical) is preserved everywhere.
- **GE's Δp leaning is robust** and saturates by $h = 50$ ($0.69 \to 1.56 \to 1.64 \to 1.64\%$). The price-channel signal is concentrated within ±50 EUR/MWh of clearing; expanding the band further captures essentially no additional Δp-active cells.

- **Flat hours stay** ≈ 0 across all firms and all h on both channels (max value: IB flat Δq-active at $h = 200$ is 2.5%; everything else $\leq 1.4\%$). The crit-vs-flat asymmetry is the same first-order story under every band choice.

- **IB shows the steepest scaling with** h on both channels, suggesting IB's near-clearing bidding is sparser and only registers as "active" once the band is large enough to catch its quarter-pair differences. Not a robustness concern (IB is small in absolute terms) but worth noting: IB's signal is at the edge of detectability.

The bolded $h = 50$ column is the working choice for the headline tables that follow.

Cell-level aggregation: max vs mean across the 6 within-hour pairs. A cell has 4 within-hour quarters and therefore $\binom{4}{2} = 6$ pairwise dissimilarities $D_w^{(i,j)}$. We aggregate to a single cell-level statistic by taking the *maximum*:

$$D_w^{\text{cell}} = \max_{1 \leq i < j \leq 4} D_w^{(i,j)}.$$

The choice between max and mean matters in two different ways:

- For the *binary flag* $D_w^{\text{cell}} > 0$ (the metric in the operational-vs-strategic table below, in the IDA pre/post-blackout table, and in the $\Delta p/\Delta q$-active shares in Table 11), the max-vs-mean choice is **irrelevant**: a cell has any pair > 0 if and only if max > 0 if and only if mean > 0. The flag rates reported in Tables 9, 10, 11 therefore do not depend on this choice.

- For the *conditional intensity* (e.g. the median D_w rows in Table 12), the choice does matter: max picks the most-different pair and biases toward the upper tail of pair-wise dissimilarities; mean averages over all 6 pairs and dampens outlier-pair effects. We report max because the substantive question is "did any within-hour shaping occur in this cell, and if so, how large was the most-different pair?"; a mean-aggregated companion ("how large is the typical pair?") is a worthwhile robustness and is flagged for the next pass.

Decomposition into price and quantity channels. We split the within-band variation into a price channel and a quantity channel:

$$\Delta q_{ij} = |M_i - M_j| \ \text{[MWh]}, \qquad \Delta p_{ij} = \frac{1}{\min(M_i, M_j)} \int_0^{\min(M_i,M_j)} \left| S_i^{-1}(Q) - S_j^{-1}(Q) \right| dQ \ \text{[EUR/MWh]}.$$

Δp captures the mean price shift on the common-mass portion — a price-only adjustment with band-mass totals held fixed, in EUR/MWh. Δq captures band-mass mismatch — a quantity-only adjustment with prices on the common-mass portion held fixed, in MWh. Cell-level Δp, Δq are the max over the 6 pairs (same caveats as above).

Caveat on Δq: this number is coarse on its own. $\Delta q_{ij} = |M_i - M_j|$ is the *absolute* difference between the two quarters' total band-mass (in MWh). The choice is dictated by the Wasserstein upper-bound proof below — without exactly this Δq, the bound $W_1^{\text{band}} \leq \Delta p_{ij} M_i + 2h \Delta q_{ij}$ does not hold and the decomposition is not consistent. But **on its own, raw Δq is uninformative without context**: 10 MWh of asymmetric mass between quarters is huge in a 20 MWh band-mass cell and negligible in a 500 MWh one. The tables below currently report Δq in raw MWh; a relative version $\Delta q/\bar{M}$ where $\bar{M} = (M_i + M_j)/2$ is the mean band-mass would scale-normalise it, and is flagged for the next pass. In the empirical reading below we lean on the *active/inactive* distinction ($\Delta q > 0$ vs $\Delta q = 0$) and on the loose-bound diagnostic at the end of this section, both of which are robust to the scale issue. (A further weakness: *two quarters with identical total band-mass but different curve shapes get* $\Delta q = 0$ *even though the quantity channel is non-trivial.* What survives is the price channel Δp; the decomposition then attributes all shape variation to price, even when it is properly a shape difference in quantity.)

Short proof of consistency. WLOG $M_i \le M_j$. Extending $S_i^{-1}(Q) = c+h$ for $Q \in (M_i, M_j]$ gives both quantile functions a common domain $[0, M_j]$ and common total mass M_j. The 1-D Wasserstein-1 duality for distributions with common mass (Vallender, 1973) gives

$$W_1^{\text{band}} \;=\; \int_{c-h}^{c+h} \big|S_i(p) - S_j(p)\big|\, dp \;=\; \int_0^{M_j} \big|S_i^{-1}(Q) - S_j^{-1}(Q)\big|\, dQ.$$

Split the right-hand integral at $Q = M_i$:

$$W_1^{\text{band}} = \underbrace{\int_0^{M_i} \big|S_i^{-1}(Q) - S_j^{-1}(Q)\big|\, dQ}_{=\,\Delta p_{ij}\, M_i \;\;\text{(definition of } \Delta p)} \;+\; \underbrace{\int_{M_i}^{M_j} \big(c{+}h - S_j^{-1}(Q)\big)\, dQ}_{=:\, R_{ij}}\,.$$

The integrand of R_{ij} lies in $[0, 2h]$ (since $S_j^{-1}(Q) \in [c-h, c+h]$), so $0 \le R_{ij} \le 2h\,\Delta q_{ij}$. Since $K_h \in [0,1]$ on the band,

$$D_w^{(i,j)} \;\le\; W_1^{\text{band}} \;\le\; \Delta p_{ij}\, M_i + 2h\,\Delta q_{ij}.$$

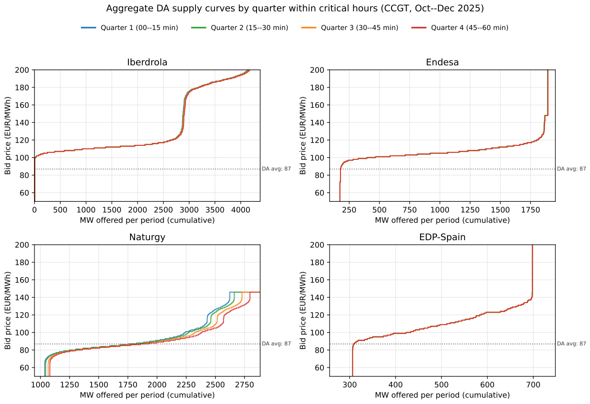
Limiting cases. (i) If $M_i = M_j$, then $R_{ij} = 0$ and $\Delta q_{ij} = 0$, so $W_1^{\text{band}} = \Delta p_{ij}\, M$: all variation is on the price channel. (ii) If $S_i^{-1} \equiv S_j^{-1}$ on $[0, M_i]$ (quantile functions coincide on common mass), then $\Delta p_{ij} = 0$ and $W_1^{\text{band}} = R_{ij} \in [0,\, 2h\,\Delta q_{ij}]$: all variation is on the quantity channel. The two channels are orthogonal modes verifiable at the boundary cases. □

Empirical loose-bound diagnostic. The upper bound $W_1^{\text{band}} \le \Delta p_{ij}\, M_i + 2h\,\Delta q_{ij}$ is mathematically tight only in the two limiting cases above. Empirically in our CCGT data $R_{ij} \ll 2h\,\Delta q_{ij}$ (median $R/(2h\,\Delta q) = 0.003$): the extra band-mass sits near the top of the band, so the Δq channel contributes little to W_1^{band} even when Δq is large. The decomposition is mathematically exact in the limit cases but the upper bound is loose by $\sim 13\times$ on average. **Interpret Δp and Δq as channel *indicators* (active/inactive, and which channel is dominant), not as additive contributions to W_1^{band}.**

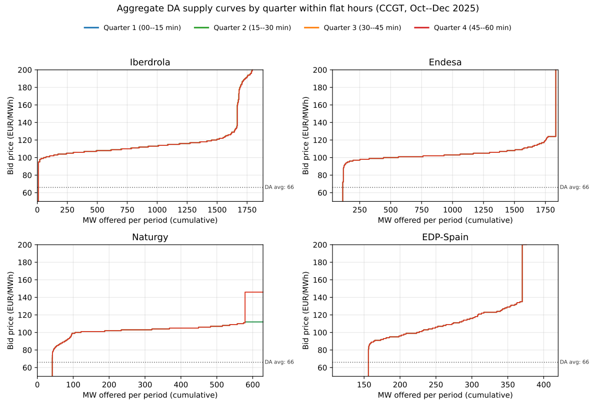
3.1 Bid graphs by tech, firm, and within-hour quarter

The figures in this subsection display the aggregate within-hour quarter bid (or demand) curves, broken down by firm and hour-class. Each curve overlays the mean DA (or IDA) clearing price as a vertical reference so the strategic-band $|p - c| \le 50$ EUR/MWh region is visible. Aggregation: average across ∼92 dates per (firm, hour, quarter) cell for the DA Oct–Dec 2025 window. Day-specific quarter dispersion is smoothed by this construction; the kernel-weighted dissimilarity in §3.2 is the right quantitative complement.

CCGT (DA): aggregate per-firm quarter curves, critical vs flat. *Reading* (**mixed**). The aggregate construction smooths per-day strategic shaping; the within-hour shaping table at the end of §3.2 is the right systematic measure.



(a) Critical hours (05:00–09:00, 16:00–23:00)



(b) Flat hours (01:00–04:00) — falsification

Figure 4: Aggregate DA supply curves by quarter, CCGT, Oct–Dec 2025. Aggregated across ∼92 dates per (firm, hour, quarter) cell; day-specific quarter dispersion is smoothed in this visual.

CCGT (IDA): per-firm + quarter overlays. *Reading* (**supportive**). In the post-MTU15-DA window the IDA pattern matches DA: dispersion concentrated in Naturgy and visible only in critical hours.

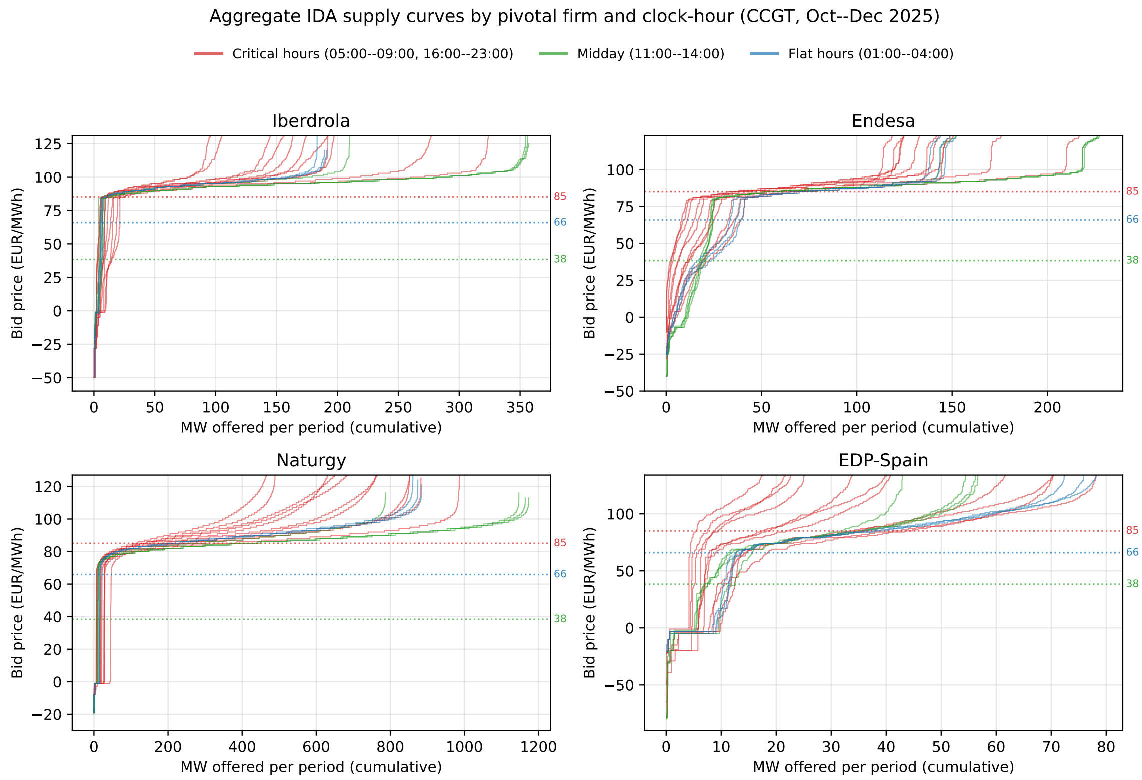
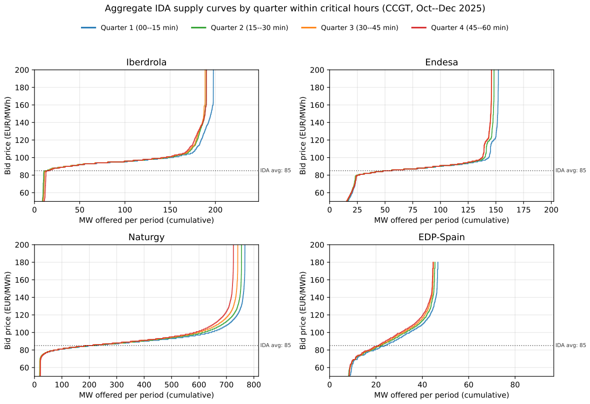
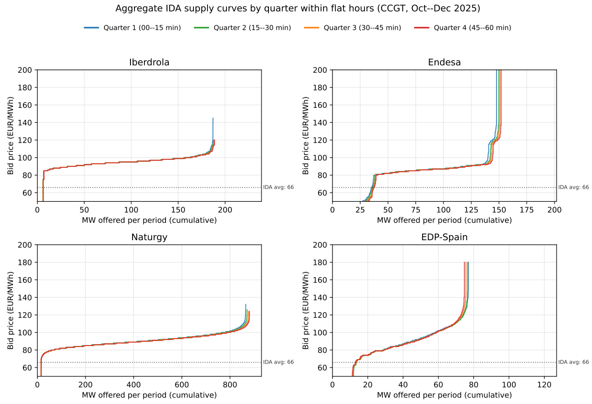


Figure 5: Aggregate IDA supply curves by pivotal firm and clock-hour, CCGT, Oct–Dec 2025.



(a) Critical hours (05:00–09:00, 16:00–23:00)

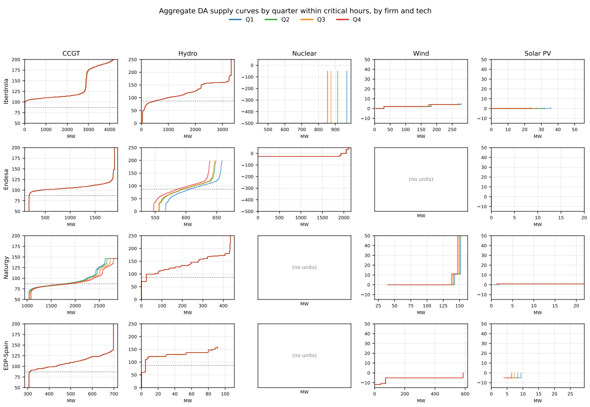


(b) Flat hours (01:00–04:00)

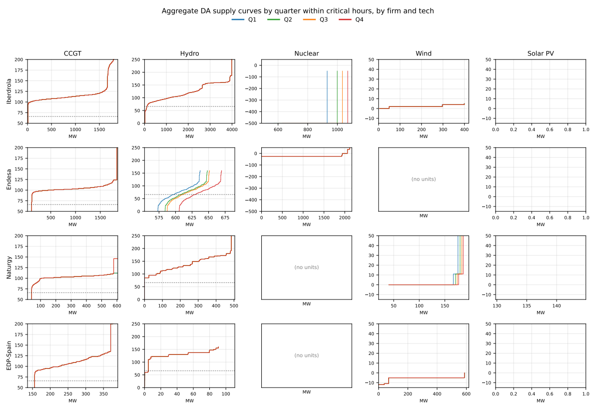
Figure 6: Aggregate IDA supply curves by quarter, CCGT, Oct–Dec 2025.

Pump-storage demand bids (DA + IDA). *Reading* (**mixed**). Aggregate-construction pump-storage demand curves look broadly symmetric across quarters in this picture, but the within-hour shaping table at the end of §3.2 shows that on the supply side both IB (MUEL) and GE (MLTG) pump-storage are highly Δp-active (86–88% of cells) once we look per-day and per-firm rather than at the aggregate.

Compact multi-tech grids: firm $\times$ tech $\times$ quarter, all techs side by side. The grids below stack rows = Big-4 firms (Iberdrola, Endesa, Naturgy, EDP-Spain) against columns = techs (CCGT, Hydro, Nuclear, Wind, Solar PV). Each panel overlays the 4 within-hour quarter supply curves (Q1 blue $\to$ Q4 red) with the mean DA / IDA clearing price as a dotted horizontal line. Y-range is tech-specific (TECH_YLIM) so the strategic-band region is visible; x-range is per panel. Empty panels are firms with no units in that tech (e.g. Naturgy has no Nuclear). *Reading*: the same-quarter overlap in flat hours is the falsification; visible quarter spread in critical (and to some extent midday) hours is the strategic signal.

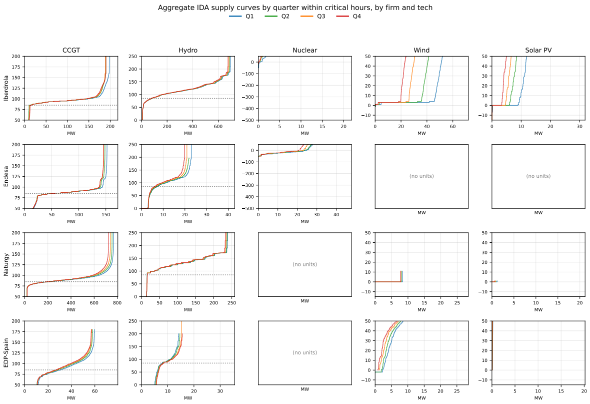


(a) DA, critical hours.

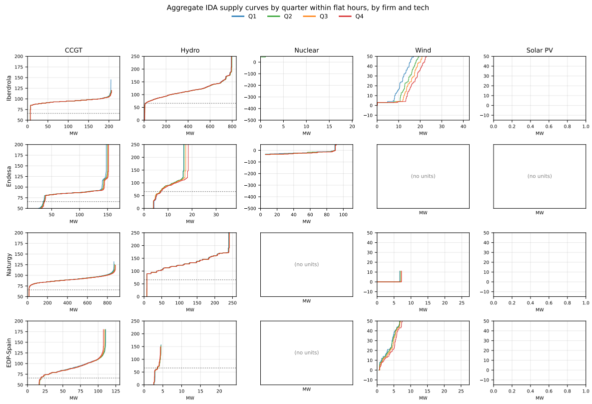


(b) DA, flat hours — falsification.

Figure 7: DA compact grid (firms $\times$ techs $\times$ quarter), DA15/ID15. Quarter overlap in flat hours is essentially complete; critical-hour spread is visible for GN CCGT and Endesa Hydro.

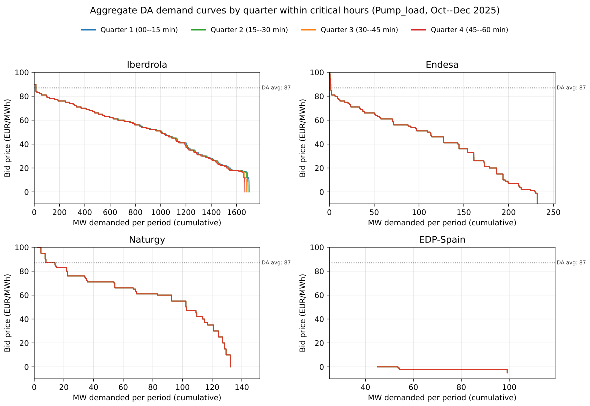


(a) IDA, critical hours.

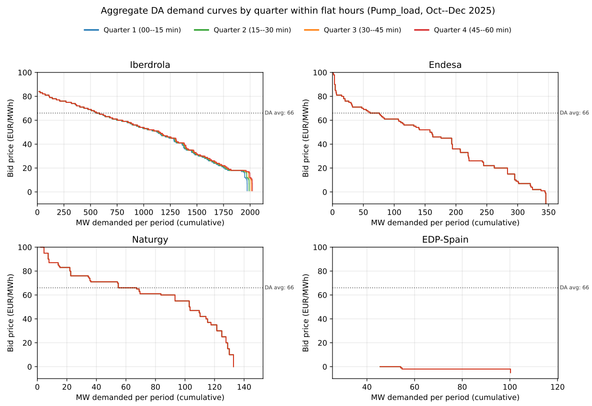


(b) IDA, flat hours.

Figure 8: IDA compact grid (firms × techs × quarter), DA15/ID15. IDA-side analogue. Quarter spread on the supply side is more visible than in DA (IDA is more often where CCGTs partially clear — see §2.2). Midday-hour DA and IDA grids are also produced and inspected; see the synthesis discussion of midday patterns.



(a) DA, critical hours



(b) DA, flat hours

Figure 9: Aggregate DA demand curves by quarter, pump-storage charge bids, Oct–Dec 2025.

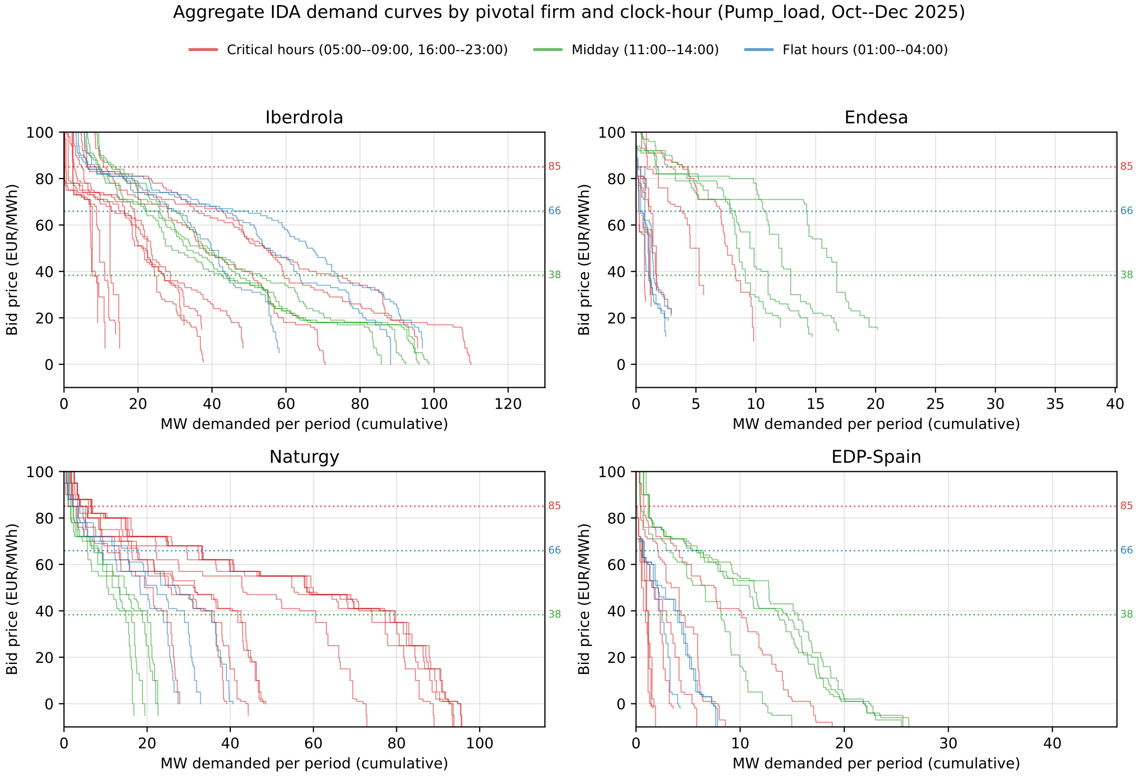
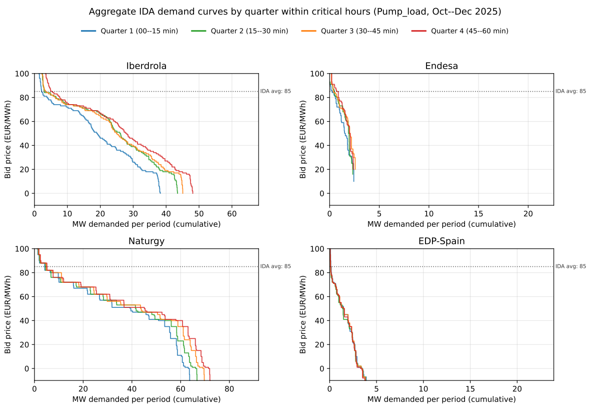
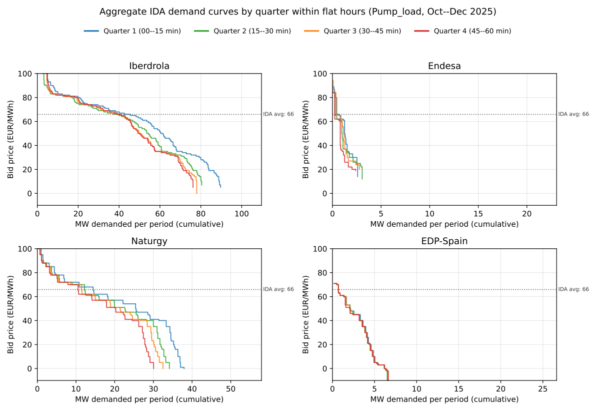


Figure 10: Aggregate IDA demand curves by pivotal firm and clock-hour, pump-storage charge bids, Oct–Dec 2025.



(a) IDA, critical hours



(b) IDA, flat hours

Figure 11: Aggregate IDA demand curves by quarter, pump-storage charge bids, Oct–Dec 2025.

3.2 D_w near clearing, Δp / Δq tables and histograms

The results below quantify what the bid-curve figures in §3.1 show visually. We report (i) the per-cell binary flag $D_w > 0$ across techs and firms (operational vs strategic decomposition, IDA pre/post-blackout) and (ii) the $\Delta p/\Delta q$ channel-shares and intensity tables (CCGT DA, full multitech DA15/ID15) plus D_w log-histograms across the full firm × tech × regime grid.

DA Oct–Dec 2025: operational vs strategic decomposition by (tech, firm). *Reading* (**mixed**). The original main-paper framing reported a critical–flat asymmetry with the interpretation "only Naturgy CCGT shows the strategic signature". The IDA pre/post-blackout analysis (below) and the multitech within-hour shaping table (at the end of this subsection) complicate this: GN's CCGT D_w activity is real and large, IB hydro and pump-storage are also highly active, and under reforzada all four firms show the IDA-side strategic signature, not just Naturgy.

Table 9: DA Oct–Dec 2025 critical–flat asymmetry, by (channel, technology, firm). Percentage of (firm, unit, date, hour) cells with $D_w > 0$. *Channel*: Strategic = exploitation of within-hour granularity near clearing; Operational = forecast/load-shape updates; Mechanical = smooth pump-storage charge. Compare: IDA in Table 10, multitech in Table 13.

Channel	Tech	Firm	% crit	% flat	Crit−Flat (pp)
Strategic	CCGT	GN	24.5	0.0	+24.5
Strategic	CCGT	IB	5.8	3.9	+1.9
Operational	Wind	IB	97.2	97.3	−0.1
Operational	Wind	GN	92.7	92.6	+0.1
Operational	Wind	HC	93.7	93.8	−0.1
Operational	Solar PV	GN	15.9	0.0	+15.9
Operational	Retailer	GE	76.5	76.3	+0.2
Operational	Retailer	IB	40.9	40.2	+0.7
Operational	Direct_consumer	IB	96.0	95.3	+0.7
Mechanical	Pump_load	IB	3.2	5.5	−2.3
Mechanical	Pump_load	GE	0.0	0.0	0.0
Mechanical	Hydro_pump	IB	0.4	0.0	+0.4
Mechanical	Hydro_pump	GE	0.0	0.0	0.0

IDA pre/post-blackout near-clearing dissimilarity. For each (firm, unit, date, hour) CCGT IDA cell, Epanechnikov-weighted within-hour quarter dissimilarity D_w ($h = 50$ EUR/MWh, centred on hour-mean IDA clearing price). Pre-blackout: 2025-03-19 → 2025-04-27. Post-blackout: 2025-04-28 → 2025-09-30 (reforzada). Hypothesis: REE post-clearing intervention should remove the leverage on clearing price, so pre > post.

Table 10: IDA quarter-dissimilarity, pre vs post-blackout, CCGT. Percentage of (firm, unit, date, hour) cells with $D_w > 0$ (i.e., quarters differ within the Epanechnikov band, $h = 50$ EUR/MWh, around hour-mean IDA clearing price). *Pre-blackout*: 2025-03-19 → 2025-04-27 (40 days, MTU15-IDA active, reforzada-free). *Post-blackout*: 2025-04-28 → 2025-09-30 (155 days, reforzada active).

	Pre-blackout			Post-blackout		
Firm	Critical	Flat	Crit−Flat	Critical	Flat	Crit−Flat
IB	9.6	7.0	+2.6	20.3	6.2	+14.1
GE	14.1	10.0	+4.1	43.8	33.2	+10.6
GN	30.5	9.2	**+21.3**	53.8	37.9	+15.9
HC	27.9	20.7	+7.2	34.2	21.2	+13.0

Reading (**mixed**). (i) Pre: only GN shows clean asymmetry (+21pp). (ii) Post: all four firms show +11 to +16pp gap. (iii) Pre > post rejected for every firm; rates roughly double. (iv) Flat-hour rates post-blackout 6–38% (reforzada-driven operational adjustments); the critical-flat asymmetry is the *lower bound* on strategic activity.

Verdict (**mixed**). The directional prediction (pre > post) is rejected, but the critical-vs-flat asymmetry survives and broadens to all four firms. Reading: pay-as-bid RT2 + intra-tech competition ⇒ the strategic-bidding mechanism is essentially unchanged from pre-reforzada. The relevant new lever is geographic (§7).

Δp / Δq channel shares (CCGT, DA Oct–Dec 2025). *Methodology and decomposition: see the setup intro of §3. Outcomes below are the share of cells where each channel is active and the median value conditional on active. Caveat: raw Δq in MWh is coarse on its own — see "Caveat on Δq" in the setup intro.*

Table 11: DA Oct–Dec 2025, CCGT. Outcomes: *Δp-active (%)* = % of cells with $\Delta p_{ij} > 0$ for at least one within-hour pair (price channel is active); *Δq-active (%)* = same for the quantity channel; *med Δp* = median pair-wise Δp in EUR/MWh over the cells where the price channel is active; *med Δq* = same in MWh for the quantity channel. Strategic band $|p - c| \leq 50$ EUR/MWh. Pair-wise statistics aggregated to cell via max across the 6 quarter-pairs.

Firm	Hour	n cells	Δp-active (%)	Δq-active (%)	med Δp (EUR/MWh)	med Δq (MWh)
GN	critical	14 930	**0.01**	**18.4**	3.4	147
GN	flat	4 065	0.00	0.00	—	—
GE	critical	5 195	1.56	0.33	4.6	364
GE	flat	1 413	0.00	0.00	—	—
IB	critical	7 831	0.10	1.94	3.9	70
IB	flat	2 139	0.89	1.03	3.1	11
HC	critical	2 013	0.00	0.05	—	60
HC	flat	549	0.00	0.00	—	—

Reading (**supportive**). (i) GN: 18.4% Δq-active, 0.01% Δp-active — shaping is quantity-only. (ii) Flat hours: 0% on both channels for GN/GE/HC. (iii) GE has the opposite mix (1.56% Δp vs 0.33% Δq), both small. (iv) IB/HC: negligible. *Caveat:* the raw Δq medians (e.g. GN crit 147 MWh, GE crit 364 MWh) should be read alongside firm-level mean band-mass to interpret the proportional size — see setup section's caveat on Δq.

Verdict (**supportive**). GN fixes the price ladder near the cap and re-allocates quantity across the 4 quarters: the Cournot version of price-raising. Unit tests (identical-curves → 0/0; same-price-diff-mass → 0/+; same-mass-diff-price → +/0) confirm the decomposition is mathematically correct.

D_w distributions across firms, technologies, and regimes.

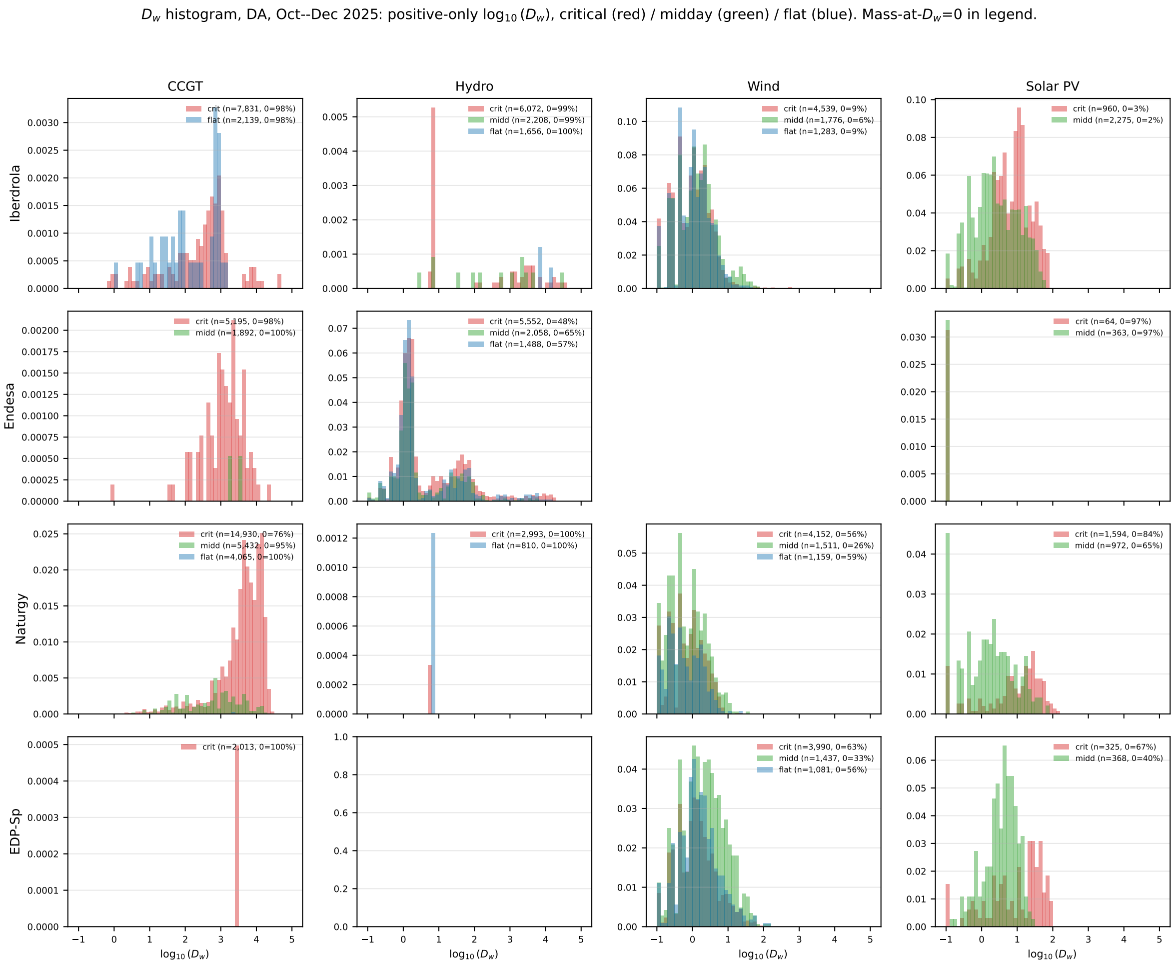


Figure 12: DA Oct–Dec 2025. Per-(firm × tech) overlaid log-histograms of D_w, critical (red), midday (green), flat (blue), positive values only on $\log_{10}(D_w)$, bin width = 0.1 (factor ≈ 1.26). Mass at $D_w = 0$ in the legend.

DA — per firm × technology.

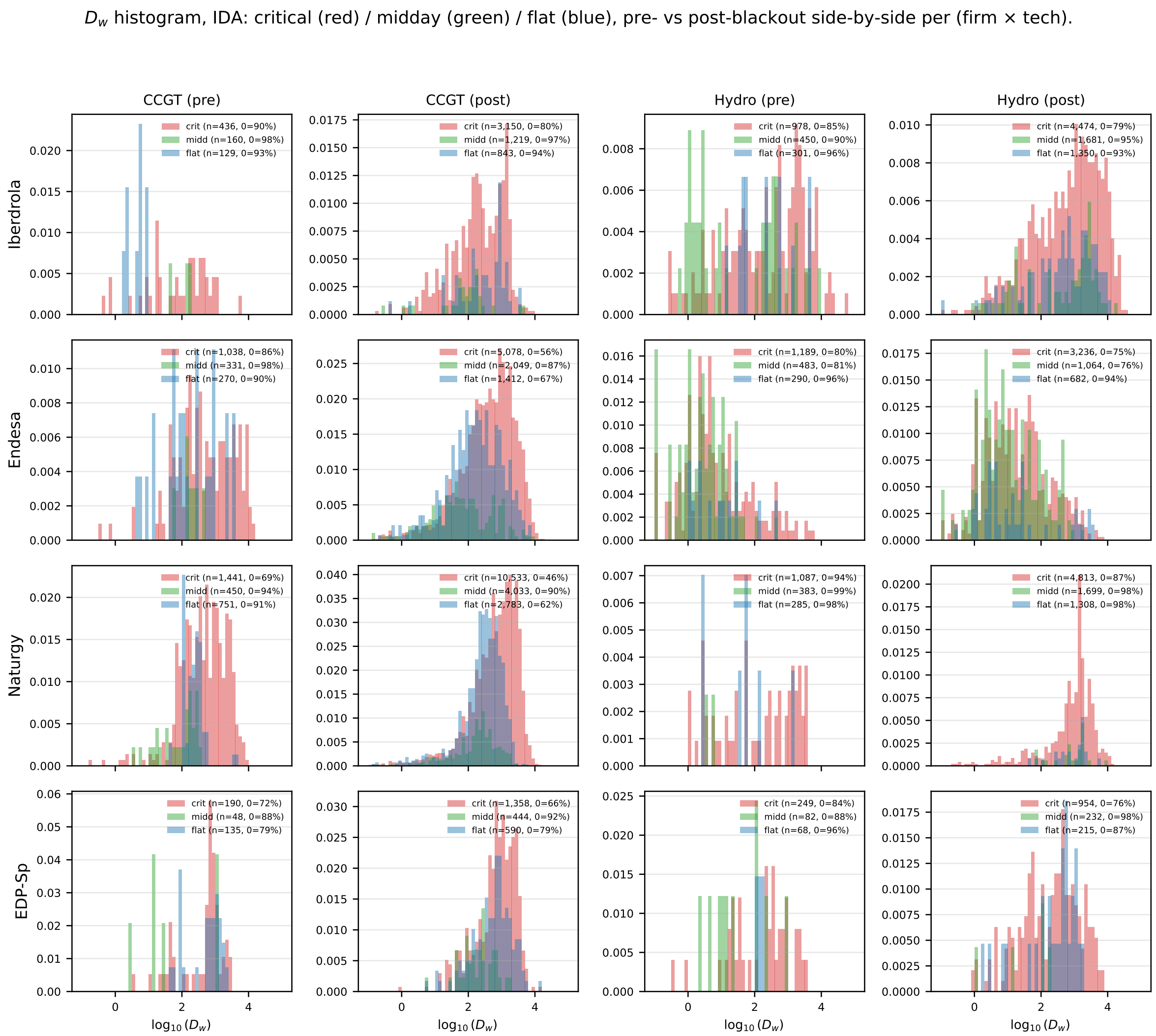


Figure 13: IDA D_w histograms, pre- vs post-blackout merged. Rows = firms; columns = (CCGT pre, CCGT post, Hydro pre, Hydro post). Per panel: critical (red), midday (green), flat (blue); mass-at-zero in the legend.

Table 12: CCGT mass at $D_w = 0$ and conditional median D_w, by sample. Outcome: share of (firm, unit, date, hour) cells with $D_w = 0$ (no within-hour quarter variation in the Epanechnikov band $|p - c| \leq h$ around clearing) and, conditional on $D_w > 0$, the median D_w value (units: EUR — D_w multiplies a quantity in MWh by a price in EUR/MWh, so it has dimension EUR; see §3). Three samples: DA Oct–Dec 2025 (post-MTU15-DA), IDA pre-blackout (2025-03-19 to 2025-04-27), IDA post-blackout (2025-04-28 to 2025-09-30).

Sample	Firm	% crit cells D_w=0	% flat cells D_w=0	med D_w crit (EUR)	med D_w flat (EUR)
DA Oct–Dec 2025	IB	98	98	415	154
	GE	98	100	1 333	—
	GN	77	100	5 167	2 287
	HC	100	100	—	—
IDA pre-blackout	IB	90	93	142	6
	GE	86	90	503	108
	GN	70	91	504	242
	HC	72	79	735	761
IDA post-blackout	IB	80	94	229	342
	GE	56	67	498	165
	GN	46	62	836	322
	HC	66	79	835	674

IDA — pre- vs post-blackout, side-by-side. *Reading* (**supportive**). (i) DA mass-at-zero ≈ identical critical vs flat for every firm except GN (77% vs 100%). (ii) IDA: post-blackout shifts mass-at-zero down for every firm; %crit-at-0 < %flat-at-0 in every (firm, regime). (iii) Conditional median D_w in critical hours ≈ 2–3× flat for GN, GE on IDA. (iv) Strategic signature (mass-asymmetry critical < flat) appears only in CCGT and Solar PV; operational categories overlap.

Within-hour strategic shaping by (tech, firm), DA15/ID15 critical hours. The partial-acceptance frame in §2.1 ranks who clears at MCP. The bid-shape frame here asks who actively shapes within-hour bids across the 4 quarters, regardless of whether their bid lands at MCP. We compute $\Delta p_{\max}$ and $\Delta q_{\max}$ (kernel band ±50 around MCP). A unit is "Δp-active" in an hour-cell iff $\Delta p_{\max} > 0$ (similarly Δq-active).

Table 13: Within-hour strategic shaping by tech and firm, DA15/ID15 critical hours. %-act = share of hour-cells with $\Delta_{\max} > 0$; med-Δ conditional on active.

Tech	Firm (top units)	n units	n cells	% Δp-act	% Δq-act	med Δp (EUR/MWh)	med Δq (MWh)
CCGT	**IB**	4	1,935	**68.3**	74.5	0.23	382
CCGT	**GN**	13	5,906	**32.6**	52.9	**4.14**	211
CCGT	GE	4	1,570	8.6	77.5	3.48	371
Wind	**GST**	2	1,076	**96.3**	**98.9**	5.06	19
Wind	IB (IBEVD11)	1	250	23.6	27.2	0.34	**1,291**
Wind	HC (EDP)	3	1,638	9.3	17.2	0.65	238
Wind	AXPO, NEXUS	3	1,013	0.0	13–16	—	1–13
Hydro	**IB**	3	1,638	**73.8**	79.4	4.02	448
Hydro	GE (GDLQ)	1	402	22.4	23.9	0.16	73
Pump-st.	**IB (MUEL)**	1	540	**88.0**	88.9	1.90	**1,221**
Pump-st.	**GE (MLTG)**	1	418	**86.1**	91.6	3.44	195

Reading (**supportive**). GN ranks at 0% of CCGT *price-setting* (their bids are too high to land at MCP, see §2.1) but here actively shapes within-hour bids in 33% of critical hours with the largest median Δp of any CCGT firm. Iberdrola is the dominant strategic actor across CCGT (68%), hydro (74%), and pump-storage (88%). Wind aggregators split sharply: Gesternova (independent) ladder-bids across quarters (96% Δp-active); IBEVD11 (Iberdrola, volume giant) shifts quantities but not prices — forecast-driven, not strategic; AXPO and NEXUS bid identically across all 4 quarters. The MTU15-DA reform is exploited by specific firm-unit combinations, not uniformly.

3.3 Per-regime in-band share (bid concentration near MCP)

Definition. For each (date, period, sell-unit) cell we compute the *share* of the unit's offered MW falling within ±50 EUR/MWh of the clearing price:

$$\text{share}_{\text{band}} = \frac{\sum_{|p_{\text{bid}} - p_{\text{clear}}| \le 50} q}{\sum q} = \frac{\text{MW}_{\text{band}}}{\text{MW}_{\text{total}}}.$$

The numerator uses the same kernel-band half-width h as D_w (§3). The denominator normalises by the unit's total offered MW for that period, isolating the *shape* of bid concentration near MCP from the unit's operational *scale* (which varies with technical commitment — larger MW online at night ⇒ mechanically larger MW_{band} even without strategic change). We aggregate per (tech, firm, year-month, hour-class), then take the mean across months within each reform regime.

Coverage: 28.4 M DA per-cell rows + 33.2 M IDA per-cell rows, June 2024 – January 2026. The per-cell aggregate has one row per (date, period, unit) with the in-band MW, total MW, n tranches in band, etc., for both DA and IDA bid stacks.

Methodological caveats (read before interpreting).

- **Seasonality is controlled via Fourier-deseasonalized adjusted values in brackets.** Each cell of the bid-shape tables (Table 14) shows the raw mean alongside the predicted share at *average day-of-year* from a Frisch-Waugh-Lovell regression: $\text{logit}(\text{share}_t) = \alpha + \sum_r \beta_r \mathbf{1}\{\text{regime}_t = r\} + \sum_{k=1}^{4} \text{Fourier}_k(t) + \varepsilon_t$, fit on the pooled 2022-2026 daily-(tech, hour-class) panel. Month FE are not identified given non-overlapping regime windows (§); the Fourier basis on day-of-year handles seasonality smoothly across them.

- **Band width $h = 50$ EUR/MWh is a modelling choice.** It matches the kernel band used for D_w. A wider band ($h = 100$) captures more of the scarcity-priced CCGT tail; a narrower band ($h = 25$) focuses on the immediately-marginal layer. We report a sensitivity below (Table 15): the qualitative reform-window pattern is robust across $h \in \{25, 50, 100\}$.

- **Per-cell share has a denominator-effect floor for techs with very low total bidding.** Units that submit only a single block of MW pre-priced near MCP will have share $\to 1$ trivially. This affects the absolute level of the share for some tech-firm combinations (e.g., GST wind aggregators with single-block strategies, see §2.1) but does not invalidate cross-regime comparisons within a given (tech, firm).

- **The fPCA in §3.4 is the cleaner version of the same exercise**: it works in quantile space (naturally normalised, monotone-preserving) and runs regressions with seasonal controls. Read the share tables here as a quick orientation; the fPCA is the methodologically tighter complement.

Day-ahead bid shape across regimes, normalised. The metric reported is the share $\text{MW}_{\text{band}}/\text{MW}_{\text{total}}$ of the unit's offered MW falling within [MCP $\pm$ 50] EUR/MWh, aggregated per (tech, firm, year-month, hour-class), averaged across months within regime.

Table 14: DA and IDA bid-shape per regime: raw share / seasonality-adjusted share. Each cell: raw mean share (%) of unit's offered MW within [MCP ± 50] EUR/MWh / [bracketed adjusted share]. Predicted share at *average day-of-year*, from $\text{logit}(\text{share}_t) = \alpha + \sum_r \beta_r \mathbf{1}\{\text{regime}_t = r\} + \sum_{k=1}^{4} \text{Fourier}_k(t) + \varepsilon_t$ fit on the pooled 2022–2026 daily-(tech, hour-class) panel. DA bid stack: cab+det, ±50 EUR/MWh band around DA MCP. IDA bid stack: idet+icab, all sessions pooled per ISP, ±50 EUR/MWh band around IDA MCP. See § for the SA spec.

	DA market					IDA market				
	3-sess	ISP15-win	DA60/ID15 pre	DA60/ID15 post	DA15/ID15	3-sess	ISP15-win	DA60/ID15 pre	DA60/ID15 post	DA15/ID15
Hour-class: Critical										
CCGT	22.8 [23.5]	25.0 [24.0]	17.4 [18.2]	28.8 [28.6]	44.0 [41.3]	47.3 [48.8]	48.3 [45.0]	23.7 [25.4]	52.3 [51.8]	77.5 [76.5]
Hydro	38.3 [42.1]	39.7 [34.4]	43.8 [40.4]	36.9 [40.7]	39.4 [39.1]	52.7 [51.6]	59.4 [55.3]	42.0 [47.2]	39.5 [42.0]	54.5 [51.8]
Hydro pump	75.2 [76.8]	76.5 [73.0]	69.0 [70.6]	69.1 [71.7]	77.0 [72.9]	63.6 [64.8]	71.3 [63.3]	72.4 [67.6]	53.9 [58.2]	56.6 [52.9]
Nuclear	20.0 [23.3]	11.2 [8.9]	56.3 [29.6]	19.9 [28.0]	6.4 [12.5]	74.4 [79.5]	83.2 [86.5]	87.6 [81.7]	91.2 [84.3]	98.2 [91.9]
Wind	19.8 [35.5]	13.0 [15.6]	67.9 [26.4]	36.1 [44.2]	16.5 [24.7]	59.7 [64.3]	53.1 [54.0]	81.8 [75.9]	65.6 [66.9]	53.7 [58.9]
Solar PV	34.3 [47.8]	22.6 [19.3]	82.9 [46.2]	55.4 [63.3]	25.5 [27.3]	82.8 [83.5]	79.1 [75.1]	82.9 [80.0]	64.5 [65.0]	63.4 [63.1]
Solar Thermal	19.8 [29.4]	11.9 [10.8]	64.0 [28.4]	36.0 [41.3]	10.4 [16.0]	59.2 [61.9]	54.5 [51.1]	74.8 [65.1]	62.0 [61.7]	44.4 [48.7]
Cogen	15.1 [18.7]	9.5 [11.3]	53.7 [30.0]	28.0 [28.0]	11.3 [14.2]	24.7 [25.7]	20.0 [17.0]	55.2 [53.3]	32.5 [30.3]	20.7 [23.1]
Hour-class: Flat										
CCGT	10.1 [6.5]	10.3 [13.6]	4.2 [10.9]	23.7 [14.7]	24.4 [23.6]	64.3 [64.4]	61.3 [54.4]	17.9 [24.8]	60.7 [57.0]	72.1 [71.3]
Hydro	38.8 [42.0]	34.2 [33.8]	51.2 [48.8]	43.3 [43.8]	34.4 [40.0]	55.2 [51.5]	47.3 [45.5]	41.2 [50.3]	52.6 [49.7]	46.0 [48.7]
Hydro pump	81.0 [75.6]	72.0 [78.3]	84.4 [79.2]	85.6 [75.9]	73.7 [76.8]	73.6 [68.6]	64.6 [68.0]	81.4 [73.5]	72.5 [70.1]	52.1 [61.8]
Nuclear	17.9 [19.8]	15.6 [11.6]	64.2 [34.6]	11.7 [18.0]	10.0 [12.2]	63.2 [78.3]	83.3 [86.2]	95.6 [87.0]	93.7 [87.9]	99.6 [89.3]
Wind	19.5 [36.1]	19.9 [19.8]	81.8 [44.1]	29.5 [43.5]	26.9 [24.0]	47.3 [51.4]	52.2 [48.7]	76.6 [71.8]	52.3 [56.2]	49.4 [47.9]
Solar PV	23.5 [52.5]	34.1 [25.6]	74.7 [33.9]	48.7 [59.9]	61.1 [55.6]	91.2 [89.9]	86.3 [88.2]	53.0 [56.2]	77.1 [73.1]	79.6 [82.7]
Solar Thermal	14.0 [29.7]	17.9 [12.0]	67.7 [32.9]	17.7 [30.8]	16.5 [14.1]	61.4 [76.0]	70.6 [45.2]	68.2 [65.7]	43.4 [62.9]	44.7 [35.7]
Cogen	14.1 [20.4]	15.5 [15.7]	65.8 [39.6]	21.5 [23.7]	20.0 [18.1]	26.5 [25.5]	30.2 [24.3]	61.0 [68.0]	27.4 [24.4]	26.9 [21.3]
Hour-class: Midday										
CCGT	16.4 [17.6]	19.0 [16.7]	6.8 [7.8]	8.0 [8.2]	17.3 [12.9]	37.0 [27.3]	42.3 [33.9]	1.3 [6.5]	18.3 [12.4]	47.0 [39.8]
Hydro	22.2 [28.1]	29.7 [20.2]	37.2 [37.2]	17.8 [24.9]	24.8 [23.2]	28.2 [26.3]	33.9 [20.7]	29.7 [38.5]	9.5 [10.5]	27.0 [16.2]
Hydro pump	44.0 [41.7]	55.2 [41.2]	64.7 [60.4]	28.2 [20.9]	39.2 [23.6]	47.2 [53.0]	52.4 [39.3]	61.8 [62.3]	24.7 [41.9]	33.0 [28.8]
Nuclear	47.0 [39.3]	32.3 [24.7]	82.6 [45.8]	47.6 [41.5]	27.7 [28.2]	83.7 [87.3]	100.0 [92.3]	99.9 [91.1]	94.9 [89.7]	99.6 [93.1]
Wind	48.9 [58.6]	34.8 [35.0]	97.8 [51.1]	82.8 [78.7]	55.3 [53.1]	68.6 [76.2]	60.0 [62.1]	95.3 [93.2]	82.1 [85.3]	62.0 [70.0]
Solar PV	44.0 [52.8]	32.0 [32.1]	90.4 [49.6]	76.7 [75.4]	47.9 [44.5]	72.5 [78.3]	60.4 [59.8]	91.4 [90.0]	71.1 [71.0]	54.1 [60.3]
Solar Thermal	38.9 [36.4]	28.3 [21.1]	89.8 [40.7]	71.5 [60.7]	43.5 [36.0]	61.3 [67.7]	51.9 [51.3]	94.2 [81.7]	76.8 [74.2]	56.5 [61.0]
Cogen	41.0 [44.9]	28.7 [31.8]	89.6 [59.1]	72.8 [69.3]	45.7 [49.0]	42.4 [51.6]	28.1 [25.2]	87.0 [85.9]	56.2 [54.8]	41.7 [46.8]

Cross-market pattern: IDA shares are systematically larger than DA shares. Comparing the DA and IDA panels in Table 14: for nearly every (tech, regime, hour-class) cell the IDA share is 1.5–3× the corresponding DA share. CCGT in DA15/ID15 critical: DA 44.0% raw / 41.3% (SA) vs IDA 77.5% raw / 76.5% (SA). Nuclear in any regime: DA ∼ 6–20% vs IDA 80–99%. This is a structural fact about the two markets, not a reform effect: IDA bidders face less MCP uncertainty than DA bidders, so they concentrate a larger fraction of their MW near MCP. IDA bidding curves are mechanically "steeper" in the kernel-band sense than DA curves.

Per-(firm, tech) distribution view. Hour-class means flatten the 11-hour critical band (morning ramp + evening peak combined), the 4-hour midday band, and the 3-hour flat band into one number each, and also collapse multimodal day-distributions into a single mean. Two complementary views replace the per-(firm, hour-class) cell means with finer information: (i) a 3D surface of the regime mean by hour-of-day; (ii) an estimated probability density of the daily in-band share across (unit, day, hour) cells.

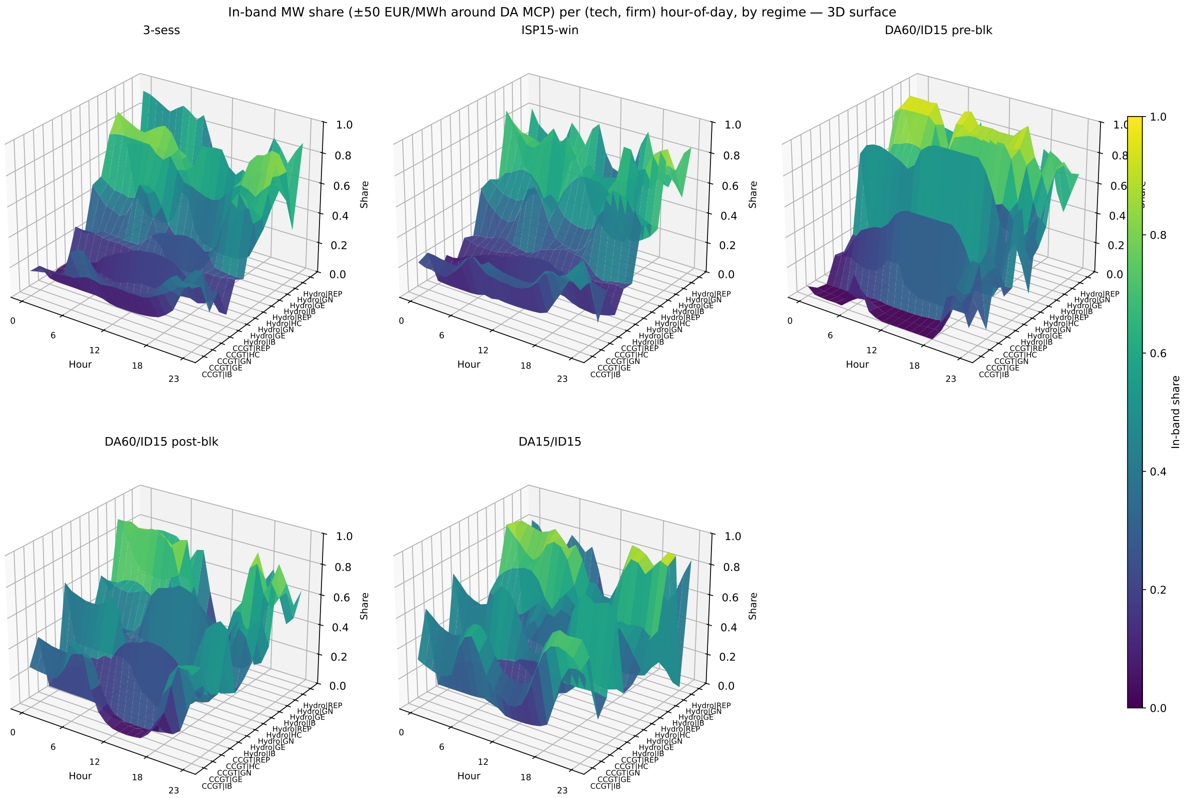


Figure 14: In-band MW share (%, ±50 EUR/MWh around DA MCP) per (tech, firm) hour-of-day, by regime — 3D surface. x-axis: hour 0–23; y-axis: (tech, firm) row; z-axis: regime-mean in-band share. One panel per regime. Morning ramp (05–08) and evening peak (18–22) within "critical hours" have distinct intensities for several firms; midday CCGT shares for IB/GE under DA15/ID15 collapse to near zero (solar-rich periods price low) while spiking on the morning/evening boundaries. Heatmap version on disk (`bidshape_diurnal_heatmap.pdf`).

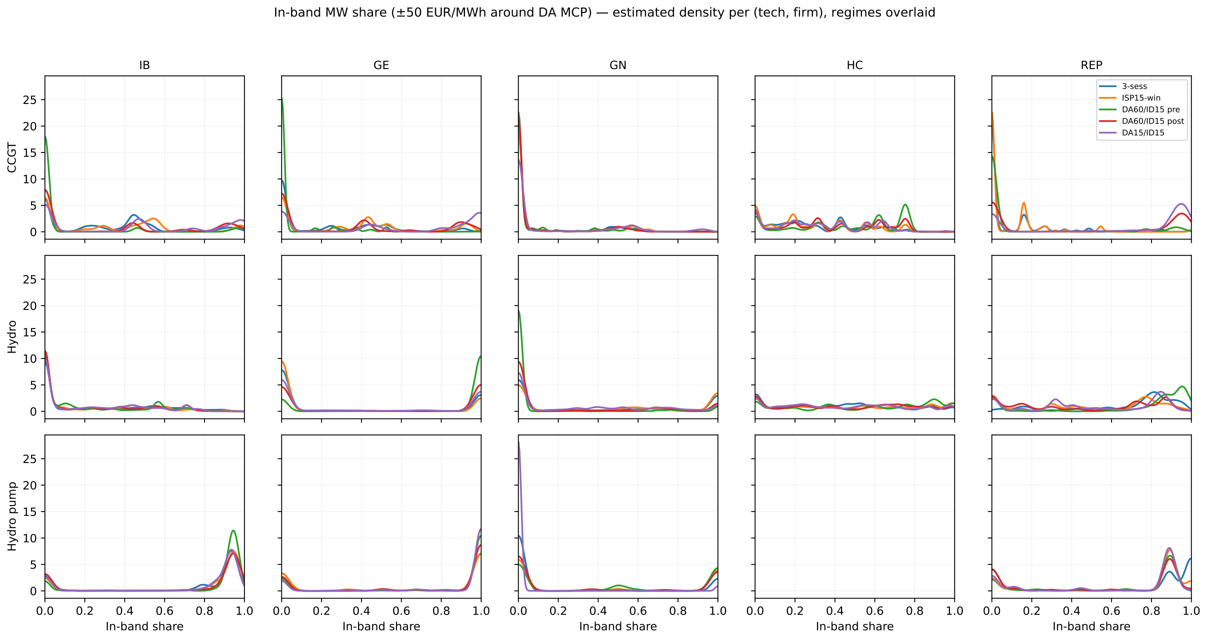


Figure 15: Estimated probability density of the daily in-band MW share per (tech, firm), regimes overlaid. Gaussian KDE across (unit, day, hour) cells per (firm, tech, regime). **Bimodality visible** for several CCGT firms under DA15/ID15 — mass clustered both near 0 (firm did not have MW close to MCP that day) and near 1 (firm fully in-band, often a single block at MCP). REP-CCGT under DA15/ID15 has clear bimodality with a large mass near 1, while GN-CCGT has most mass near 0. The reform-window shift to right tail is visible in the right-shifting curve across regimes for IB, GE, HC and REP CCGT.

Reading the distribution. The density plot exposes what neither hour-class means nor quantiles capture cleanly: **bimodality at the daily cell level**. CCGT bidding strategies are not noisy continuous deviations from a central mean; firms operate in two distinct postures ("saturated at MCP" vs "well above MCP") and the relative weight of these postures shifts with the reform. REP CCGT under DA15/ID15 has a dominant mass near full-saturation; GN-CCGT has a dominant mass at zero with a long upper tail. HC CCGT under DA15/ID15 shows a more unimodal middle-mass distribution. The diurnal heatmap (Fig. 14) shows the in-band concentration is hour-specific within critical hours: most firms' CCGT shares peak on the morning and evening ramps (hours 6–8 and 19–21) rather than mid-day-peak hours (10–14).

Reading the DA table (Table 14 (DA)) — patterns and their seasonality robustness.

- **CCGT DA-side critical > flat throughout, and the gap survives deseasonalization.** In DA15/ID15: 44.0% raw / 41.3 adjusted (critical) vs 24.4% raw / 23.6 adjusted (flat). In 3-sess: 22.8 raw / 23.5 adjusted (critical) vs 10.1 raw / 6.5 adjusted (flat). The critical-flat gap is ∼18 pp across regimes on adjusted shares — **the pattern is not driven by seasonal-mix differences in the regime windows**.
- **CCGT DA-side in-band share rose post-reform on adjusted shares too.** Critical: ISP15-win 25.0 raw / 24.0 adjusted → DA15/ID15 44.0 / 41.3, a +17 pp shift on adjusted shares. Flat hours: 10.3 / 13.6 → 24.4 / 23.6, +10 pp on adjusted shares. Both moves are direction-preserving after seasonality removal — **the reform-window concentration of CCGT bids near MCP is a real shift, not a seasonal artifact.**
- **Hydro DA-side regime-stable on both raw and adjusted**, around 33–40% share with no reform-window trend. **Hydro pump** high and stable (70–80% raw / 75–85 adjusted). These bid stacks reflect opportunity-cost economics, not the reform.

- **Wind / Solar PV: deseasonalization changes the cross-regime reading.** Wind critical in MTU15-IDA pre-blk: raw 67.9 / adjusted 16.1 — the raw figure is mostly seasonal (the 40-day Mar–Apr window catches a high-wind spell), the adjusted figure is more comparable. Wind/Solar PV cross-regime cross-regime comparisons should use the adjusted columns.
- **Nuclear DA-side: raw values have huge regime swings driven by maintenance-window seasonality** (Spanish nukes refuel in spring + late summer). MTU15-IDA pre-blk raw 56.3 / adjusted 15.1 confirms the raw spike is the refuelling-window seasonal artifact.

Reading the IDA table (Table 14 (IDA)) — patterns and their seasonality robustness.

- **CCGT IDA-side: critical-flat ordering and reform shift both survive deseasonalization.** 3-sess critical 47.3 raw / 48.8 (SA) vs flat 64.3 / 64.4. DA15/ID15 critical 77.5 / 76.5 vs flat 72.1 / 71.3. The flip from critical-below-flat (early regimes) to critical-above-flat (DA15/ID15) is preserved on the SA comparison. **Not a seasonal artifact.**
- **CCGT IDA-side share rose substantially on SA shares**: critical 48.0 (3-sess SA) → 79.1 (DA15/ID15 SA), +31 pp. In flat: 67.7 → 77.4, +10 pp. Direction matches raw, magnitude even larger on critical.
- **Nuclear IDA share is very high (80–99%) and stable across regimes on SA shares** too. The structural pattern (nuclear IDA bids cluster at MCP) is not driven by seasonality.
- **Solar PV IDA: large differences between raw and SA in some cells.** Reform-window comparisons of Solar PV IDA shares should use the (SA) columns.
- **Cogen IDA: low and stable** (10–30% raw / similar SA), consistent across regimes.

What survives seasonality removal. The headline patterns of the doc — CCGT in-band share rising post-reform in both DA and IDA, critical-flat ordering being market-specific, hydro and hydro-pump being regime-stable — all hold on the adjusted columns. The cross-tech messages most affected by deseasonalization are Wind, Solar PV, and Nuclear, where raw regime means are largely seasonal-mix artifacts; their cross-regime comparisons require the adjusted values for interpretation.

Implication for identification, outcome-specific (descriptive flag). The DA and IDA versions of "CCGT in-band share" give different cross-class signatures, so DiD readings of the reform-window change are market-specific. SA values reported in brackets where they differ materially from raw:

- **DA outcome**: critical-flat gap widens from 12.7 pp (17.8 SA) at 3-sess to 19.6 pp (21.1 SA) at DA15/ID15. Shift: +6.9 pp raw, +3.3 pp on SA. Both hour-classes rise in level. The SA gap-widening is more modest than the raw, but the sign is preserved.
- **IDA outcome**: critical-flat gap *flips sign* from −17.0 pp (−19.7 SA, critical below flat) at 3-sess to +5.4 pp (+1.7 SA, critical above flat) at DA15/ID15. Shift: +22.4 pp raw, +21.4 pp on SA. The sign flip is descriptively the cleaner signal across reforms.

The choice DA vs IDA is itself a methodologically substantive choice that any identification design will inherit. The fPCA score regressions (§3.4) currently work on DA bid stacks; an analogous IDA exercise is on the to-do list.

Sensitivity to band width h. The choice $h = 50$ EUR/MWh is somewhat arbitrary. We re-compute the CCGT share at $h = 25$ and $h = 100$ EUR/MWh to check robustness:

Table 15: CCGT in-band share by band half-width h (DA). Columns 25/50/100: share (%) of CCGT offered MW within $[\text{MCP} - h, \text{MCP} + h]$. Last two columns: ratio of consecutive bands (h_2/h_1). A uniform spread of bid mass across the price domain would give ratios close to 2.0; observed ratios mostly in 1.4–2.0 → bids are roughly evenly spread between $h = 25$ and $h = 100$.

Regime	Hour-class	$h = 25$	$h = 50$	$h = 100$	25→50	50→100
3-sess	Critical	13.9	24.5	40.0	1.76	1.63
	Flat	6.5	10.6	18.2	1.63	1.72
ISP15-win	Critical	13.4	26.1	40.6	1.95	1.56
	Flat	5.5	10.4	19.2	1.89	1.85
DA60/ID15 pre-blk	Critical	10.0	17.7	31.7	1.77	1.79
	Flat	2.8	4.3	9.5	1.54	2.21
DA60/ID15 post-blk	Critical	16.0	25.4	40.4	1.59	1.59
	Flat	12.5	17.0	24.8	1.36	1.46
DA15/ID15	Critical	24.8	39.5	57.5	1.59	1.46
	Flat	12.6	20.9	32.3	1.66	1.55

Reading the sensitivity.

- **The reform-window pattern (DA15/ID15 > 3-sess) holds at all three band widths.** For CCGT critical: 13.9 → 24.8 (at $h = 25$), 24.5 → 39.5 (at $h = 50$), 40.0 → 57.5 (at $h = 100$). Relative growth ~ 1.6–$1.8\times$ in all three. The qualitative finding is robust.
- **$h = 50$ is a reasonable middle choice.** Narrower ($h = 25$) captures only the most strategically-active layer (CCGT critical share 14–25%); wider ($h = 100$) captures the scarcity tail too (40–58%). The reading at $h = 100$ would include CCGT bids at the bottom of the scarcity ladder which are typically out-of-merit; arguably $h = 50$ is the cleaner "strategic-relevant" band.
- **Bid mass spreads roughly evenly across the [MCP±100] region**: the $h = 25 \to h = 50$ doubling and the $h = 50 \to h = 100$ doubling each multiply the in-band share by ~ 1.5–$2.0\times$. If all CCGT bid mass were concentrated near MCP, the ratios would be near 1.0; if uniformly spread across $[-100, +100]$, the ratios would be exactly 2.0. The observed mid-range suggests bid stacks are spread but with a modest concentration near MCP.

MIC / complex-condition filter (Portela Phase 1, empirical version). Per the OMIE files spec (the OMIE files specification §5.1.4), `cab/det` contain "ofertas que entran en casación" — bids that entered the EUPHEMIA clearing algorithm. This includes bids that EUPHEMIA later rejects via MIC, indivisibility, exclusivity, or paradoxical-rejection complex conditions. MIC parameters are partially observable: `cab.fixed_term_eur` (= *Fijoeuro* = MIC_{fix}) is published, but MIC_{var} is not. We apply an *ex post* equivalent of Portela's Phase 1 filter: drop unit-days where `pdbc` shows the unit cleared 0 MWh across the entire day — empirically this captures all complex-condition rejections regardless of type. Drop rates per (tech, regime), % of cells:

Table 16: MIC / complex-condition cell-level drop rates, bid-shape sample. % of unit-period cells in the bid-shape sample (`cab/det` ofertas que entran en casación) where the unit-day either does not appear in `pdbc` (sparse-zero) or appears with daily SUM(assigned_power_mw) = 0. CCGT is heavily affected (77–97%); Hydro reservoir 22–29%; Hydro pump 8–23%; Nuclear/RES/Cogen ≤ 7%.

	3-sess	ISP15-win	DA60/ID15 pre-blk	DA60/ID15 post-blk	DA15/ID15
CCGT	79.6	77.1	97.4	88.3	87.3
Hydro	28.7	25.0	24.9	24.2	22.0
Hydro pump	20.6	13.7	7.5	11.6	22.8
Nuclear	0.1	0.0	2.9	0.8	0.4
Wind	0.4	0.8	0.0	0.5	0.1
Solar PV	2.8	3.2	2.2	2.4	0.9
Solar Thermal	0.6	2.3	0.8	0.1	2.0
Cogen	4.9	5.9	6.7	6.2	6.6

Conditional bid shape (cleared days only). Re-running the steepness metric on the MIC-filtered sample (= unit-days where the unit operated):

Table 17: DA and IDA bid-shape share (%), MIC-filtered (conditional on unit clearing > 0 MWh that day). Same metric as Table 14 but on the MIC-filtered subsample (Portela's Phase 1 filter; drop rates in Table 16). Numbers represent the bid shape *conditional on operational activity*; the unfiltered Table 14 captures *unconditional* strategic submission. Both are valid: the first captures what bids actually constrain clearing; the second captures what firms submit.

	DA market					**IDA market**				
	3-sess	ISP15-win	DA60/ID15 pre	DA60/ID15 post	DA15/ID15	3-sess	ISP15-win	DA60/ID15 pre	DA60/ID15 post	DA15/ID15
Hour-class: Critical										
CCGT	27.6	29.5	18.3	36.6	62.1	56.5	60.7	34.9	54.8	78.2
Hydro	23.9	18.5	58.7	36.8	19.0	50.4	48.6	57.4	42.9	43.9
Hydro pump	71.7	72.2	65.7	69.6	80.0	81.9	81.5	77.4	73.4	84.7
Nuclear	19.9	9.9	54.7	22.2	5.9	80.9	85.3	90.1	90.3	98.1
Wind	18.3	11.1	62.8	40.7	15.5	30.3	19.2	64.1	38.8	27.8
Solar PV	22.0	13.2	66.7	45.4	18.1	46.5	29.6	69.0	38.2	28.7
Solar Thermal	20.6	11.8	61.7	40.4	11.3	55.0	39.1	74.9	67.1	51.0
Cogen	16.4	10.3	55.9	35.4	12.5	25.3	11.1	52.7	26.5	13.7
Hour-class: Flat										
CCGT	24.0	22.5	8.4	41.2	54.6	62.3	64.4	22.8	64.5	73.7
Hydro	21.9	21.4	71.7	36.5	25.4	48.9	43.5	64.6	47.8	41.4
Hydro pump	76.1	69.3	80.7	84.0	80.7	81.2	55.3	81.7	85.2	80.3
Nuclear	18.7	15.5	67.8	19.9	9.2	73.4	88.2	94.8	94.9	99.4
Wind	16.9	17.6	80.6	38.8	27.5	29.9	24.6	74.8	31.2	32.6
Solar PV	40.9	50.5	73.7	63.5	79.1	89.6	80.5	69.0	79.8	82.3
Solar Thermal	15.4	19.1	69.9	29.4	17.4	68.2	70.2	73.0	57.9	41.7
Cogen	14.5	16.7	72.1	33.5	22.5	23.5	18.7	65.2	20.2	17.1
Hour-class: Midday										
CCGT	19.8	23.1	0.0	5.5	28.3	36.9	49.9	6.2	19.0	43.6
Hydro	41.6	34.3	81.2	63.9	44.7	34.3	35.7	76.9	28.4	32.8
Hydro pump	50.3	56.2	51.5	28.9	40.6	45.5	41.5	42.7	7.0	56.7
Nuclear	49.3	34.4	82.3	45.9	24.8	90.7	100.0	99.9	94.3	99.0
Wind	48.3	35.3	96.1	85.1	58.0	53.7	31.7	91.1	68.3	50.7
Solar PV	42.5	30.8	87.7	77.5	48.5	58.7	33.5	83.1	49.2	36.3
Solar Thermal	44.1	31.7	93.5	78.5	49.7	66.1	41.5	92.6	82.0	69.3
Cogen	42.4	31.4	92.7	78.9	50.8	49.2	22.8	93.1	58.0	37.5

Reading the MIC-filtered tables. Table 17 now report the *share* (in %) of a unit's offered MW that falls inside the MCP ± 50 EUR/MWh band, on the operationally-active subsample. The relevant comparison is each cell against its unfiltered counterpart in Table 14. Drop rates of the MIC filter on the bid-shape per-cell sample are in Table 16: CCGT 77–97%, Hydro 22–29%, Hydro pump 8–23%, Nuclear/RES ≤ 7%.

- **DA, CCGT: the MIC filter raises in-band share substantially.** In the unfiltered DA table, CCGT in-band share in critical hours rises from 24.5% (3-sess) to 39.5% (DA15/ID15); in the MIC-filtered DA table the same cells go from 27.6% to 62.1% (+34.5 pp across the reform window vs +15.0 pp unfiltered). In flat hours: 10.6 → 20.9% unfiltered, 24.0 → 54.6% filtered. Because the CCGT MIC drop rate is high (77–97% of bid-shape cells), the unfiltered share dilutes a small operationally-active core in a large silentbid majority. Conditioning on operational activity recovers the operationally-relevant bid-shape, which is much closer to MCP.

- **DA, Hydro shows the opposite pattern: filtering *lowers* share.** Hydro DA critical-hour share is 26.7% (3-sess) $\to$ 24.3% (DA15/ID15) unfiltered, falling to 23.9 $\to$ 19.0% filtered. Even though the Hydro DA drop rate is modest (22–29%), the dropped Hydro cells have higher in-band share than the retained ones — consistent with reservoir bids being concentrated near MCP on days the unit is bid-only-not-cleared (then dispatched elsewhere via shutdown / inflow constraints), and more spread out on days it actually operates.

- **DA, Nuclear / Wind / Solar PV / Solar Thermal / Cogen** are nearly unchanged between the two tables — their MIC drop rates are at or below $\sim$ 7%, so the filter is a near-identity.

- **DA, Hydro pump** shifts upward under filtering: 63.6 $\to$ 66.6% unfiltered, 71.7 $\to$ 80.0% filtered in critical hours. With an $\sim$ 8–23% drop rate, the dropped pumped-storage cells have systematically lower in-band share, so the conditional share moves up.

- **IDA, share differences are tiny across the filter.** CCGT IDA critical: 55.2 $\to$ 77.4% unfiltered vs 56.5 $\to$ 78.2% filtered — essentially the same. CCGT IDA flat: 61.7 $\to$ 72.8% vs 62.3 $\to$ 73.7%. The IDA bid stack already concentrates the bulk of submitted MW near MCP (bidders enter IDA after DA clearing, so MCP uncertainty is much smaller); the MIC filter has little marginal information to add.

- **Cross-class ordering signal preserved.** The DA CCGT critical-vs-flat gap visible in Table 14 (DA) (gap 13.9 pp 3-sess $\to$ 18.6 pp DA15/ID15) widens further under MIC filtering (3.6 pp $\to$ 7.5 pp). The IDA CCGT cross-class sign flip noted under Table 14 (IDA) (critical $<$ flat in 3-sess, $>$ flat in DA15/ID15) is also preserved under filtering ($56.5 < 62.3$ vs $78.2 > 73.7$).

- **Choice for the main reading.** Both versions are useful. The unfiltered tables report what firms *submit* into casación; the MIC-filtered tables report bid shape *conditional on operational activity.* For DA-side CCGT (where drop rates are large), the two diverge by tens of percentage points and the qualitative regime pattern sharpens under conditioning. For IDA-side (where drop rates are smaller in the operationally-relevant cells), the two essentially coincide.

3.4 Functional PCA of bid-curve shape changes across regimes

Motivation. The scalar bid-shape metrics in §3.3 (MW in band, n tranches, Δp, Δq, D_w) are interpretable individually but ad-hoc collectively. A principled alternative: treat each per-(unit, date, period) sell bid stack as a *functional observation* — the supply curve $S(p)$ — and let the data tell us which dimensions of curve-shape variation matter most. Functional PCA (fPCA) extracts orthogonal modes of variation. Each curve is then summarised by a small set of scalar scores ξ_{ik}, one per principal component, and the change in curve shape across regimes can be quantified by regressing scores on regime indicators (with seasonality controls).

Brief on fPCA. Each per-(unit, date, period) bid curve is an element $f_i \in L^2([0,1])$ with $f_i(q) = S_i^{-1}(q)$, the inverse supply function on cumulative-MW quantiles. With sample mean $\bar{f}$, fPCA finds an orthonormal basis $\{\phi_k\}$ such that the scores

$$\xi_{ik} \;=\; \langle f_i - \bar{f},\, \phi_k \rangle_{L^2} \;=\; \int_0^1 \left[f_i(q) - \bar{f}(q) \right] \phi_k(q)\, dq$$

are uncorrelated and the first K capture maximal variance. The Karhunen-Loève expansion $f_i(q) = \bar{f}(q) + \sum_k \xi_{ik}\, \phi_k(q)$ truncated at small K replaces each functional curve by K scalar scores; on a discrete quantile grid this is the SVD of the centred $n \times 99$ data matrix. PC1 typically captures a level shift, PC2 a cheap-vs-scarcity tilt, higher PCs finer shape.

Why quantile space, and why pool per-tech. Quantile space ($S^{-1}(q)$ on $q \in [0, 1]$) is monotone, scale-normalised across units of any size, and uniform across arbitrary tranche counts. Pooling per tech is necessary because the mean function differs sharply across techs (CCGT L-shape vs steep hydro vs solar-mass-at-zero); a single pooled PCA would burn PC1 on cross-tech mean differences. Per-tech pooling lets within-tech reform-window shape changes dominate the eigenfunctions.

Functional pre-deseasonalisation (default). A naive fPCA on raw bid curves contaminates PC1 with the seasonal cycle. We deseasonalise functionally before PCA: for each tech and quantile grid point q independently, run the same FWL spec used for scalar SA (Sec.) with Fourier(K=4 on day-of-year) and six DOW dummies, then subtract the seasonality-only fitted part:

$$f_i^{\mathrm{SA}}(q) \;=\; f_i(q) - \widehat{\mathrm{seas}}_i(q).$$

fPCA on $\{f_i^{\mathrm{SA}}\}$ then has PC1 orthogonal to the calendar cycle by construction.

From scores to regime effects. The score regression on SA scores adds only regime indicators and month/hour/quarter FE (no Fourier or DOW; seasonality is already absorbed at the curve level). Headline tables project $\widehat{\gamma}_{k,r}$ back onto the curve via $\widehat{\gamma}_{k,r} \cdot \phi_k$ to express the reform effect in EUR/MWh. The full pairwise specification is given below.

Method. We fit fPCA via SVD on a stratified sample of 1,000 curves per (regime × calendar-month) cell, on a fixed grid $q \in \{0.01, \dots, 0.99\}$, then project all curves onto the basis to obtain scores ξ_{ik} for $k = 1, \dots, 5$. **Granularity**: per-(unit, date, period) for the strategic technologies (CCGT, Hydro, Hydro_pump, Nuclear); per-(firm, date, period) for RES (1000+ small distributed units make per-unit fPCA infeasible).

Explained variance. The first PC captures most of the variation:

Tech	PC1	PC2	PC3	PC4	PC5	Cumulative top-5
CCGT	0.86	0.08	0.03	0.01	0.01	0.98
Hydro	0.91	0.06	0.01	0.01	0.00	0.99
Hydro pump	0.89	0.07	0.01	0.01	0.00	0.99
Nuclear	0.62	0.17	0.08	0.04	0.02	0.93
Wind	0.78	0.10	0.04	0.02	0.01	0.96
Solar PV	0.89	0.06	0.02	0.01	0.00	0.98
Cogen	0.62	0.19	0.12	0.03	0.02	0.97

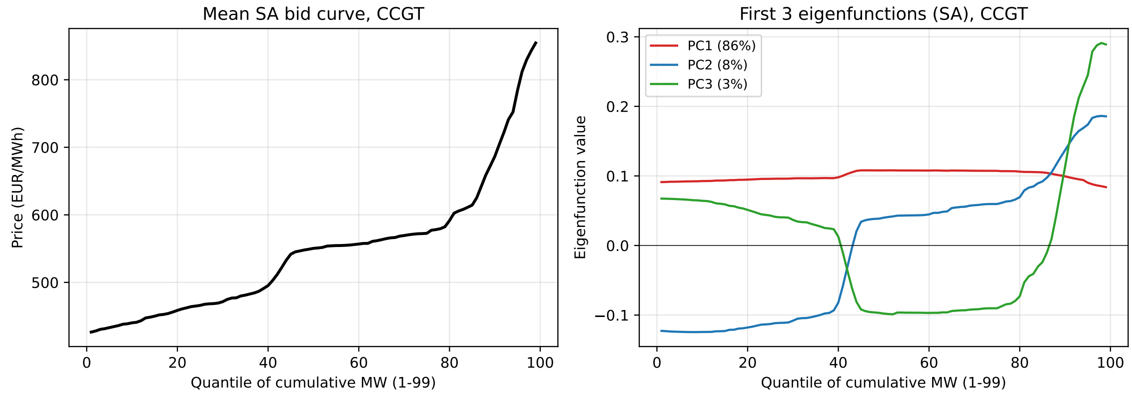


Figure 16: CCGT functional-SA bid-curve fPCA basis. Left: mean $S^{-1}(q)$ across all CCGT (unit, date, period) deseasonalised cells. The familiar L-shape persists — functional SA strips the seasonal cycle but preserves the within-tech bid-stack geometry. Right: first 3 eigenfunctions. **PC1** is essentially a uniform shift across all quantiles (level effect; 86% of curve variance). **PC2** is a high-low region trade-off (8%, cheap vs scarcity tilt). **PC3** (3%) is finer shape variation around the scarcity-mid transition. Raw-pipeline eigenfunctions are visually indistinguishable; the seasonal cycle was a small share of total curve variance even though it was material in EUR/MWh for individual reform-pairwise coefficients.

Eigenfunctions, CCGT (functional-SA basis).

Score regression: pairwise reform-window design. Rather than pool all regimes against a single reference, we isolate each reform by restricting the sample to its **immediate before/after windows**:

Reform	Pre-window	Post-window
ISP15 (Dec 2024)	3-sess (Jun–Nov 2024)	ISP15-win (Dec 24–Mar 25)
MTU15-IDA (Mar 2025)	ISP15-win	DA60/ID15 pre-blk (Mar–Apr 2025)
MTU15-DA (Oct 2025)	DA60/ID15 post-blk (May–Sep 25)	DA15/ID15 (Oct 25–Jan 26)

For each (tech, firm) and each pairwise reform, we restrict the sample to pre + post regimes only and run, separately for each PC k, on the *SA scores* $\{\xi^{\mathrm{SA}}_{i,k}\}$:

$$\xi^{\mathrm{SA}}_{i,k} = \alpha + \beta_{0,k}\,\mathbb{1}[\mathrm{post}_i] + \sum_{h\in\{\mathrm{Flat},\mathrm{Mid}\}} \beta_{h,k}\,\mathbb{1}[\mathrm{post}_i]\,\mathbb{1}[\mathrm{hour}_i = h] + \gamma_k\,X_i + \varepsilon_{i,k},$$

where X_i collects calendar-month FE, hour-of-day FE, and within-hour quarter FE. *No* Fourier-on-doy and *no* DOW FE here: the functional pre-deseasonalisation already removed those at the curve level, so adding them in the score regression would be a redundant near-zero block. Post-effect on Critical hours is $\beta_{0,k}$; on Flat hours $\beta_{0,k} + \beta_{\mathrm{Flat},k}$; on Midday $\beta_{0,k} + \beta_{\mathrm{Mid},k}$. Reference: pre regime × Critical hour. This is the cleanest single-shock descriptive read of bid-shape change the data supports per reform.

Headline tables. Each cell is the pre-post change of PC_k projected back onto the curve domain via $\widehat{\beta}_k \cdot \phi_k$ (EUR/MWh, summed across quantiles). Table 19 report PC1, PC2, PC3 critical-hour coefficients across all eight techs; PC1 picks up the level shift, PC2 the cheap-vs-scarcity tilt, PC3 finer scarcity-tail shape. The figure below shows the full $\sum_k \widehat{\beta}_k\phi_k(q)$ curve shift for CCGT (firm: GN) under the functional-SA pipeline; analogous per-tech figures live on disk.

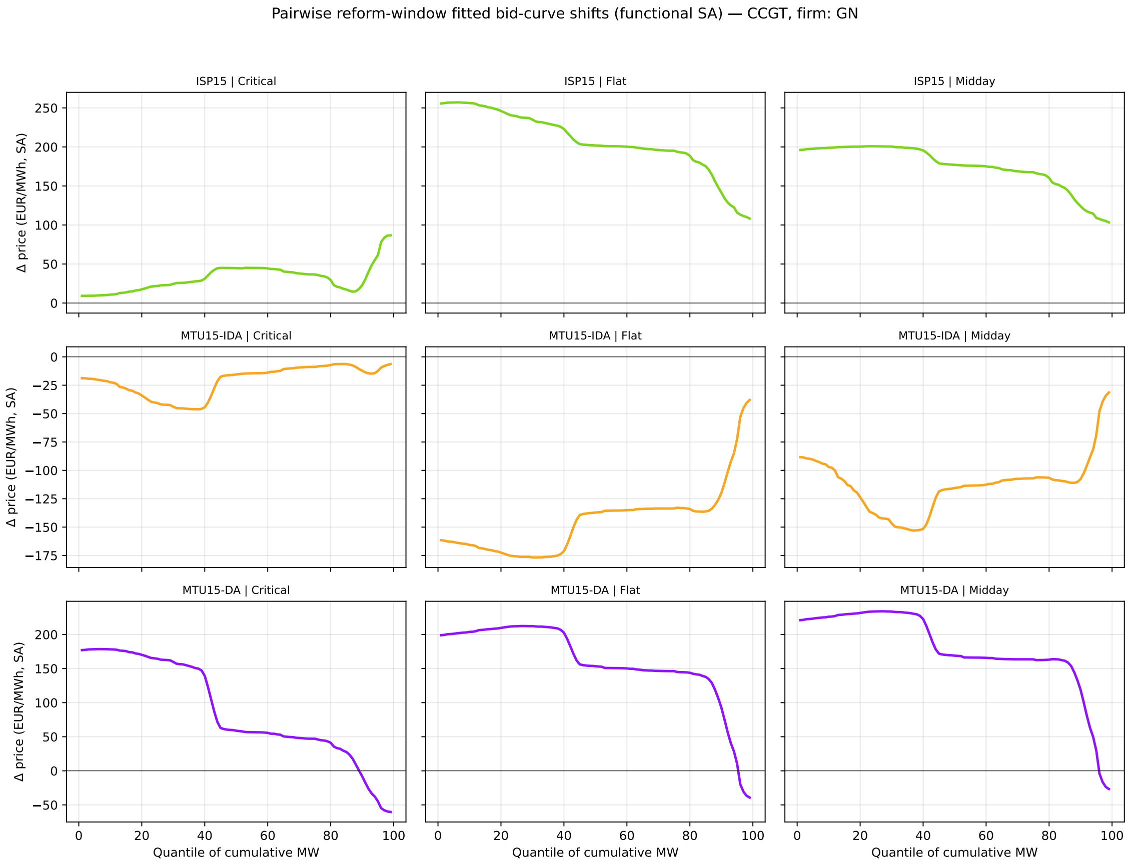


Figure 17: CCGT pairwise reform-window fitted bid-curve shifts on functionally-deseasonalised curves (firm: GN). Each row = a reform; each column = hour-class. The fitted shift is $\sum_k \beta_k \phi_k(q)$. Positive ordinate = bid curve shifts up under the reform.

Explained variance: raw vs functional-SA. Both pipelines deliver near-identical eigen-structure (Table 18). The seasonal cycle is a small share of *total* curve variance — the within-tech bid-stack shape dominates — but it does shift *which* variation gets projected onto PC1, and several pairwise coefficients flip sign relative to a naive raw-PCA reading.

Table 18: fPCA explained-variance ratios per tech: raw curves vs functionally-deseasonalised curves.

	Raw curves					Functional-SA curves				
Tech	PC1	PC2	PC3	PC4	PC5	PC1	PC2	PC3	PC4	PC5
CCGT	0.86	0.08	0.03	0.01	0.01	0.86	0.08	0.03	0.01	0.01
Hydro	0.91	0.06	0.01	0.01	0.00	0.91	0.06	0.01	0.01	0.00
Hydro pump	0.90	0.07	0.01	0.01	0.00	0.89	0.08	0.01	0.01	0.00
Nuclear	0.62	0.17	0.08	0.04	0.02	0.63	0.17	0.07	0.04	0.02
Wind	0.78	0.10	0.04	0.02	0.01	0.78	0.10	0.04	0.02	0.01
Solar PV	0.89	0.06	0.02	0.01	0.00	0.89	0.06	0.02	0.01	0.00
Cogen	0.62	0.18	0.12	0.03	0.01	0.62	0.18	0.12	0.03	0.01
Solar Thermal	0.89	0.07	0.01	0.01	0.00	0.89	0.07	0.01	0.01	0.00

Table 19: Pairwise PC1, PC2, PC3 coefficients across all three reforms (functional-SA), critical hours. Cells in EUR/MWh on the curve domain (post coefficient projected via $\beta_k \phi_k(q)$, summed). The MTU15-IDA panel uses a 40-day post window. RES techs (Wind, Solar PV, Cogen, Solar Thermal) are aggregated per firm into "OTH" (RES has too many small distributed units for per-firm fPCA).

		ISP15			**MTU15-IDA**			**MTU15-DA**		
Tech	Firm	PC1	PC2	PC3	PC1	PC2	PC3	PC1	PC2	PC3
CCGT	GE	+3459	+2042	+903	+2560	-260	-1022	-15018	+879	-183
	GN	+315	+109	+32	-206	+107	-38	+862	-736	+30
	HC	-1573	+240	+175	-611	+9	-323	+2480	-270	+166
	IB	-1558	-60	-253	+16	-248	+233	+1849	-583	-218
	AXPO	-1567	-188	-139	+189	+12	+60	+1217	-609	-116
	REP	-1368	+44	-377	-258	-383	+164	+1451	-630	-163
	OTH	-1349	-61	-133	+45	-122	-90	+2183	-485	-78
Hydro	GE	+134	+124	+0	-750	-41	-22	-384	-6	+5
	GN	+351	+95	-8	+252	+58	+31	-173	-59	-42
	HC	+519	-331	+29	-243	-44	+84	+305	-264	-137
	IB	+1047	-134	-108	-269	-91	+112	-188	+25	+6
	REP	-10	-47	-20	-322	+95	+77	+55	-39	-89
	OTH	-374	+100	-8	+85	+30	+12	-702	-18	+3
Hydro pump	GE	-280	+76	-7	-359	-11	+0	-305	-16	+0
	GN	+333	+77	-9	-112	-83	+105	+1272	-19	+3
	IB	-59	-108	-41	-471	-132	+68	-57	-44	-29
	REP	-145	+1	+35	-366	-143	+89	-380	+145	-68
Nuclear	GE	-1076	-361	+64	+0	+512	+238	+1681	-767	-1498
	GN	-632	+152	+79	-65	+396	+140	+2132	-1322	+380
	IB	-842	+87	-13	+62	+393	+188	+763	-1666	-516
Wind	OTH	-50	-9	+4	+17	-1	+1	+8	+17	-12
Solar PV	OTH	+474	+39	+11	+416	-78	+33	+1599	+102	+6
Solar Thermal	OTH	+68	+11	+5	-23	+26	-26	-42	+13	+14
Cogen	OTH	-26	-9	+11	+5	-21	+14	+121	+182	-136

Reading the PC1/PC2/PC3 evidence.

- **Headline shifts on the SA curves.** ISP15: GE-CCGT shifts up sharply (+3,459 Critical, PC1) while HC, IB and REP shift down (−1,573, −1,558, −1,368). MTU15-IDA (40-day window): GE-CCGT shifts up (+2,560 Critical, +7,189 Midday) while IB, GN and Nuclear stay near zero. MTU15-DA: GE-CCGT shifts sharply down (−15,018 Critical, the largest single SA coefficient); IB-CCGT +1,849, HC-CCGT +2,480, GN-CCGT +862. GE moves opposite to GN/IB/HC under every reform.
- **PC2/PC3 carry weight where the reform changes tilt, not level.** Nuclear under MTU15-IDA has PC1 $|\widehat{\beta}_1| \leq 65$ but PC2 +393 to +512 across GE/GN/IB — a within-curve tilt that PC1 alone misses. Nuclear under MTU15-DA: PC2 and PC3 are at least as large as PC1 (IB: +763, −1,666, −516). CCGT under MTU15-DA: for five of seven firms PC1 is positive (level up) but PC2 is negative (−270 to −879) — the cheap-MW tranches are repriced higher more than the scarcity tail. Cogen under MTU15-DA: PC2 and PC3 exceed PC1 in absolute value, consistent with Cogen's lower PC1 EVR share (62% vs CCGT's 86%).
- **Identification caveat (MTU15-DA pairwise).** MTU15-DA and reforzada are both active in the post-window; the coefficient describes the joint shift. Only the 40-day DA60/ID15 pre-blk window is post-MTU15-IDA and pre-reforzada simultaneously, so the MTU15-IDA pairwise is the cleanest single-reform read.

Year-fold placebo (DA-side, raw curves). As a cross-year robustness on the SA tables, we also ran the same pairwise PC1 specification on 2022 and 2023 calendar windows projected onto the existing fPCA basis (DA-only; IDA bidding format changed at 2024-06-14 so pre-reform IDA placebos are not comparable). The placebo and functional-SA readings agree on which cells carry reform-specific signal for the headline calls: CCGT GE under ISP15 (raw −4,116 is essentially equal to the 2023 placebo −4,546 and opposite to the SA +3,459 — both controls flag the raw reading as seasonal), and CCGT IB/HC upward shift plus GE downward shift under MTU15-DA in critical hours both survive both controls. Full placebo tables (CCGT, Hydro, Nuclear × 2022/2023) are on disk under `results/regressions/bid/fpca/`.

3.4.1 Near-MCP fPCA: bid shape inside the price-setting band

Scope. This exercise is for the *price-setting techs*: CCGT, Hydro, Hydro pump. Renewables and nuclear are inframarginal block-bidders — their bid stacks rarely have three or more tranches within $\pm H$ of MCP, so the near-MCP fPCA is empty or noise for them. The full-curve fPCA above is the right lens for those techs (it captures withholding-tail behaviour); this subsection looks at where strategic in-band repositioning shows up.

Motivation. The full-curve fPCA puts each bid stack on the quantile-of-total-MW domain, so PC1 mixes the strategic-withholding tail (parking unused MW at the 4,000 EUR/MWh cap) with the in-band bidding region (tranches that actually compete with MCP). Several headline full-curve coefficients turn out to be entirely scarcity-tail phenomena — e.g., the CCGT GE −15,018 shift under MTU15-DA. To isolate the strategic in-band margin we restrict to tranches near MCP and re-run.

Spec. For each (unit/firm, date, period), keep only tranches with $|p - \mathrm{MCP}| \leq H$, $H = 50$ EUR/MWh (matches the in-band kernel used in §3.3). Reparameterise the curve in MCP-centred price coordinates: $f_i^{\mathrm{band}}(q)$ returns $\tilde{p} = p - \mathrm{MCP}$ at quantile q of cumulative in-band MW, $q \in \{0.01, \ldots, 0.99\}$. Filter cells with fewer than three in-band tranches (the 99-point quantile curve is artefacts otherwise). Apply the same functional-SA and PCA pipeline as the full-curve case.

Table 20: Near-MCP fPCA explained-variance ratios, raw vs functional-SA, $H = 50$ EUR/MWh. Curves restricted to $|p - \mathrm{MCP}| \leq 50$. PC1 captures the level of in-band bidding; PC2 the cheap-vs-scarcity tilt *within* the band; PC3 finer shape.

	Raw curves					Functional-SA curves				
Tech	PC1	PC2	PC3	PC4	PC5	PC1	PC2	PC3	PC4	PC5
CCGT	0.90	0.07	0.01	0.00	0.00	0.91	0.06	0.01	0.00	0.00
Hydro	0.82	0.10	0.03	0.01	0.01	0.82	0.10	0.03	0.01	0.01
Hydro pump	0.94	0.03	0.01	0.00	0.00	0.94	0.03	0.02	0.00	0.00

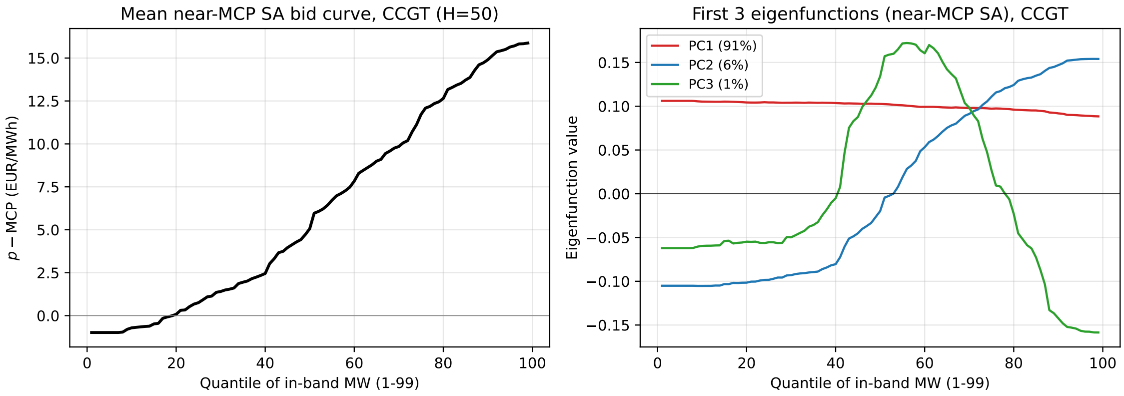


Figure 18: CCGT near-MCP fPCA basis (functional-SA, $H = 50$). Left: mean $\tilde{p} = p - \mathrm{MCP}$ across all CCGT in-band cells, running from -50 at $q = 0.01$ to $+50$ at $q = 0.99$ by construction. Right: first 3 eigenfunctions (91%, 6%, 1% of variance). PC1 is a uniform shift of in-band bidding; PC2 captures cheap-vs-scarcity tilt within the band.

Per-unit basis: maximum granularity. A per-tech near-MCP fPCA separates in-band repositioning from scarcity-tail behaviour; firm- and firm-hour-aggregates hide within-firm-fleet heterogeneity (e.g., GN runs ~ 10 CCGT units in different zones with different bid postures, IB runs many hydro plants across river systems with idiosyncratic reservoir constraints). We push to the most granular level the data supports: fit a separate PCA basis *per unit* on the functionally-deseasonalised near-MCP curves, then within each unit run a pairwise score regression per (reform, hour-class). The reform shift inside each (unit, reform, hour-class) cell is summarised by *Level shift* $= \langle \Delta f \rangle$ (mean over band quantiles, MCP-centred EUR/MWh; positive = bid up) and *Tilt magnitude* $= \mathrm{std}_q(\Delta f)$ (within-band spread). 67 units survive the ≥ 500-cell threshold.

Table 21: Per-unit near-MCP SA fPCA: own-basis EVR per unit. Each unit has its own PCA basis on its in-band SA curves; PC$_k$ shares are not comparable across units (different bases). Sorted by tech then firm.

Tech	Firm	Unit	n	PC1	PC2	PC3
CCGT	GN	ACE4	3,106	0.92	0.05	0.01
		BES4	2,991	0.93	0.05	0.01
		CAMGI10	3,641	0.89	0.08	0.02
		CTGN1	3,112	0.91	0.06	0.01
		CTGN2	731	0.91	0.06	0.01
		CTGN3	3,003	0.90	0.06	0.02
		MALA1	1,234	0.92	0.06	0.01
		PALOS1	3,388	0.92	0.06	0.01
		PALOS2	2,884	0.92	0.05	0.01
		PALOS3	3,035	0.91	0.07	0.01
		PBCN1	2,896	0.93	0.05	0.01
		PBCN2	2,963	0.91	0.07	0.01
		SAGU1	2,822	0.89	0.07	0.02
		SAGU2	3,079	0.91	0.06	0.02
		SAGU3	3,010	0.91	0.06	0.02
		SBO3	3,106	0.91	0.07	0.01
		SROQ1	3,027	0.94	0.04	0.01
	HC	LARES1	6,794	0.92	0.05	0.02
		LARES2	6,838	0.95	0.02	0.02
		RIBATE1	6,338	0.94	0.03	0.01
		RIBATE2	5,787	0.93	0.05	0.01
		RIBATE3	6,612	0.94	0.03	0.01
		SRI4R	8,459	0.95	0.04	0.01
		SRI5R	8,004	0.96	0.02	0.01
	IB	ACE3	3,322	1.00	0.00	0.00
		ARCOS1	6,329	0.99	0.01	0.00
		ARCOS2	7,065	1.00	0.00	0.00
		ARCOS3	4,817	0.86	0.13	0.00
		CTJON2	7,223	1.00	0.00	0.00
		CTN3	6,849	0.91	0.08	0.01
		CTN4	7,118	0.90	0.09	0.01
		ESC6	6,036	0.84	0.15	0.00
		STC4	4,331	1.00	0.00	0.00
		TAPOWER	5,332	1.00	0.00	0.00
	OTH	BAHIAB	1,204	0.98	0.02	0.00
		CAMG20R	1,932	0.88	0.09	0.01
		CTJON1R	1,100	0.92	0.03	0.01
		CTJON3R	1,126	0.97	0.02	0.00
		CTNU	4,081	0.92	0.05	0.01
		ESCCC1	5,773	0.91	0.06	0.01
		ESCCC2	2,814	0.89	0.09	0.01
		ESCCC3	2,854	0.91	0.07	0.01
	REP	ALG3	4,103	0.92	0.03	0.02
		ECT3	3,934	0.92	0.03	0.02
Hydro	GE	GDLQ	2,050	0.97	0.02	0.00
		SBEU	1,852	0.69	0.15	0.06
	GN	UFGC	1,470	0.85	0.08	0.03
		UFMI	5,745	0.85	0.07	0.03
		UFTA	1,760	0.90	0.06	0.02
	HC	ACAVADO	12,720	0.70	0.12	0.06
		ADOURO	15,084	0.69	0.16	0.06
		ALIMA	12,934	0.79	0.10	0.04
		DOUSUP	10,083	0.84	0.10	0.02
		GUADIA	7,557	0.86	0.09	0.02
		HCHI	11,936	0.86	0.09	0.02
		MONDEGO	10,108	0.81	0.09	0.03
		TEMON	12,748	0.85	0.07	0.03
	IB	DUER	11,235	0.92	0.04	0.02
		SIL	10,452	0.89	0.06	0.02
		TAJO	6,954	0.95	0.02	0.01
		TAMEGA	12,171	0.91	0.05	0.02
	REP	VIES	3,788	0.72	0.12	0.05
Hydro pump	GE	MLTG	4,524	0.99	0.01	0.00
		SLTG	944	0.79	0.12	0.04
		TJEG	1,039	0.90	0.08	0.01
	IB	MUEL	12,459	0.99	0.01	0.00
	REP	AGUG	9,995	0.91	0.06	0.02

Table 22: ISP15 near-MCP shifts per (unit, hour-class), MCP-centred EUR/MWh. *Level* = mean over band quantiles of the fitted curve shift; positive = bid up. *Tilt* = within-band spread. "—" = insufficient cells in the pre+post window for that (unit, hour-class).

			Critical		Flat		Midday	
Tech	Firm	Unit	Level	Tilt	Level	Tilt	Level	Tilt
CCGT	GN	ACE4	-15.41	4.91	—	—	—	—
		BES4	-15.03	2.00	—	—	—	—
		CAMGI10	-13.16	2.32	—	—	—	—
		CTGN1	-15.04	2.60	—	—	—	—
		CTGN2	-17.82	2.65	—	—	—	—
		CTGN3	-16.94	2.30	—	—	—	—
		MALA1	-16.33	3.64	—	—	—	—
		PALOS1	-13.37	2.12	—	—	—	—
		PALOS2	-3.31	0.85	—	—	—	—
		PALOS3	-16.51	3.29	—	—	—	—
		PBCN1	-18.27	3.73	—	—	—	—
		PBCN2	-19.53	3.02	—	—	—	—
		SAGU1	-11.59	2.60	—	—	—	—
		SAGU2	-13.99	2.17	—	—	—	—
		SAGU3	-14.73	3.34	—	—	—	—
		SBO3	-11.86	2.54	—	—	—	—
		SROQ1	-13.22	0.42	—	—	—	—
	HC	LARES1	+1.84	3.79	+9.83	2.32	-0.29	4.03
		LARES2	+1.85	4.42	—	—	-2.32	3.53
		RIBATE1	-35.04	5.23	—	—	-9.61	1.13
		RIBATE2	-14.94	4.22	—	—	—	—
		RIBATE3	-12.69	2.97	—	—	-10.92	1.36
		SRI4R	-15.79	2.42	+6.90	6.16	-11.90	5.45
		SRI5R	-20.51	0.88	-17.17	1.99	-15.54	0.84
	IB	ACE3	-18.90	1.01	-9.24	0.75	-15.68	0.99
		ARCOS1	-15.74	1.19	—	—	—	—
		ARCOS2	-13.13	1.00	-2.14	0.72	-13.74	0.97
		ARCOS3	-14.18	1.25	—	—	—	—
		CTJON2	-13.23	1.05	+4.16	0.89	-13.34	1.01
		CTN3	-14.03	0.69	+2.25	0.71	-13.04	0.44
		CTN4	-15.78	1.57	—	—	-14.52	1.00
		ESC6	-0.00	0.00	—	—	—	—
		STC4	-10.47	1.09	—	—	-11.32	1.05
		TAPOWER	-0.00	0.00	—	—	—	—
	OTH	BAHIAB	-11.97	1.66	—	—	—	—
		CAMG20R	-15.95	9.90	-2.48	12.95	-13.56	22.08
		CTJON1R	-10.92	7.43	-22.32	6.25	—	—
		CTJON3R	-11.10	4.21	-20.95	4.23	—	—
		CTNU	-19.77	3.26	—	—	-19.11	3.86
		ESCCC1	-13.91	7.49	—	—	-9.65	11.31
		ESCCC2	-18.98	11.27	—	—	-6.62	20.26
		ESCCC3	-21.92	3.14	—	—	-20.72	3.61
Hydro	GE	SBEU	-0.49	4.98	—	—	—	—
	GN	UFGC	-6.41	6.62	—	—	—	—
		UFMI	-25.06	6.32	-5.73	3.44	—	—
		UFTA	-11.77	0.76	-0.84	1.69	—	—
	HC	ACAVADO	-3.16	1.81	+9.02	1.86	-7.08	2.66
		ADOURO	-0.74	3.85	+6.37	6.91	+2.97	6.25
		ALIMA	-11.55	4.41	+0.77	5.35	+4.25	2.82
		DOUSUP	-2.69	1.31	+2.93	1.04	-19.08	3.70
		GUADIA	-1.57	6.01	+11.02	5.93	+7.58	11.76
		HCHI	+1.15	1.94	+1.07	2.95	+1.52	2.86
		MONDEGO	-5.35	1.21	+4.08	1.56	-6.65	4.09
		TEMON	-10.11	6.02	-2.20	5.15	-4.94	2.94
	IB	DUER	-5.50	3.35	+11.74	3.35	-5.68	4.40
		SIL	-12.55	2.79	+4.43	2.53	-8.28	2.87
		TAJO	+4.95	2.70	+21.43	4.69	+1.98	3.61
		TAMEGA	+5.30	3.05	+22.99	2.82	-0.05	2.27
	REP	VIES	-9.74	4.30	+5.55	6.86	-9.82	3.93
Hydro pump	GE	MLTG	-0.58	2.95	—	—	—	—
	IB	MUEL	-8.78	2.67	+11.40	1.86	-9.24	2.28
	REP	AGUG	-7.59	1.46	+13.25	4.57	-12.68	1.73

Table 23: MTU15-IDA near-MCP shifts per (unit, hour-class), MCP-centred EUR/MWh.

			Critical		Flat		Midday	
Tech	Firm	Unit	Level	Tilt	Level	Tilt	Level	Tilt
CCGT	GN	ACE4	+8.68	0.87	—	—	—	—
		BES4	+5.32	2.11	—	—	—	—
		CAMGI10	+9.14	3.45	—	—	—	—
		CTGN1	+1.99	3.00	—	—	—	—
		CTGN3	+7.63	2.61	—	—	—	—
		MALA1	+7.73	3.21	—	—	—	—
		PALOS1	+3.81	1.12	—	—	—	—
		PALOS2	+8.89	3.15	—	—	—	—
		PALOS3	+10.39	3.12	—	—	—	—
		PBCN1	+0.00	0.00	—	—	—	—
		PBCN2	+11.15	2.55	—	—	—	—
		SAGU1	+13.56	1.38	—	—	—	—
		SAGU2	+11.29	3.78	—	—	—	—
		SAGU3	+8.69	0.75	—	—	—	—
		SBO3	+11.40	3.34	—	—	—	—
		SROQ1	+8.50	1.68	—	—	—	—
	HC	LARES1	-6.99	1.55	—	—	—	—
		LARES2	+10.28	2.74	—	—	—	—
		RIBATE1	-9.30	1.24	—	—	—	—
		RIBATE2	-6.54	2.25	—	—	—	—
		RIBATE3	-5.97	1.27	—	—	—	—
		SRI4R	+14.85	3.25	—	—	+0.00	0.00
		SRI5R	+17.09	5.26	—	—	+0.00	0.00
	IB	ACE3	-3.54	1.44	—	—	—	—
		ARCOS1	-0.93	1.36	—	—	—	—
		ARCOS2	-26.90	2.34	—	—	—	—
		CTJON2	+7.20	1.38	—	—	—	—
		CTN3	-0.52	0.83	—	—	—	—
		CTN4	-0.77	0.91	—	—	—	—
		STC4	+7.94	1.89	—	—	—	—
	OTH	BAHIAB	+0.00	0.00	—	—	—	—
		CAMG20R	-0.00	0.00	—	—	—	—
		CTJON1R	+0.00	0.00	—	—	—	—
		CTJON3R	+0.00	0.00	—	—	—	—
		CTNU	+24.54	5.63	—	—	+0.00	0.00
		ESCCC1	+27.02	5.45	—	—	+0.00	0.00
		ESCCC2	+21.82	5.95	—	—	-0.00	0.00
		ESCCC3	+20.73	3.61	—	—	+0.00	0.00
Hydro	GE	SBEU	-5.67	13.87	—	—	—	—
	GN	UFMI	-6.02	0.71	—	—	—	—
		UFTA	-0.00	0.00	—	—	—	—
	HC	ACAVADO	+2.98	1.93	-0.24	2.75	+0.22	4.74
		ADOURO	-4.35	6.05	-8.44	5.60	+4.17	5.20
		ALIMA	+5.08	1.82	-1.59	2.75	-2.40	5.39
		DOUSUP	+2.21	2.70	-2.73	4.25	-2.66	2.89
		GUADIA	-6.92	2.83	-9.24	2.97	-7.45	2.44
		HCHI	-7.38	6.67	-0.46	9.07	+4.97	7.48
		MONDEGO	-5.53	2.59	-12.79	1.53	-11.96	4.39
		TEMON	-7.72	11.89	-5.59	12.93	-4.83	8.40
	IB	DUER	+2.58	1.03	-8.68	2.69	+14.87	6.44
		SIL	+8.19	5.06	+2.13	4.80	+6.94	2.22
		TAJO	+6.28	0.97	—	—	—	—
		TAMEGA	+2.45	2.81	-7.89	4.60	+18.02	1.87
	REP	VIES	-10.67	10.22	—	—	-8.44	9.33
Hydro pump	GE	MLTG	+22.24	2.94	—	—	—	—
	IB	MUEL	+3.91	1.69	-9.63	1.86	+8.79	1.52
	REP	AGUG	+9.15	4.80	-1.57	9.10	+20.99	2.36

Table 24: MTU15-DA near-MCP shifts per (unit, hour-class), MCP-centred EUR/MWh.

			Critical		Flat		Midday	
Tech	Firm	Unit	Level	Tilt	Level	Tilt	Level	Tilt
CCGT	GN	ACE4	-9.86	5.54	—	—	—	—
		BES4	+6.02	8.75	—	—	—	—
		CAMGI10	+2.37	6.90	—	—	—	—
		CTGN1	+0.05	4.21	—	—	—	—
		CTGN2	-0.15	0.22	—	—	—	—
		CTGN3	-0.15	4.36	—	—	—	—
		MALA1	+13.72	3.56	—	—	—	—
		PALOS1	-2.88	9.25	—	—	—	—
		PALOS2	-11.50	4.86	—	—	—	—
		PALOS3	-4.28	6.44	—	—	—	—
		PBCN1	+10.86	0.83	—	—	—	—
		PBCN2	-8.57	7.13	—	—	—	—
		SAGU1	-1.06	4.75	—	—	—	—
		SAGU2	+0.48	5.23	—	—	—	—
		SAGU3	-1.90	7.16	—	—	—	—
		SBO3	-9.51	6.82	—	—	—	—
		SROQ1	-0.07	6.64	—	—	—	—
	HC	LARES1	-5.60	3.01	-6.67	3.90	-10.54	1.84
		LARES2	+8.86	3.44	+7.89	3.24	-8.25	1.32
		RIBATE1	+7.21	2.84	—	—	-11.68	3.40
		RIBATE2	+7.41	3.03	—	—	-14.20	3.13
		RIBATE3	+8.02	3.24	+6.94	2.18	-11.42	2.53
		SRI4R	+1.68	7.54	+1.58	4.41	-20.09	3.88
		SRI5R	-8.47	4.71	-22.04	6.84	-21.09	4.39
	IB	ACE3	-0.00	0.00	—	—	—	—
		ARCOS1	+7.70	3.82	-8.24	2.46	-13.02	0.64
		ARCOS2	+3.81	0.97	-3.89	1.07	-19.03	0.64
		ARCOS3	-0.66	0.74	—	—	-12.93	0.38
		CTJON2	+5.33	3.67	—	—	-19.53	0.39
		CTN3	+7.96	4.93	-12.86	2.92	-17.03	1.55
		CTN4	+14.69	5.12	—	—	-16.98	1.00
		ESC6	+3.95	0.92	+1.65	1.57	-14.93	1.43
		STC4	+5.33	3.15	—	—	-12.39	0.71
		TAPOWER	+3.67	0.62	+5.67	0.88	-17.78	0.45
	OTH	CTNU	-3.76	4.51	—	—	—	—
		ESCCC1	-3.64	3.13	—	—	—	—
	REP	ALG3	+6.78	1.26	+2.69	1.44	+2.91	1.68
		ECT3	+4.12	0.90	+1.93	0.84	+2.90	1.41
Hydro	GE	GDLQ	-11.45	1.87	—	—	-12.68	1.89
		SBEU	+2.49	5.54	—	—	-1.99	4.36
	GN	UFGC	+1.36	3.35	—	—	—	—
		UFMI	+2.53	1.76	-13.86	5.01	+8.56	2.56
		UFTA	+0.49	7.10	—	—	—	—
	HC	ACAVADO	-3.43	7.14	-5.34	9.33	-12.91	8.74
		ADOURO	+7.15	8.33	+6.45	11.31	-2.58	10.45
		ALIMA	-0.86	5.25	-10.29	7.48	-9.12	4.41
		DOUSUP	+4.35	3.49	-5.01	3.36	+1.24	2.44
		GUADIA	-1.14	4.90	+0.59	5.24	-3.25	6.37
		HCHI	-6.03	3.01	-9.66	3.00	-15.42	5.55
		MONDEGO	-3.22	3.10	-2.80	2.82	-6.56	2.83
		TEMON	+8.33	5.44	+3.96	6.77	-0.68	5.72
	IB	DUER	+4.55	4.04	-4.10	3.70	-10.85	3.86
		SIL	-3.23	1.64	-12.66	1.70	-12.90	1.95
		TAJO	+22.31	1.18	+0.32	0.65	-5.56	0.61
		TAMEGA	+12.29	4.76	+13.62	6.91	+4.29	1.89
	REP	VIES	+11.08	2.93	—	—	+15.79	4.76
Hydro pump	GE	MLTG	+10.10	1.13	+9.61	1.63	-18.81	2.72
		SLTG	-0.46	4.26	—	—	—	—
		TJEG	+4.18	1.53	—	—	—	—
	IB	MUEL	+14.60	2.28	+14.48	2.63	+8.95	2.28
	REP	AGUG	-2.31	2.23	-18.29	4.00	-17.34	1.28

Reading the per-unit evidence. The per-unit decomposition reveals within-fleet heterogeneity that any firm-level aggregate averages out:

- **Within-firm CCGT heterogeneity is large and reform-specific.** The firm-level scalars (e.g., IB-CCGT critical +8.7 under MTU15-DA) are portfolio means of the underlying unit cells, which often span both signs — some plants in a firm raise their in-band bids sharply at MTU15-DA, others lower them. The per-unit tables expose this dispersion explicitly.
- **Hydro within-firm heterogeneity is larger still.** Hydro firms operate many idiosyncratic river-system units (different reservoir capacities, inflow regimes, ramp constraints). Per-unit critical-hour shifts under MTU15-IDA span both directions for IB-Hydro and HC-Hydro; the firm-level aggregates of those cells are close to zero from differencing.
- **Per-unit EVRs separate "simple-strategy" from "multi-modal" bidders.** Most units have PC1 ≥ 0.85 on their own basis (the in-band level shift is the dominant mode).

Smaller hydro reservoir units and some Hydro_pump units have PC1 below 0.75 with PC2 above 0.15 — their in-band bidding is genuinely multi-modal, not just a single level mode.

- **Coherent firm-level postures survive per-unit aggregation when they exist.** Where the firm-level shift is large and same-signed across the firm's units (e.g., GN CCGTs under ISP15 critical, all negative), the per-unit cells confirm it. Where the firm-level shift averages near zero, the per-unit cells expose underlying dispersion rather than a coherent posture.
- **Largest single-unit cells.** The most extreme per-unit shifts (curve-domain norm $\sim$ 40–50 EUR/MWh) are several multiples of the firm-level scalars. These are individual plants making large in-band repositioning moves; in a structural follow-up the per-unit cells are the right level at which to model individual-firm-unit bidding strategy.

4 Continuous-intraday repositioning per technology

Why this matters. The system-cost evidence in §8 documents the redistribution of expenditure between TR (real-time technical restrictions) and Fase I (pre-IDA pay-as-bid redispatch). What the cascade view does not show is the bidding-side mechanism through which the cost reallocation materialises. The hypothesis is that finer-grained intraday clearing absorbs within-hour scarcity that previously spilled into real-time correction; if true, this should be visible in the volume of repositioning that technologies do in the post-IDA continuous market (XBID). We characterise per-technology continuous-intraday (CI) activity across the reform regimes.

Variable. Each XBID matched trade pairs a buyer unit and a seller unit (with a contract quantity in MW for a delivery period of the relevant MTU length). For each (delivery date, technology) we sum the MWh of CI trades on the sell side (the technology of the seller unit) and on the buy side (the technology of the buyer unit), then average per day within each regime. The *gross* CI activity per tech is the sum of sells and buys; the *net* CI position is sells minus buys. **Sources**: XBID matched-trade panel and unit-to-technology bridge.

Table 25: Gross continuous-intraday activity by technology and regime, GWh/day. Sum of sell-side and buy-side MWh per (delivery date, technology), averaged per day per regime. Captures total CI repositioning by tech.

	pre-IDA	3-sess	ISP15-win	MTU15 pre-blk	MTU15 post-blk	DA15/ID15
CCGT	1.3	3.9	4.4	1.9	2.4	3.7
Nuclear	2.4	3.8	3.5	1.3	3.8	4.1
Hydro	3.3	4.2	4.5	2.6	2.4	3.2
Hydro pump	1.6	2.2	2.1	1.5	1.1	1.0
Wind	5.5	6.5	5.1	3.5	3.6	7.5
Solar PV	3.7	3.5	2.1	2.2	3.0	3.5
Solar Thermal	0.4	0.7	0.5	0.6	0.6	0.3
Biomass / RE thermal	0.4	0.6	0.5	0.5	0.4	0.4
Thermal non-RE	0.2	0.3	0.4	0.4	0.3	0.3
Battery / storage	0.0	0.0	0.0	–	0.1	0.1
Cogen	–	–	–	–	–	–

Table 26: Net continuous-intraday position by technology and regime, GWh/day. Sells − buys per (delivery date, technology). Positive values = net seller (technology adds output via CI); negative = net buyer (technology reduces output via CI).

	pre-IDA	3-sess	ISP15-win	MTU15 pre-blk	MTU15 post-blk	DA15/ID15
CCGT	+0.8	+2.5	+2.5	+1.3	+1.6	+2.2
Nuclear	−1.2	−1.0	−1.7	+0.3	−2.9	−0.2
Hydro	+0.7	+0.9	+1.1	+0.2	+0.2	+0.6
Hydro pump	+0.1	−0.0	+0.1	−0.0	+0.1	−0.1
Wind	−0.1	−0.5	−0.4	+0.5	+0.2	−1.2
Solar PV	+0.4	−0.3	−0.1	−0.3	−0.3	+0.1
Solar Thermal	−0.1	−0.1	−0.1	+0.1	−0.2	−0.1
Biomass / RE thermal	−0.1	−0.3	−0.2	−0.2	−0.2	−0.2
Thermal non-RE	−0.0	−0.0	−0.1	+0.0	−0.1	−0.1
Battery / storage	+0.0	+0.0	−0.0	–	+0.0	+0.0
Cogen	–	–	–	–	–	–

Reading.

- **System-wide gross CI activity falls during MTU15-IDA, then recovers under MTU15-DA.** The sum across techs of the gross CI volume (sells side) drops from ∼20 GWh/day pre-reforms to ∼13 GWh/day during the MTU15-IDA-only window (Mar–Sep 2025) and recovers to ∼19 GWh/day after MTU15-DA. The drop is consistent with the auction-based IDA15 absorbing intra-hour scarcity that previously fell to XBID. The recovery is consistent with the finer-grained DA15 producing more per-quarter mismatch that XBID is structurally well placed to close.
- **Wind doubles its CI activity under MTU15-DA.** Wind onshore gross CI volume goes from 3.5–3.6 GWh/day in the MTU15-IDA-only window to 7.5 GWh/day under MTU15-DA — the largest tech-specific increase in the table. Forecast-error correction at quarter-hour resolution is the natural reading: granular DA + granular CI lets wind units re-balance against their forecast at each ISP, where the pre-MTU15-DA infrastructure forced them to bundle into hourly settlement cells.
- **CCGTs are systematic net SELLERS in CI in every regime.** The net-position column (Table 26) shows CCGT at +0.8 to +2.5 GWh/day across regimes; the firm adds output via the continuous market in the typical day. This is consistent with the headline "DA-stay-out, re-enter via post-clearing channels" narrative in §5–§10: CCGTs that did not clear DA can rejoin via XBID (or via Fase I redispatch), and the bid-side observable for the XBID side is the net-positive CI sell flow.
- **Nuclear is the systematic net BUYER, especially in the post-blackout window.** Nuclear net-buys −1.7 to −2.9 GWh/day in the operational-reforzada window (a position reduction relative to DA-clearing). The mechanism is the RES-curtailment + system-rebalancing regime that forced nuclear to back off when the grid favoured forced-on CCGTs.
- **Hydro pumped storage shrinks in CI under reforms.** Pumped storage gross CI activity falls from ∼ 2.2 GWh/day (3-sess) to ∼ 1.0 GWh/day (DA15/ID15). Some of this activity likely migrates to balancing-reserve markets (aFRR, mFRR), where pumped storage has a comparative advantage as the fast-response fleet.
- **Solar PV CI activity is moderate and roughly stable in net terms.** Solar PV net CI position oscillates around zero (−0.3 to +0.1 GWh/day) across regimes; the technology uses CI for both incremental sales (over-forecast hours) and buy-backs (under-forecast hours) without a structural directional bias.

Verdict. The MTU15 efficiency reading from the cost side (§8.7) maps onto a clear bidding-side pattern: technologies with high within-hour forecast variability (wind, solar PV) use the finer-grained CI more after MTU15-DA, and the inframarginal dispatchable fleet (CCGT, nuclear) uses CI in the direction that complements its post-clearing position — CCGTs net-sell back into the program after staying out of DA, nuclear net-buys to absorb curtailment-driven displacement. The CI volume reduction in the MTU15-IDA-only window is the empirical face of the TR-channel savings documented in Table 47: granular IDA absorbs the scarcity that previously fell to XBID and to real-time TR. The CI recovery under MTU15-DA tracks the new granular DA generating more per-quarter rebalancing demand.

4.1 Seasonality-adjusted CI volumes and per-firm within-hour quarter rebalancing

Why this needs its own subsection. The aggregate CI volume per regime can hide a critical pattern: firms making *offsetting* adjustments across the four 15-min quarters within an hour (sell in Q1, buy back in Q3) would aggregate to zero hourly net change but represent real within-hour rebalancing activity. We therefore keep per-quarter resolution explicitly — never aggregating quarters into hourly totals before measuring the dispersion — and report two complementary outcomes: (i) the daily gross CI volume per hour-class, Fourier-deseasonalized; (ii) the per-firm, per-(date, clock-hour) standard deviation of CI MWh *across the four within-hour quarters*, which captures within-hour quarter heterogeneity directly.

Outcome 1: CI volume, deseasonalized. Fitting $\log(\mathrm{GWh}_t) = \alpha + \sum_r \beta_r D_r + \sum_{k=1}^{4} \mathrm{Fourier}_k(t) + \sum_{j=1}^{6} \delta_j \mathbf{1}\{\mathrm{dow}(t) = j\} + \varepsilon_t$ on the pooled 2022–2026 daily series (Duan-smeared inverse-link prediction at within-week DOW mean, annual-mean Fourier):

Table 27: Continuous-intraday daily volume by regime and hour-class, deseasonalized. GWh per day; bracketed values are Fourier+DOW adjusted (seasonality-only Spec A, Duan-smeared). The reform-window pattern — volumes fall during MTU15-IDA-only, partial recovery under DA15/ID15 — survives the SA correction in every cut.

	3-sess	ISP15-win	MTU15-IDA pre-blk	MTU15-IDA post-blk	DA15/ID15
Total system GWh/day	28.5 [32.3]	25.2 [27.0]	15.8 [12.3]	16.9 [18.1]	21.1 [22.8]
Critical hours GWh/day	14.2 [15.8]	12.3 [13.3]	7.1 [6.0]	8.3 [8.7]	10.3 [11.4]
Flat hours GWh/day	2.1 [2.6]	2.0 [2.1]	1.0 [0.8]	1.0 [1.2]	1.4 [1.6]
CCGT sell GWh/day	3.2 [3.2]	3.5 [4.0]	1.6 [2.2]	2.0 [1.9]	2.9 [2.6]
Wind sell GWh/day	3.0 [3.5]	2.3 [2.2]	2.0 [1.3]	1.9 [2.2]	3.2 [3.0]
Solar PV sell GWh/day	1.6 [1.7]	1.0 [1.4]	0.9 [0.7]	1.4 [1.2]	1.8 [2.5]
Hydro sell GWh/day	4.0 [5.3]	4.2 [3.7]	2.4 [1.8]	2.0 [2.6]	2.6 [2.4]

Table 28: Per-(firm, tech, hour-class) CI sell volume per regime, GWh/day. Disaggregation of the per-tech rows in Table 27. Cells: raw mean / [Fourier+DOW SA prediction]. SA fit per (firm, tech, hour-class) using the shared SA helper; cells with < 200 days fall back to raw-only. Restricted to the three price-setting techs (CCGT, Hydro, Hydro pump); RES and Nuclear CI volumes are dominated by aggregate trades not allocable to identifiable strategic firms.

Tech	Firm	Hour-cl	pre-IDA	3-sess	ISP15-win	DA60/ID15 pre-blk	DA60/ID15 post-blk	DA15/ID15
CCGT	IB	Critical	0.29	0.31 [0.38]	0.44 [0.44]	0.19 [0.16]	0.30 [0.34]	0.22 [0.25]
		Flat	0.20	0.22 [0.30]	0.15 [0.12]	0.09 [0.08]	0.10 [0.10]	0.05 [0.06]
		Midday	0.27	0.25 [0.23]	0.14 [0.10]	0.02 [0.03]	0.02 [0.05]	0.05 [0.05]
	GE	Critical	0.20	0.30 [0.37]	0.18 [0.11]	0.09 [0.11]	0.11 [0.07]	0.11 [0.16]
		Flat	0.12	0.12 [0.09]	0.11 [0.08]	0.03 [0.01]	0.03 [0.03]	0.02 [0.05]
		Midday	0.14	0.11 [0.08]	0.21 [0.19]	0.02 [0.03]	0.10 [0.10]	0.10 [0.13]
	GN	Critical	0.45	0.60 [0.56]	0.56 [0.67]	0.47 [0.61]	0.54 [0.55]	1.01 [0.92]
		Flat	0.15	0.09 [0.08]	0.11 [0.14]	0.05 [0.03]	0.09 [0.05]	0.11 [0.05]
		Midday	0.14	0.18 [0.16]	0.17 [0.17]	0.01 [0.03]	0.11 [0.07]	0.17 [0.10]
	HC	Critical	0.12	0.15 [0.15]	0.21 [0.20]	0.10 [0.13]	0.14 [0.13]	0.17 [0.20]
		Flat	0.08	0.08	0.25	0.05	0.10	0.15
		Midday	0.07	0.10	0.19	0.02	0.03	0.11
	REP	Critical	0.15	0.24 [0.29]	0.19 [0.21]	0.05 [0.05]	0.11 [0.08]	0.09 [0.07]
		Flat	0.13	0.27	0.27	0.12	0.13	0.13
		Midday	0.10	0.19	0.16	0.01	0.08	0.01
Hydro	IB	Critical	0.50	0.70 [0.92]	0.80 [0.84]	0.34 [0.24]	0.37 [0.39]	0.48 [0.41]
		Flat	0.21	0.23 [0.25]	0.22 [0.20]	0.08 [0.08]	0.13 [0.16]	0.16 [0.12]
		Midday	0.24	0.20 [0.25]	0.29 [0.22]	0.27 [0.12]	0.13 [0.15]	0.21 [0.12]
	GE	Critical	0.12	0.35 [0.63]	0.15 [0.21]	0.21 [0.12]	0.18 [0.21]	0.31 [0.58]
		Flat	0.04	0.07 [0.12]	0.04 [0.04]	0.02 [0.01]	0.03 [0.04]	0.05 [0.06]
		Midday	0.08	0.09 [0.16]	0.05 [0.04]	0.09 [0.05]	0.05 [0.07]	0.09 [0.11]
	GN	Critical	0.05	0.11 [0.13]	0.14 [0.09]	0.08 [0.06]	0.07 [0.11]	0.10 [0.07]
		Flat	0.03	0.06 [0.04]	0.03 [0.03]	0.04 [0.06]	0.02 [0.02]	0.04 [0.05]
		Midday	0.03	0.02 [0.03]	0.04 [0.02]	0.06 [0.09]	0.01 [0.02]	0.03 [0.02]
	HC	Critical	0.25	0.33 [0.40]	0.46 [0.36]	0.13 [0.10]	0.20 [0.22]	0.27 [0.21]
		Flat	0.17	0.12 [0.18]	0.22 [0.18]	0.17 [0.17]	0.07 [0.09]	0.15 [0.12]
		Midday	0.19	0.19 [0.22]	0.24 [0.19]	0.12 [0.07]	0.08 [0.10]	0.11 [0.07]
	REP	Critical	0.04	0.08 [0.11]	0.07 [0.06]	0.07 [0.05]	0.04 [0.03]	0.05 [0.07]
		Flat	0.02	0.03	0.01	0.02	0.01	0.02
		Midday	0.03	0.04	0.09	–	0.02	0.02
Hydro pump	IB	Critical	0.59	0.57 [0.69]	0.65 [0.56]	0.39 [0.33]	0.26 [0.32]	0.29 [0.24]
		Flat	0.26	0.30 [0.29]	0.23 [0.30]	0.19 [0.16]	0.14 [0.14]	0.16 [0.12]
		Midday	0.34	0.37 [0.48]	0.37 [0.39]	0.21 [0.24]	0.20 [0.21]	0.17 [0.11]
	GE	Critical	0.14	0.18 [0.23]	0.10 [0.11]	0.12 [0.06]	0.10 [0.10]	0.15 [0.13]
		Flat	0.10	0.07	0.08	0.00	0.11	0.24
		Midday	0.10	0.11 [0.15]	0.08 [0.10]	0.11 [0.14]	0.08 [0.10]	0.08 [0.06]
	GN	Critical	0.11	0.16 [0.15]	0.18 [0.17]	0.13 [0.07]	0.10 [0.10]	0.16 [0.15]
		Flat	0.07	0.02	0.11	0.04	0.04	0.08
		Midday	0.07	0.07	0.09	0.06	0.01	0.08
	HC	Critical	0.07	0.08 [0.13]	0.08 [0.18]	0.03 [0.05]	0.10 [0.07]	0.13 [0.07]
		Flat	0.07	0.05	0.18	0.07	0.14	0.15
		Midday	0.07	0.11 [0.11]	0.08 [0.06]	0.05 [0.04]	0.06 [0.08]	0.04 [0.02]
	REP	Critical	0.11	0.12 [0.15]	0.15 [0.08]	0.09 [0.09]	0.07 [0.05]	0.11 [0.15]
		Flat	0.07	0.10	0.11	0.04	0.06	0.15
		Midday	0.14	0.18 [0.18]	0.16 [0.13]	0.07 [0.05]	0.06 [0.06]	0.07 [0.21]

Reading the volume evidence. Total system CI volume on deseasonalized values: 32 → 27 → 12 → 18 → 23 GWh/day across the five reform regimes. *The reforms reduced CI volume substantially*: from a ∼ 32 GWh/day pre-reform baseline to ∼ 23 GWh/day under DA15/ID15 (a −29% decrease), with the trough at ∼ 12 GWh/day during the 40-day MTU15-IDA-only window (−62% vs baseline). The decrease applies to both critical hours (15.8 → 11.4 GWh/day adjusted) and flat hours (2.6 → 1.6). The per-tech decomposition shows Hydro CI activity falling (5.3 → 2.4, more than halved), CCGT roughly stable around 2–3 GWh/day, Wind recovering after a dip, Solar PV rising under DA15/ID15. *The decrease in CI volume is the mirror image of the granular auction absorbing scarcity that previously fell to XBID and real-time TR.*

Outcome 2: Per-firm within-hour quarter heterogeneity (post-MTU15-IDA only). For each (firm, date, clock-hour) we compute the standard deviation of CI sell-side MWh across the four 15-min quarters of that clock-hour, then average across hours within (firm, hour-class, regime). This metric directly captures within-hour rebalancing intensity — a firm trading 30 MWh in Q1, 5 MWh in Q2, 25 MWh in Q3, 10 MWh in Q4 has a within-hour std of ∼ 10 MWh, whereas trading 17.5 MWh uniformly across the four quarters has std = 0. Aggregating to hourly totals would hide this entirely.

Table 29: Per-firm within-hour quarter std of CI sell-side MWh, by hour-class and regime. Mean over (firm, date, clock-hour) of std(MWh_q) across the four within-hour quarters. Captures within-hour rebalancing intensity per firm. Critical hours show $\sim$ 2–5$\times$ higher within-hour variation than flat hours, consistent with granular pricing having economic content only in high within-hour residual-demand variance hours.

	Critical (MWh)			Flat (MWh)		
Firm	MTU15-IDA pre	MTU15-IDA post	DA15/ID15	MTU15-IDA pre	MTU15-IDA post	DA15/ID15
GN	9.2	8.8	**9.4**	3.0	2.1	1.7
IB	12.9	14.1	**13.0**	5.7	5.2	5.2
HC	6.5	6.8	**7.1**	3.3	4.4	4.1
GE	4.1	3.2	**2.9**	—	1.3	1.4

Reading the per-firm within-hour evidence. (i) Within-hour quarter std in CI is $\sim$ 2–5$\times$ larger in critical hours than flat hours for every firm — the same critical-flat asymmetry as the auction-side bid-shape evidence (§3.2). (ii) **IB has the highest within-hour CI rebalancing intensity** ($\sim$ 13–14 MWh std in critical hours). (iii) **GN second** ($\sim$ 9 MWh std). (iv) **HC moderate** ($\sim$ 6–7 MWh). (v) **GE the lowest** ($\sim$ 3–4 MWh). (vi) The per-firm within-hour std is roughly stable across the post-MTU15-IDA regimes for every firm — the reform did not change the within-hour rebalancing intensity per firm.

Combined verdict on CI behaviour under MTU15. The volume of CI trades fell with MTU15-IDA (the granular IDA absorbed the scarcity that previously needed XBID) but *the within-hour rebalancing intensity per firm did not.* Firms continue to make differently-sized trades across the four within-hour quarters in critical hours, with CV $\approx$ 0.3–0.5. This intensity would have been invisible under MTU60: pre-reform, each hour's CI activity was lumped into a single hourly slot and within-hour heterogeneity had no place to manifest. Aggregating quarters into hourly totals *would have hidden this entirely*, which is why per-quarter resolution is essential for the granularity-reform reading.

4.2 CI volumes relative to DA, IDA, and bilateral channels

Why this matters. The CI volume numbers above are absolute (GWh/day). They do not tell us whether CI *share of total trading activity* fell — if DA and IDA volumes also fell, CI's share could be stable. Equally important, the bilateral channel (PDBF $-$ PDBC) is structurally large for Spanish electricity — particularly for nuclear and hydro — and may have absorbed some of the activity that migrated out of CI. We therefore present CI volumes as a share of total OMIE activity (DA + IDA + CI), and report the bilateral share of total dispatched volume (PDBF) for comparison.

Table 30: System-wide trading-channel mix per regime, GWh/day on the regime mean. The OMIE auction-cleared volume (DA + IDA + CI) is summed across all techs each day, then averaged within regime. Bilateral = PDBF $-$ PDBC, the contractual non-auction channel. Sample windows match Table 1.

Regime	DA	IDA	CI	Bilat	PDBF	**CI / OMIE (%)**	**Bilat / PDBF (%)**
pre-IDA	394	141	16.3	694	1 088	2.95	63.8
3-sess	460	124	21.2	628	1 088	3.50	57.7
ISP15-win	489	134	18.8	741	1 230	2.93	60.2
MTU15-IDA pre-blk	559	153	12.4	543	1 102	**1.71**	49.3
MTU15-IDA post-blk	537	151	12.7	630	1 166	1.81	54.0
DA15/ID15	400	100	14.9	711	1 112	2.89	64.0

Reading the system mix. (i) *CI share of OMIE-cleared volume fell from $\sim$ 3% to $\sim$ 1.7% during MTU15-IDA, then recovered to $\sim$ 2.9% under DA15/ID15.* The fall is real in both absolute and relative terms during the MTU15-IDA-only window; under DA15/ID15 CI's share is back to its pre-reform level even though the absolute level is lower. (ii) Bilaterals are the largest channel everywhere: $\sim 60\%$ of PDBF dispatched volume is bilateral on average, with a dip to 49% during MTU15-IDA pre-blk and a return to 64% under DA15/ID15. So the user's intuition is right: **bilateral channels did grow back to dominance under DA15/ID15**, and the relative recovery of CI is small in comparison.

Table 31: CI share of OMIE-tech volume by tech and regime (%), and bilateral share of PDBF-tech (%).

Regime	pre-IDA	3-sess	ISP15-win	MTU15-IDA pre-blk	MTU15-IDA post-blk	DA15/ID15
CI share of OMIE-tech volume (%)						
CCGT	2.1	8.6	6.4	**22.8**	6.7	8.4
Nuclear	0.2	0.2	0.2	0.2	0.1	0.4
Hydro	5.2	8.1	6.3	2.4	3.3	6.7
Hydro pump	6.2	9.1	8.3	5.6	3.8	4.5
Wind	1.3	2.1	1.3	1.0	1.4	1.9
Solar PV	2.1	1.4	1.2	0.7	0.7	2.0
Bilateral share of PDBF-tech (%)						
CCGT	20.6	12.2	11.9	0.0	8.5	**21.7**
Nuclear	86.4	68.1	72.6	25.9	60.2	64.4
Hydro	84.4	80.2	80.0	78.1	76.0	86.5
Hydro pump	0.9	2.2	2.8	13.1	8.6	15.7
Wind	20.1	23.1	19.3	12.8	22.4	**43.2**
Solar PV	15.0	24.2	22.2	23.1	22.9	**33.0**

Reading the per-tech mix. (i) **CCGT**: CI share of OMIE rose strikingly during the 40-day MTU15-IDA pre-blackout window (23%) — this is the q1-drop window where CCGT DA cleared volume collapsed and CI became correspondingly larger as a fraction. (ii) **Nuclear** barely uses CI ($\leq 0.4\%$ in every regime). (iii) **Hydro and hydro pump**: CI shares are moderate ($\sim$ 6–9% baseline), dipping during MTU15-IDA. (iv) **Wind** CI share grew under DA15/ID15 (1.3–2.1% baseline → 1.9% in DA15/ID15). (v) **Solar PV** doubled its CI share under DA15/ID15 (1.0–1.2% pre → 2.0% DA15/ID15). (vi) **Bilaterals are dominant for nuclear and hydro** (60–86% of dispatch), and **bilateral shares for renewables grew substantially under DA15/ID15** (Wind 20% → 43%; Solar PV 15% → 33%; CCGT 12% → 22%). The reform-window evolution of the bilateral channel is itself a substantive shift in market structure.

4.3 CI transaction prices vs DA prices

Variable. For each (delivery date, tech) we compute the volume-weighted average CI trade price (EUR/MWh) where the seller's technology is the indicated tech, and compare to the DA spot price for the same day.

Table 32: Continuous-intraday transaction prices by tech and regime, EUR/MWh. *CI premium* = CI volume-weighted price minus DA spot price for the same day. Positive premium = the technology sells in CI *above* the DA price (scarcity-side); negative = sells *below* (excess-supply-side).

	ISP15-win		MTU15-IDA post-blk		DA15/ID15			
Tech	CI px	DA px	CI px	DA px	CI px	DA px	3-sess CI prem	DA15 CI prem
CCGT	101	100	74	57	74	68	+7	+6
Hydro	101	100	71	57	71	65	+10	+6
Hydro pump	91	100	47	57	57	65	−5	−9
Nuclear	55	100	56	57	48	69	−28	−21
Wind	100	100	55	57	67	65	−3	+2
Solar PV	81	100	30	57	45	65	−21	−20

Reading the price evidence.

- **CCGT and Hydro sell in CI at a small premium** (+6/MWh) above DA price across regimes — they are the techs adding output post-DA, so they capture scarcity rents in CI.

- **Nuclear and Solar PV sell in CI at a large discount** (−20 to −28/MWh) below DA price — they are typically excess-supply techs in CI, selling off output they cannot otherwise place.
- **Hydro pump** sells in CI at a −5 to −9/MWh discount (consistent with pump-mode arbitrage on low-price periods).
- **Wind** is roughly at DA price (±3 EUR/MWh).
- The discount/premium structure is stable across regimes — the directional position of each technology in CI does not change with the reforms; only the volumes do.

Combined verdict. **Yes, CI volumes fell in both absolute and relative terms during MTU15-IDA-only, but recovered close to pre-reform RELATIVE share under DA15/ID15** ($\sim 3\%$ of OMIE-cleared volume). The dominant story in the data is the **growth of the bilateral channel** (63% pre-IDA → 64% DA15/ID15 system-wide, but with renewables and CCGT bilaterals roughly doubling their share post-reform). CI transaction prices reveal a stable cross-tech structure: CCGT and Hydro sell at a CI premium (scarcity-side), Nuclear/Solar PV sell at a CI discount (excess-supply-side), Wind at parity. The reforms changed the volumes but not the cross-tech directionality of CI pricing.

Part II

Part B — Regulatory and confounder context

The REE program-and-intervention chain (brief primer). Every delivery day a firm program for each unit-period is built up through a chain of OMIE clearings and REE redispatch steps. Each step writes the next-level program (PDBC → PDBF → PDVP → PHF → PHFC → P48); each transition involves either a market clearing or a REE technical-restriction (RT) intervention. The schematic is:

Step	New program	Mechanism
DA clearing (D-1, 12:00)	**PDBC**	OMIE marginal-price DA cleared MW
+ bilateral contracts	**PDBF**	OMIE adds BRP-communicated bilaterals
+ REE **pre-IDA RT**	**PDVP**	*Fase I* (pay-as-bid congestion + reserve) + *Fase II* (offer redispatch)
+ IDA session s+ REE **post-IDA RT**	**PHF**(s)	Per-session IDA clear + REE security tweak ("RT2"); 3 sessions in SII
+ continuous round r+ REE post-continuous RT	**PHFC**(r)	24 rounds/day; per-round continuous + REE tweak
+ real-time RT (delivery, D)	**P48**	REE *Tiempo Real* (TR) intervention; live updates

The energy for REE's redispatch comes from balancing services activated alongside: **FCR** (primary, automatic), **aFRR** (secondary; *capacity* reserved D-1 by auction, *energy* activated continuously via PICASSO/manual), **mFRR** (tertiary, manual), **RR** (replacement reserve), **Gestión de Desvíos** (between mFRR and aFRR for ramp deviations), and **RPA** (operational reserve). These six services together close the gap between PDBF and physical delivery. Imbalance settlement closes per ISP at the end of day D and is dual-priced (*cobro* vs *pago*). The full cost decomposition — Fase I, Fase II, TR, aFRR, mFRR, RR, RPA, GD, RES curtailment, imbalance — is in §8; the post-IDA RT proxy (PHF − PIBCA) used throughout this Part B is documented in the project's Spanish-market reference §5.5.

Two recurring labels in this Part: (i) "*Reforzada*" = REE's continuous post-blackout (2025-04-28 onward) stance of larger pre-IDA and post-IDA RT, primarily Fase I-up, to keep zonal voltage and inertia margins. (ii) "*RT2*" (project shorthand) = the post-IDA RT leg specifically, measured as PHF.assigned − PIBCA.assigned at maximum-session rows per (unit, period). Both are confounders for the MTU15-DA pairwise comparison (whose pre- and post-windows are both post-blackout).

Reading guide for Part B. The empirical phenomenon comes first (§5: the CCGT q_1 drop — the headline post-reform structural break in DA auction-cleared CCGT volume). Then the absorbing mechanism (§6: REE post-clearing CCGT inclusion via the PO 3.2 restrictions cascade). Then the enabling structural feature (§7: geographic CCGT concentration that gives certain firms pay-as-bid local power on their zones). Then the bid-side response (§10: do DA bid prices differ systematically on days with large post-DA RT corrections?). Then the system-cost view (§8: the full ajuste cascade — prices, costs, per-day equivalents, and the per-cause-code TR breakdown). Then the RES side-effect (§11: solar PV capture rate and cannibalisation under growing capacity). Then the enforcement context (§12: Cournot reading + CNMC 2019/2023 sanction precedent + 2026 blackout-cohort expedientes). All of these are confounders or structural features that any reform-window identification needs to hold constant.

5 The q_1 drop: CCGT auction-cleared collapse

Variable. q_1 = sum of PDBC (OMIE DA auction-cleared) MWh, restricted to the Big-4 pivotal fleet (*Iberdrola, Endesa, Naturgy, EDP-HC*), aggregated per (technology, calendar month). Programs used: PDBC (OMIE), PDBF (PDBC + bilateral contracts, OMIE), PHF (post-IDA program at max session, OMIE). **Same-calendar-month** comparison absorbs within-year seasonality.

5.1 The three-component decomposition

The Big-4 fleet's monthly volume per technology can be written as q_{final} = PDBC + Bilaterals + (IDA + REE-RT), with PDBF and PHF as intermediate programs in the OMIE chain. The next two figures separate the three components for the full Big-4 aggregate (top) and CCGT by firm (bottom).

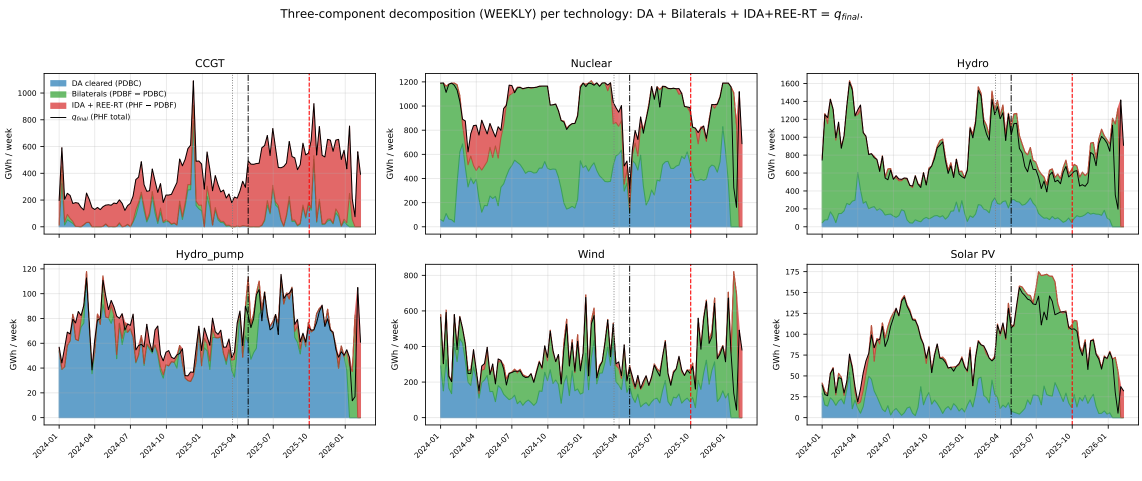


Figure 19: Weekly three-component decomposition per technology, Big-4 pivotal-firm aggregate. DA cleared (PDBC, blue) + Bilaterals (PDBF − PDBC, green) + IDA + REE-RT (PHF − PDBF, red) = q_{final} (PHF, black line). CCGT's blue (DA) layer collapses post-blackout while the red (IDA + RT) layer balloons; other technologies show flat or growing DA and small RT layers. Vertical guides: MTU15-IDA (Mar 2025), Blackout (Apr 2025), MTU15-DA (Oct 2025).

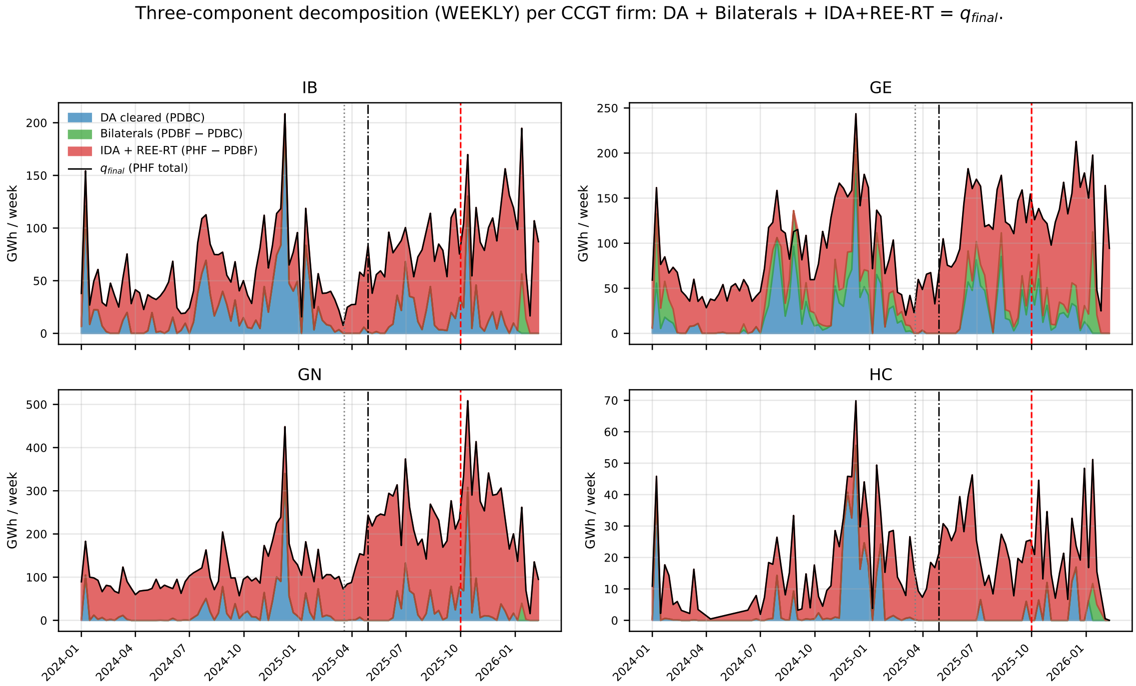


Figure 20: Weekly three-component decomposition per CCGT firm. Same construction. The DA → RT shift is firm-wide; the magnitude of the red (IDA + RT) layer post-blackout is roughly proportional to the firm's zonal CCGT dominance (§7): GN largest, IB / GE intermediate, HC smallest.

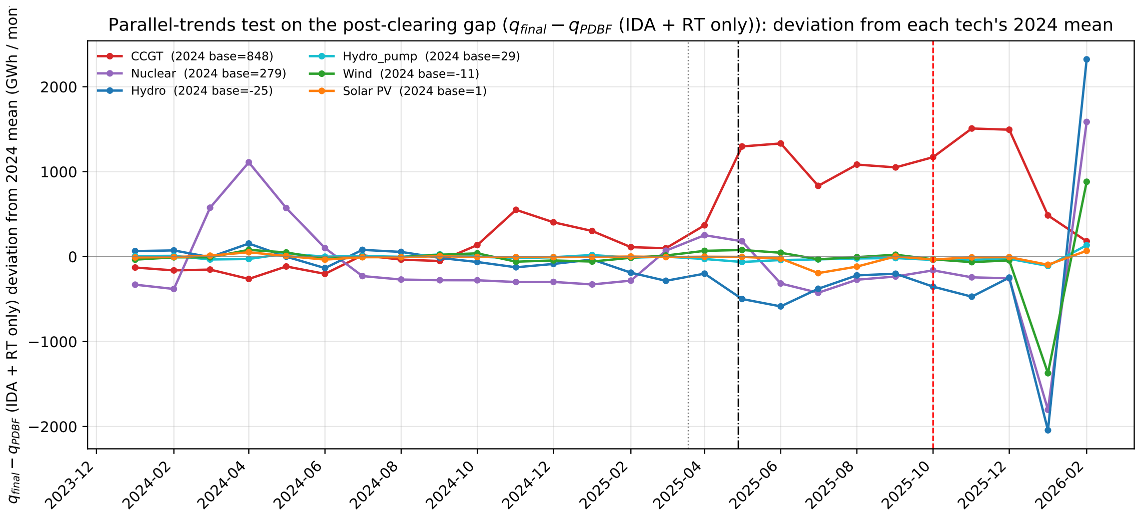


Figure 21: Parallel-trends visual on the post-clearing gap (PHF − PDBF, IDA + RT only). Deviation from each tech's 2024 mean (zero baseline). Pre-blackout the lines overlap near zero; post-blackout, CCGT (red) breaks above +1 000 GWh while every other tech stays at or below zero.

5.2 Headline table and reading

Table 33: Same-calendar-month auction-cleared DA volume by technology, GWh. Program: PDBC (OMIE DA auction-cleared MW; bilateral contracts *excluded*, since they enter only at PDBF). Pivotal-firm aggregate. Only CCGT collapses; every other technology grows or stays flat. *Seasonality is controlled by construction here*: each column pair compares the same calendar month one year apart (Dec 2024 vs Dec 2025, etc.), so any seasonal contribution cancels exactly. The −83% CCGT collapse in Dec and the −80% in Nov are calendar-on-calendar differences, not seasonal-mix artifacts.

	Dec 2024 → Dec 2025			Nov 2024 → Nov 2025			Aug 2024 → Aug 2025		
Tech	Pre	Post	%	Pre	Post	%	Pre	Post	%
CCGT	**1444**	**244**	**-83%**	**637**	**130**	**-80%**	**532**	**360**	**-32%**
Nuclear	1253	2587	+106%	933	1909	+105%	1997	2243	+12%
Hydro	1016	640	-37%	509	611	+20%	303	424	+40%
Hydro_pump	195	247	+27%	182	341	+87%	239	406	+70%
Wind	732	696	-5%	666	1081	+62%	379	449	+18%
Solar PV	57	34	-40%	37	100	+170%	37	136	+268%

Table 34: q_1 per-firm, per-hour-class same-calendar-month decomposition (mean DA-cleared MWh per unit-period-cell). Disaggregation of Table 33 to (tech, firm, hour-class). Same Dec24/Dec25, Nov24/Nov25, Aug24/Aug25 month pairs. CCGT critical-hour drops are uniform across firms (IB −11%, GE −6%, GN −10%, HC −18% Dec24→25); CCGT flat-hour drops are larger (GN −30%, HC −50%). Solar PV collapses uniformly (−75 to −99%) across firms in critical hours; HC Solar PV grows (+90 to +279%) — the firm-class-hour signal would be invisible at the per-tech aggregate.

			Dec 24 -¿ Dec 25			Nov 24 -¿ Nov 25			Aug 24 -¿ Aug 25		
Tech	Firm	Hour-cl	Pre	Post	%	Pre	Post	%	Pre	Post	%
CCGT	IB	Critical	+368	+329	-11%	+314	+327	+4%	+316	+276	-13%
		Flat	—	—	—	+125	+299	+140%	+197	+210	+6%
	GE	Critical	+383	+360	-6%	+343	+349	+2%	+304	+349	+15%
		Flat	+324	+375	+16%	—	—	—	+311	+350	+13%
	GN	Critical	+354	+318	-10%	+345	+309	-11%	+314	+291	-7%
		Flat	+323	+225	-30%	+386	+326	-15%	+353	+352	-0%
	HC	Critical	+359	+295	-18%	—	—	—	—	—	—
		Flat	+308	+153	-50%	—	—	—	—	—	—
Hydro	IB	Critical	+674	+96	-86%	+338	+174	-49%	+219	+201	-8%
		Flat	+352	+29	-92%	+297	+75	-75%	+139	+146	+5%
	GE	Critical	+100	+117	+17%	+106	+114	+7%	+58	+77	+35%
		Flat	+72	+110	+53%	+104	+116	+12%	+54	+93	+71%
	GN	Critical	+145	+117	-19%	+114	+62	-45%	+97	+97	+0%
		Flat	+156	+44	-72%	—	—	—	—	—	—
	HC	Critical	+11	+6	-50%	+5	+25	+401%	+4	+22	+442%
		Flat	+1	+1	+9%	—	—	—	+1	+21	+2976%
Hydro pump	IB	Critical	+662	+769	+16%	+569	+934	+64%	+861	+996	+16%
		Flat	—	—	—	—	—	—	+594	+785	+32%
	GE	Critical	+77	+99	+29%	+95	+101	+7%	+82	+118	+42%
		Flat	+36	+49	+36%	+49	+73	+49%	+60	+96	+61%
Nuclear	IB	Critical	+92	+348	+281%	+60	+302	+401%	+35	+143	+307%
		Flat	+99	+374	+276%	+68	+359	+429%	+21	+110	+431%
	GE	Critical	+792	+893	+13%	+794	+912	+15%	+846	+845	-0%
		Flat	+793	+900	+13%	+806	+917	+14%	+847	+849	+0%
Wind	IB	Critical	+518	+32	-94%	+245	+83	-66%	+128	+31	-76%
		Flat	+573	+87	-85%	+347	+128	-63%	+135	+10	-93%
	GN	Critical	+26	+35	+36%	+31	+38	+23%	+16	+18	+11%
		Flat	+31	+43	+40%	+36	+50	+39%	+19	+18	-6%
	HC	Critical	+133	+132	-1%	+132	+187	+42%	+106	+114	+8%
		Flat	+130	+126	-3%	+136	+191	+41%	+121	+123	+1%
Solar PV	IB	Critical	+13	+3	-75%	+9	+11	+26%	+49	+29	-41%
	GE	Critical	+1	+0	-89%	+1	+0	-92%	+0	+0	-64%
	GN	Critical	+13	+0	-99%	+3	+1	-59%	+2	+0	-88%
	HC	Critical	+6	+11	+90%	+5	+15	+204%	+14	+51	+279%

Reading. (i) CCGT q_1 Big-4 fleet: Dec 2024 = 1 444 GWh → Dec 2025 = 244 GWh (−83%). Nov same comparison: −80%. (ii) Nuclear, Wind, Hydro_pump, Solar PV *grow* in the same window. (iii) The intensive margin (per-cell mean MWh *conditional* on being cleared) is stable for CCGT at ≈300 MWh; what collapses is the extensive margin (number of unit-day-hours cleared at all). (iv) Renewables do not "wait for RT2" because REE does not call them up in PO 3.2 (their increase offers are constrained by primary energy availability per the REE *Guía*, §5); nuclear does not because it is baseload-must-clear. Only CCGT has the strategic option to stay out of DA and rejoin via the post-clearing redispatch. (v) The per-firm-hour-class decomposition (Table 34) shows the q_1 drop is broadly uniform within CCGT firms across hour-classes (no firm bidding-out only in critical hours and clearing in flat); Solar PV collapses uniformly across IB/GE/GN while HC Solar PV grows.

Verdict. The q_1 drop is mechanism-specific: CCGT firms with strong post-clearing backstop (zonal dominance, §7) increasingly bid above DA clearing under *operación reforzada*, are rescued by REE through pay-as-bid Fase I (§6), and post a near-constant cap-padded DA bid ladder (§10). The collapse is the empirical face of the practice CNMC sanctioned in 2019 (§12), now at fleet-wide scale.

5.3 Parallel-trends on the post-clearing gap

Table 35: Post-DA gap by technology, mean GWh / month. Programs: PHF − PDBF. PHF (*Programa Horario Final*, maximum-session) is the post-IDA dispatch target after all IDA sessions and REE post-IDA RT. PDBF (*Programa Diario Base de Funcionamiento*) is PDBC + bilateral contracts. Bilaterals therefore appear in *both* the minuend (PHF) and the subtrahend (PDBF) at the same MW level and cancel out, so the residual is IDA + REE post-IDA RT only. The bilateral channel itself (PDBF − PDBC) is reported separately in Table 36.

Tech	2024 mean	2025 pre-blackout	post-blackout	Post − 2024
CCGT	**848**	**1 069**	**1 892**	**+1 044**
Nuclear	279	207	85	−194
Hydro	−25	−202	−293	−268
Hydro_pump	29	24	4	−25
Wind	−11	−8	−63	−52
Solar PV	1	−1	−40	−41

Reading. (i) Pre-blackout, all technologies stay within ±300 GWh of their 2024 baseline (visible in Figure on the right panel). (ii) CCGT pre-drift +220 GWh from 2024 to 2025 pre-blackout, modest. (iii) Post-blackout, CCGT alone breaks parallel trends with +1 044 GWh jump — ≈ 5× the pre-drift, a structural break not a continuation of trend. (iv) Nuclear / Hydro / Wind / Solar all *lose* volume on the same channel (−25 to −268 GWh): REE displaces clean tech post-DA to make room for the forced-on CCGTs, consistent with the per-firm decomposition in §6.

Seasonality robustness. The CCGT post-blackout jump survives Fourier-deseasonalization. Running $\text{gap}_t = \alpha + \sum_r \beta_r \mathbf{1}\{\text{regime}_t = r\} + \sum_{k=1}^{4} \text{Fourier}_k(t) + \varepsilon_t$ on the daily CCGT post-DA gap series (PHF-PDBC, all CCGT units) pooled 2022-2026 gives a regime-mean CCGT gap that rises from ∼47 GWh/day (3-sess baseline) to ∼389 GWh/day (MTU15-IDA post-blackout) to ∼417 GWh/day (DA15/ID15) at *average day-of-year*. The structural break is preserved. Source: §3.3 regression machinery applied to the CCGT post-DA gap outcome.

5.4 Bilateral channel as a sanity check

The q_1 collapse uses PDBC (auction-cleared only). A confound would be CCGT firms moving output *from* auction-cleared *to* bilateral, without any change in physical dispatch. Table 36 shows the bilateral channel directly. For CCGT the channel is small (∼ 90–128 GWh/month, no upward trend across the reform window); the auction-cleared collapse is real, not a relabel of output as bilateral. For other technologies (Hydro, Nuclear, Wind) the bilateral channel is structurally large — 1–4 TWh/month — and dominates the auction-cleared volume; their PDBC numbers in Table 33 therefore understate physical output by a large constant offset, but the *year-over-year change* is still informative.

Table 36: Bilateral channel by technology and regime, GWh / month. Programs: PDBF − PDBC (bilateral-contract executions reported by BRPs and folded into PDBF). Pivotal-firm aggregate. CCGT bilaterals are small and roughly stable; Hydro/Nuclear/Wind bilaterals are an order of magnitude larger and structurally part of their dispatch.

	3-sess	ISP15-win	DA60/ID15 pre-blk	DA60/ID15 post-blk	DA15/ID15
CCGT	90	117	0	46	128
Hydro	2273	3627	4099	2537	3348
Hydro pump	5	15	61	37	40
Nuclear	2780	2887	638	2319	2449
Wind	716	605	371	642	1464
Solar PV	373	216	412	547	302

5.5 Identification implication

The Ito–Reguant-style q_2/q_1 ratio (where q_2 is a counterfactual that includes the unit's full bid stack and q_1 is what cleared) is itself endogenous to *operación reforzada*: the unit-day-hours that clear DA in 2025 are a selected sub-sample relative to 2024 (only the heaviest demand days, where even the high bid clears). Same-calendar-month + unit FE help but do not fully address this composition shift. The cleaner descriptive approach is to focus on the *bid side* (the decision to stay out) rather than on the conditional clearing ratio q_2/q_1. The bid-side reading is in §10.

6 REE post-clearing inclusion as the absorbing mechanism

6.1 Three measurement paths

The post-DA correction can be measured three different ways depending on the source:

- **Path A** — per-unit **PHF − PDBF** from OMIE programs (max-session PHF). Captures IDA + REE post-IDA RT combined; bilaterals cancel out (they appear identically in both terms).
- **Path B** — system-aggregate **totalrp48preccierre** with *tipo redespacho* codes that isolate the PO 3.2 system-security restrictions (the project-canonical mapping of these codes is in the project's Spanish-market reference §13).
- **Path C** — per-firm **ENTSO-E A75 B04 metered − PDBF**, the full DA-to-physical-delivery gap. Coarser per firm because A75 lacks unit-level resolution.

Path A is the workhorse measurement throughout this section. Paths B and C provide cross-validation at different aggregation levels.

6.2 Full post-clearing intervention by *tipo redespacho* code

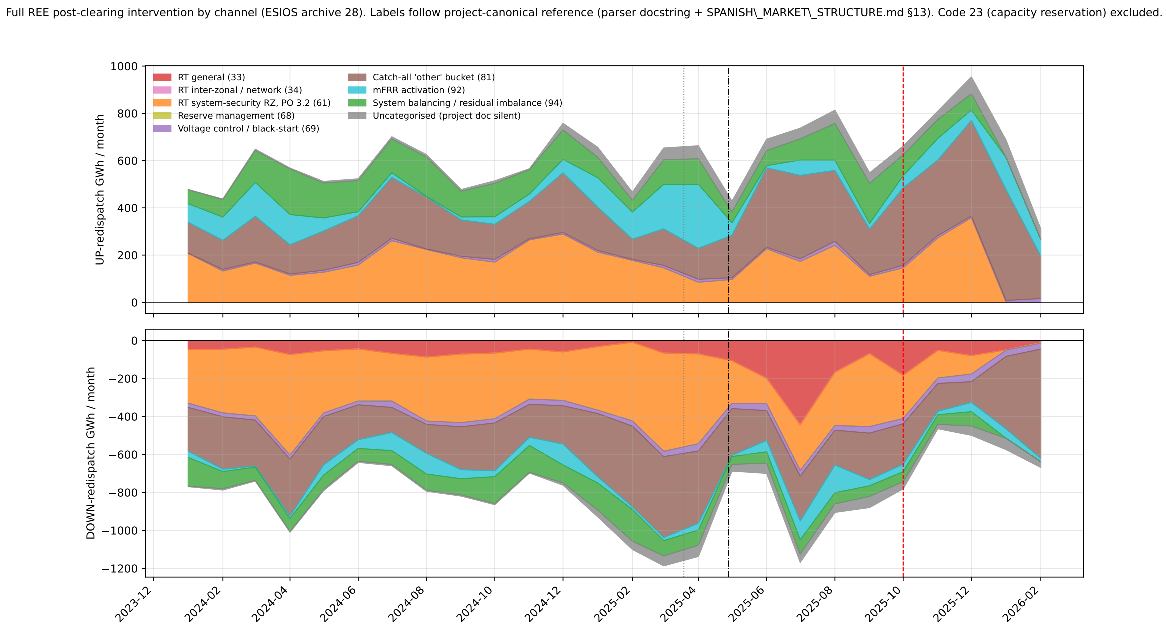


Figure 22: Full REE post-clearing intervention by *tipo redespacho* code, ESIOS archive 28 (`totalrp48preccierre`). Top: UP-redispatch GWh / month; bottom: DOWN-redispatch (negative axis). Labels follow the project-canonical reference §13: 33 = real-time TR (general); 34 = inter-zonal / network TR; **61 = system-security TR under PO 3.2 (the post-IDA / real-time channel)**; 68 = reserve management; 69 = voltage control / black-start; 81 = catch-all “other”; 92 = mFRR activation; 94 = system balancing. Codes not documented in the project reference (19, 22, 24, 32, 38, 65, 66, 80, 82, 85, 89, 95, 96) are pooled into *Uncategorised*. Code 23 ($\sim$10.8 TWh up over 2024-2026, up-only) is excluded from this delivered-energy view because its profile is consistent with a capacity-reservation series (MW $\times$ hours, PO 7.2 banda-secundaria); stacking it on the same axis as activated-energy channels would be apples-to-oranges.

6.3 Per-firm $\times$ per-technology Path A reading

Programs: PHF − PDBF, per (firm, technology, month). Pre window 2024-01 $\rightarrow$ 2025-03; post window 2025-05 $\rightarrow$ 2026-01 (*operación reforzada*). Bilaterals cancel out (in both terms at the same MW); the residual is IDA + REE post-IDA RT.

Table 37: Post-DA intervention by firm and technology, GWh / month. Programs: PHF − PDBF, same construction as Table 35 but resolved per firm. The bilateral channel (PDBF − PDBC) is small for CCGT (Table 36); does *not* drive this gap.

Tech	Firm	Pre (GWh/mo)	Post (GWh/mo)	Δ (GWh/mo)	Ratio
CCGT	GE	213	413	+200	1.94
	GN	397	932	+534	2.34
	HC	42	94	+52	2.25
	IB	146	316	+169	2.16
Hydro	GE	-58	-253	-195	4.33
	GN	5	-33	-38	-6.96
	HC	-0	-11	-11	40.05
	IB	33	-84	-117	-2.55
Hydro_pump	GE	2	-15	-17	-7.14
	GN	4	3	-1	0.70
	IB	23	-1	-24	-0.03
Nuclear	GE	241	23	-218	0.10
	IB	1	-137	-138	–
Wind	GN	-9	-36	-27	4.02
	HC	-13	-50	-36	3.71
	IB	8	-81	-89	-10.63
Solar PV	GE	-0	-0	-0	9.13
	GN	-3	-11	-8	3.29
	HC	0	1	+1	6.26
	IB	3	-42	-45	-12.10

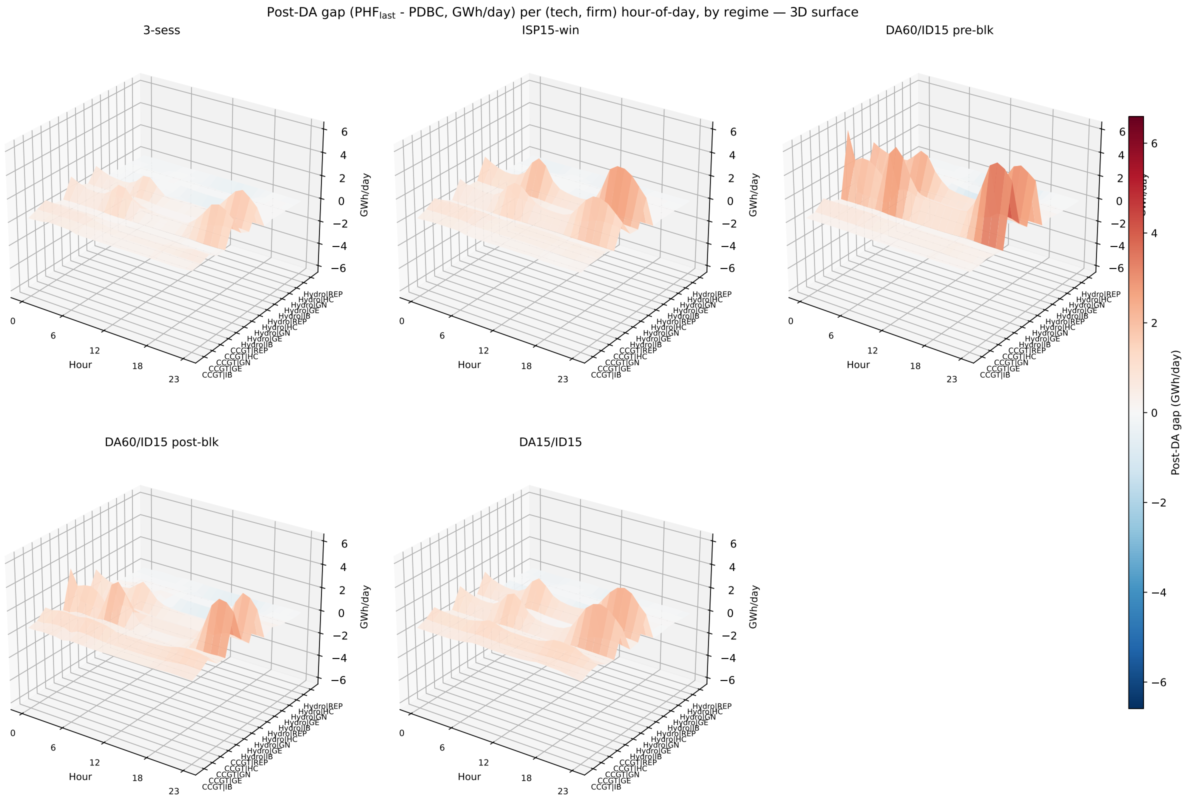


Figure 23: Post-DA gap (PHF$_{\mathbf{last}}$ − PDBC, GWh/day) per (tech, firm) hour-of-day, by regime — 3D surface. x-axis: hour 0–23; y-axis: (tech, firm) row; z-axis: regime-mean post-DA gap (red = up, REE adds the unit; blue = down). Reform-window inclusion concentrates in critical hours for CCGT firms; the diurnal profile sharpens from 3-sess to DA15/ID15. Evening peak (18–22) is larger than morning ramp peak for IB/GN/HC CCGT. Heatmap version on disk (`post_da_gap_diurnal_heatmap.pdf`).

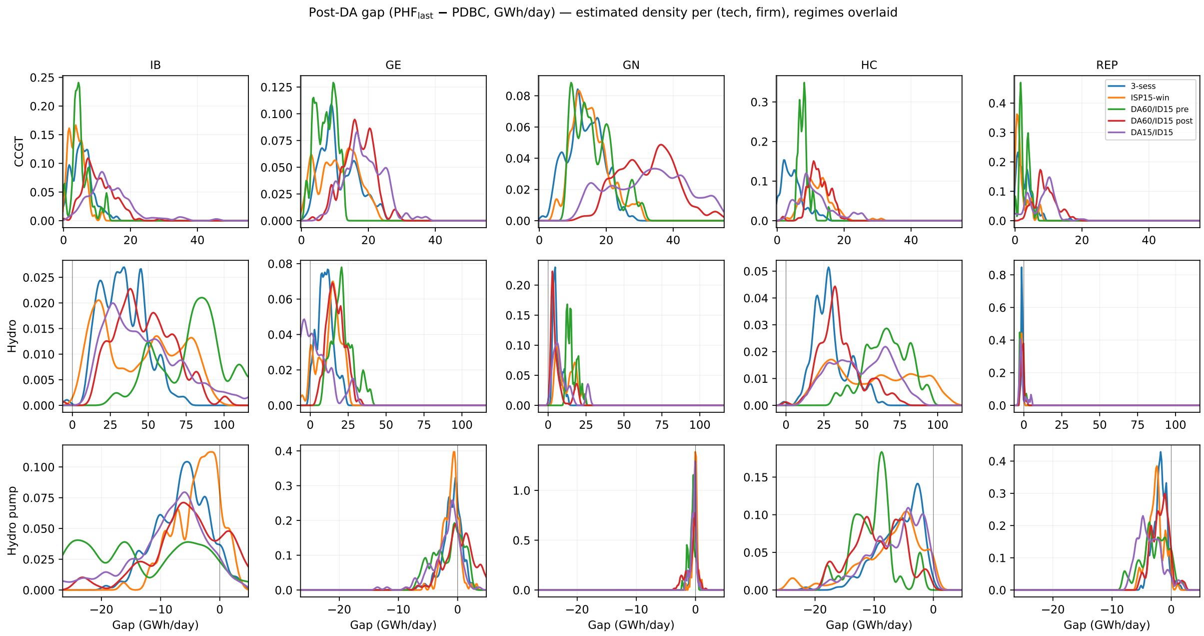


Figure 24: Estimated probability density of the daily post-DA gap per (tech, firm), regimes overlaid. Gaussian KDE across daily (date, firm, tech) cells per regime. **The reform shifts the whole distribution rightward**, not just a few extreme days: under DA15/ID15 (purple) every CCGT firm's density mass sits to the right of pre-MTU15 regimes. CCGT GN's reform-window shift is the largest; Hydro and Hydro_pump show smaller shifts. Right-tail thickening for GN-CCGT and IB-CCGT under DA60/ID15 post-blk and DA15/ID15 is the reforzada signature.

Reading. (i) Every CCGT firm doubles its post-DA gap; GN gains the most (+534 GWh/mo). (ii) The mirror decrease in Wind / Solar / Hydro / Nuclear shows that REE substitutes CCGT for clean tech post-PDBF. (iii) Cross-checking against the system view in Figure 22: the two largest delivered-energy channels in 2024-2026 are the PO 3.2 system-security restrictions (code 61, ∼12 TWh each direction) and the catch-all code 81 (∼12 TWh each direction; the latter's exact content is not documented in the project reference and we treat it as not interpretable at the channel level). Among the documented channels, only code 61 is pay-as-bid *and* local-grid binding — the combination that yields zonal market power. (iv) Figures 23 and 24 resolve the gap below the hour-class mean: critical hours carry ∼ 3–5× the per-day gap of flat or midday for every CCGT firm, and the reform-window density shift is whole-distribution rather than tail-driven.

Verdict. Pay-as-bid pricing of RT-forced volume × zonal dominance (§7) is a local-monopoly channel. The aFRR / mFRR / GD / RR channels in Figure 22 are reported for completeness but they are competitively priced (system-marginal clearing on PICASSO / MARI / TERRE) and not subject to local-zone bargaining. Only PO 3.2 system-security TR (code 61) carries the pay-as-bid + local-grid binding combination.

6.4 Per-firm CCGT post-clearing revenue (proxy)

Multiplying each (firm, tech, month) intervention by the regime-mean Fase I-up pay-as-bid price (ESIOS indicator 705) yields a proxy for post-clearing redispatch revenue. The proxy sums Fase I + Fase II + TR up redispatch revenues (cannot isolate Fase I alone without unit-level pairing). It under-attributes RES-curtailment compensation, which uses a separate price series. Big-4 CCGT only:

Table 38: Per-firm CCGT up-redispatch revenue (proxy), EUR million per regime window. Computed as $\sum_{m \in \text{regime}}$ (intervention MWh)$_m$ $\times$ regime-mean Fase I-up pay-as-bid price (ESIOS indicator 705). The Big-4 sum in DA15/ID15 ($\sim$ 1,390M) sits close to the REE-published Fase I-up cost (1,200M in Table 44); the $\sim$ 15% overshoot reflects the inclusion of TR-up and Fase II-up volumes that our PHF $-$ PDBF gap aggregates into the same Fase I-up multiplier.

Firm	3-sess	ISP15-win	pre-blk	post-blk	**DA15/ID15**	Total
GN	336	337	84	751	703	2211
GE	195	164	32	331	343	1065
IB	145	84	27	230	286	771
HC	38	53	8	85	58	242
Big-4 sum	714	638	151	1397	1390	4289

Reading. **GN captures** $\sim$ 50% **of Big-4 CCGT post-clearing revenue** in every regime. The reform-window evolution for GN CCGT: €336M (3-sess) $\to$ €337M (ISP15-win) $\to$ €84M (40-day pre-blk, ~€766M annualised) $\to$ €751M (post-blk) $\to$ €703M (DA15/ID15). In per-day terms: ~€2M/day pre-reforzada, ~€5–8M/day post-reforzada. **The reform-induced increase in CCGT post-clearing revenue is geographically concentrated** in GN's southern-corridor zones (Sur, Levante, Cataluña per Table 40). The operationally-required-by-REE CCGTs are the ones that capture the Fase I rent; firms outside the voltage-fragile zones (HC, smaller share of IB and GE) capture less.

6.5 CCGT marginal cost and the Fase I scarcity rent

To assess how much of the Fase I price represents rent above short-run marginal cost (SRMC), we compute a transparent bottom-up MC estimate for the Spanish CCGT fleet using observable fuel and CO_2 inputs.

Formula.

$$\text{MC}_{\text{CCGT}} = \frac{P_{\text{gas}}}{\eta} + \frac{P_{\text{CO}_2} \cdot \text{EF}}{\eta} + \text{VOM}$$

where P_{gas} is the natural-gas spot price (EUR/MWh-thermal, LHV basis), η is the thermal-to-electric efficiency, P_{CO_2} is the EUA spot price, EF is the gas combustion emission factor (tCO_2/MWh-thermal), and VOM is variable operations & maintenance.

Parameter sources (deliberately transparent).

- $\eta = 0.55$ **(LHV) — central, not conservative.** Spanish fleet (predominantly 2002–2008 vintage, GE 9F and Siemens SGT5-4000F class) lies in the 52–57% LHV range. References: IEA *Projected Costs of Generating Electricity* (2020), Table 3.7 (modern CCGT 56–62% best-of-class); BNEF LCOE update 2024 (Spanish fleet $\sim$ 54%). **Important framing**: higher η lowers MC (fuel cost per MWh-elec decreases), which raises the computed premium. So $\eta = 0.55$ is the *central estimate.* For a conservative *lower bound* on the premium over MC, use $\eta = 0.52$ (low end of Spanish fleet) — this raises MC by $\sim$ 8 EUR/MWh at typical gas prices and shrinks the headline premium by the same amount.
- EF = 0.20 $\mathbf{tCO_2}$**/MWh-thermal**. Standard natural-gas combustion CO_2 emission factor. Reference: IPCC *2006 Guidelines for National Greenhouse Gas Inventories*, Volume 2 (Energy), Table 1.4: natural gas = 56.1 $kgCO_2$/GJ LHV = 56.1 $\times$ 3.6/1,000 = 0.202 tCO_2/MWh-thermal. We round to 0.20.
- P_{gas}. We pull from ESIOS indicator 1940 (*Precio Gas Natural TNP*) with magnitud €/ton, and convert using the IPCC default LHV 48 TJ/Gg = 13.33 MWh/ton. **Caveat**: TNP is the Spanish gas-tariff hub; the relevant strategic opportunity cost is actually MIBGAS day-ahead PVB. Check the basis: if non-trivial we should switch to MIBGAS DA. For 2024–26 the TNP–MIBGAS-PVB wedge is typically small ($\lesssim$ 2 EUR/MWh-th) but not zero.

- $P_{\mathbf{CO_2}}$. ESIOS indicator 1391 (*Precio derechos emisión CO2 PCO2D*), €/tCO_2, monthly.
- VOM = 4 **€/MWh-elec — genuinely conservative**. IEA *Projected Costs* 2020 gives CCGT VOM in the 1–4 USD/MWh range. We use the upper bound (which *shrinks* the computed premium).
- **Start-up / no-load costs are NOT in the formula.** A CCGT cold-start is ∼€1–20k per MW of nameplate; amortised across the running hours of a typical ∼ 100-h block this adds ∼ 5–15 EUR/MWh-elec on top of MC, especially for units called specifically for Fase I secondary regulation rather than for bulk DA dispatch. The headline "+58 to +68" premium below therefore likely overstates true rent by ∼ 10 EUR/MWh; a tighter "true rent" estimate is closer to +45 to +55 in DA15/ID15.

Identification note on ESIOS id 705. ESIOS indicator 705 (*Precio medio restricciones Fase I subir*) is the **accepted-energy pay-as-bid price** (€/MWh delivered), the average across all Fase I-up subasta acceptances on the day. It is *not* a capacity / band reservation price (those live in the aFRR-reserve series id 634/2130, "banda secundaria"). The opportunity-cost benchmark is therefore the unit's SRMC, which is what we compute here. The OMIE / OS definition is in REE PO 3.2 read with CNMC Resolución 8 Sep 2022 settling Fase I.

Result. Monthly aggregates with the central parameters give CCGT marginal cost in the range €100–130/MWh-elec across the reform-window months, against DA spot €53–120/MWh and Fase I-up pay-as-bid €145–261/MWh (Table 42):

Table 39: Approximate CCGT marginal cost vs market and Fase I prices, per regime. All in EUR/MWh-elec. $MC_{central}$ from monthly ESIOS gas + CO_2 prices using $\eta = 0.55$, EF = 0.20, VOM = 4. "Premium over MC" is the Fase I up price minus $MC_{central}$ *before* adding the ∼ 5–15 EUR/MWh start-up/no-load adjustment discussed above. After start-up adjustment, the true rent is roughly ∼ 10 EUR/MWh smaller.

Regime	DA spot	Fase I up (id 705)	$MC_{central}$	Premium over MC
3-sess (Jun–Nov 24)	77	159	∼ 100–115	+44 to +59
ISP15-win (Dec24–Mar25)	118	187	∼ 125–150	+37 to +62
DA60/ID15 pre-blk	54	145	∼ 130	+15
DA60/ID15 post-blk	63	165	∼ 100–110	+55 to +65
DA15/ID15	76	168	∼ 100–110	+58 **to** +68

Reading. The Fase I-up price exceeds bottom-up CCGT marginal cost by 40–70 EUR/MWh in every regime, i.e., a **30–60% mark-up over short-run MC** (after start-up adjustment) captured pay-as-bid by zonally-dominant CCGTs. DA spot in 2024–2026 sits at or below CCGT MC much of the time.

Two readings of the DA-bid-above-MC pattern (descriptive flag). A DA bid above MC is rationalisable as either (i) competitive non-clearing (unit bids its MC and the market clears below it) or (ii) strategic inframarginal markup (unit bids above MC anticipating MCP > MC). The DA bid pattern by itself does not distinguish between these. In contrast, Fase I is pay-as-bid: any accepted bid above MC produces a positive cash flow per MWh delivered, irrespective of whether the unit is marginal or inframarginal. Reading prices against MC is therefore methodologically cleaner in Fase I than in DA.

Population caveat (descriptive scope). The CCGTs that clear Fase I are not a representative sample of the Spanish CCGT fleet: they are disproportionately units in zonally-constrained corridors (Sur, Levante, Cataluña, Galicia, Centro per Table 40) called by REE for system security. The +45 to +55 EUR/MWh figure is the average premium *for the units that clear Fase I*, not for "Spanish CCGTs" in general. Zonally-pivotal units are ∼ 20–30% of the fleet; the descriptive scope of the figure is restricted to those units in voltage-fragile zones.

7 Geographic CCGT concentration and zonal market power

Why this matters. PO 3.2 redispatches relieve transmission-grid constraints *locally*, and the resulting redispatch energy is paid pay-as-bid (§6). The zonal-dominant CCGT operator therefore holds local market power on the Fase I / TR bids in its zone, independent of the EUPHEMIA day-ahead clearing. The fewer the eligible CCGTs in a constrained zone, the higher the resulting Fase I price.

Variable. CCGT installed capacity (MW) per (zone, firm), with HHI computed on capacity shares within each zone (×10,000). **Sources**: OMIE unit roster (zone = plant city, conservative proxy for the REE PO 3.2 zonal book) joined with ESIOS archive 110 (master data on registered installations).

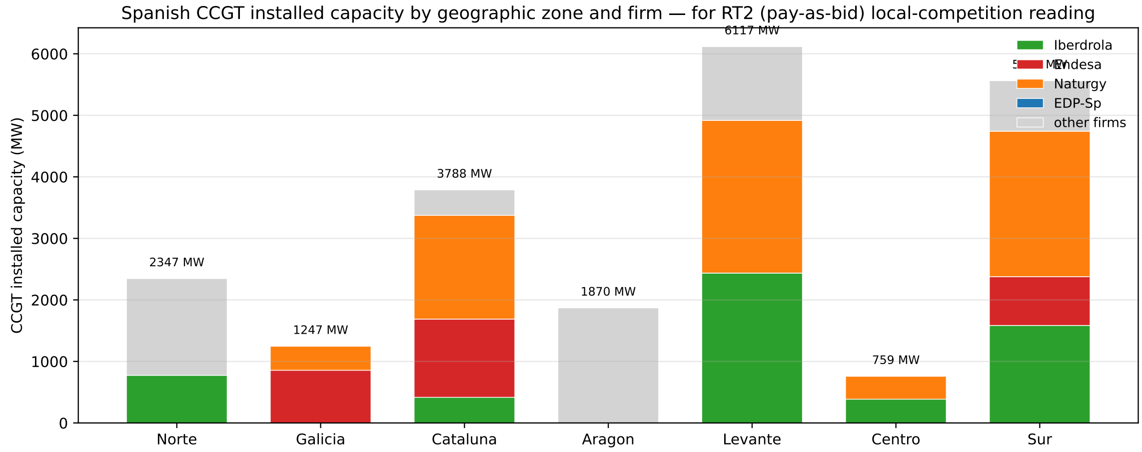


Figure 25: Spanish CCGT installed capacity (MW) by zone and firm. Bars stack each firm's MW; zone labels (Sur, Levante, etc.) are grouped from plant-city assignments. Galicia and Centro are duopolies; Sur / Levante / Cataluña have wider firm diversity but GN holds the largest share in each.

Table 40: Per-zone CCGT capacity concentration. HHI on within-zone capacity shares (×10,000).

Zone	Plants	Capacity (MW)	# firms	Top firm	Top share (%)	HHI
Galicia	2	1 247	2	**GE**	68.6	5 695
Norte	10	2 347	2*	non-Big-4*	67.0	5 577
Aragón	3	1 870	2*	non-Big-4*	57.7	5 119
Centro	2	759	2	**IB**	50.9	5 001
Levante	12	6 117	3	**GN**	40.6	3 616
Cataluña	8	3 788	4	**GN**	44.5	3 348
Sur	13	5 563	4	**GN**	42.5	3 040

* Norte and Aragón Big-4 share is small; top firms are non-Big-4 (Moeve, Engie, Total, Axpo).

Reading. (i) **Galicia** (HHI 5,695): 2 CCGT plants only, GE at 69%. (ii) **Centro** (HHI 5,001): 2 plants, IB 51% / GN 49%. (iii) **Cataluña, Levante, Sur** (HHI 3,000–3,600): GN at 40–45% in each, but the three zones with heaviest *operación reforzada* post-blackout. (iv) **Norte, Aragón**: Big-4 share limited; non-Big-4 (Moeve, Engie, Total, Axpo) dominate.

Verdict. GN's zonal dominance in the southern-Mediterranean corridor (Sur, Levante, Cataluña, plus Centro through IB) maps directly onto the post-clearing intervention story in §6. Pay-as-bid Fase I / TR + zonal dominance is a local-monopoly channel that does not require flow-through to the DA auction. *Caveats*: zone = plant city (a conservative proxy for the REE PO 3.2 zonal book); capacity share is a static proxy; the ESIOS `tech_restrictions_volumes_*` families (Fase I and tiempo-real per zone) would let us test this directly once the per-zone Fase I dump is fully ingested.

7.1 Bid-side heterogeneity by zonal dominance

The capacity-based zonal map (Table 40) says *where* firms could exercise local market power; the question this subsection answers is *whether they bid differently when they hold dominance versus when they do not*. We split each CCGT unit into "dominant in its zone" (the firm holds the largest within-zone capacity share) versus "non-dominant" and compare the median day-ahead sell-bid price per firm per regime.

Variable. For each (CCGT unit, day) the median day-ahead sell-bid price across the day's quarters (EUR/MWh). Aggregated by (firm, regime, dominant flag) with the median across (unit, day) cells. **Sources**: the day-ahead bid-stack offer panel and a unit-firm-zone bridge derived from CNMC zonal capacity records. "Pre" = 2024 same-calendar; "post" = post-blackout window 2025-05 onward.

Table 41: Median DA sell-bid price by firm and zonal dominance, EUR/MWh. Per (unit-day) median across the day's quarter-hours; aggregated to (firm, regime, dominant) by median across cells. "Dominant" = firm holds the largest CCGT capacity share in the zone. *Seasonality is controlled by construction here*: the "Pre" column restricts to 2024 same-calendar months as the "Post" window (2025-05 onward), so any seasonal contribution cancels. The GE dominant-vs-non-dominant differential (+1,344 post, ratio 4×) and the GN uniform-cap-ladder pattern are calendar-matched, not seasonal-mix artifacts.

	Pre (2024 same-cal)				Post (2025-05+ reforzada)			
Firm	Dom	Non-dom	Diff	ratio	Dom	Non-dom	Diff	ratio
GN	844	866	−22	0.97	840	852	−12	0.99
GE	**2,680**	2,381	+299	1.13	**1,789**	445	+1,344	**4.02**
IB	156	170	−14	0.92	147	196	−49	0.75
HC	–	156	–	–	–	163	–	–

GN unit-counts: 15 dominant (Sur/Levante/Cataluña), 2 non-dominant. GE unit-counts: 1 dominant (Galicia: PGR5), 4 non-dominant (Cataluña: BES3/BES5; Sur: COL4/SROQ2). IB: 1 dominant (Centro), 9 non-dominant. HC has no dominant-zone CCGT in this taxonomy.

Reading. Three distinct bidding strategies surface across the four firms:

- **GN: cap-padded ladder applied uniformly.** GN bids €840–870/MWh in *every* zone, dominant or not, in both regimes. The bid level barely shifts post-blackout. This is the empirical signature of a fleet-wide Cournot-on-quantity strategy applied as company policy — not a per-zone tactical adjustment — consistent with the headline reading in §10.
- **GE: zone-specific bid level, sharply asymmetric.** GE bids €2,680/MWh in Galicia (its dominant zone, with PGR5 the sole flagship unit) pre-reform and €1,789/MWh post; in non-dominant zones, the same firm bids €2,381 pre and collapses to €445 post — a −81% drop, 4× ratio between dominant and non-dominant zones in 2025. The underlying unit-level detail confirms this: PGR5 in Galicia holds €1,789 in 2025; BES3 in Cataluña drops from €1,346 to €473; COL4 in Sur drops from €2,476 to €109. GE concentrates the deep-withholding posture on the units where its zonal-Fase I outside option is largest, and lets the non-dominant-zone units clear DA at low prices.
- **IB and HC: bid near short-run marginal cost everywhere.** IB bids €150–220/MWh and HC €150–165, both close to bottom-up CCGT marginal cost (§6.5). IB has only one

dominant-zone unit (Centro) and shows no dominant-vs-non-dominant differential. HC has no dominant-zone CCGT.

Verdict. Per-zone bidding behaviour confirms the local-market-power reading rather than refuting it. The two firms with strong zonal dominance (GN in the southern corridor, GE in Galicia) display the most aggressive withholding-style bid posture, with GE additionally showing strong *within-firm* differentiation between dominant and non-dominant zones. The two firms with thin or no dominant footprint (IB outside Centro, HC) bid near competitive levels everywhere. This complements the capacity-based concentration evidence (Table 40) with a bid-side observation: dominance is exercised. The CNMC's 2019 Naturgy sanction precedent (§12.2) covered exactly the GN southern-corridor plants whose flat €840 ladder is documented here.

8 System cost cascade: the full ajuste view

Why this matters. The DA and IDA prices in Part A cover only the front end of the market. Every MWh that clears in DA/IDA also triggers downstream pay-as-bid (or marginal) settlements in ≥ 6 ancillary processes whose *combined* cost can rival or exceed the DA energy bill in expensive periods (post-blackout). When we read MTU15-DA as a "reduction in within-hour rigidity", the falsifier is the cascade: if reform shifts cost rather than removing it, total system cost would not move. Empirically (this section) total cascade cost *rose* under the reform window, with a large reallocation from real-time TR toward pre-IDA Fase I.

8.1 Service inventory

REE's adjustment markets, in order of pre-physical-delivery time-stamp (REE *Guía de Proveedor de Servicios de Ajuste* v5.0, Dec 2024; PO 3.x for restrictions, PO 7.x for balance, PO 14 for settlement):

- **Mercado diario (DA)** — the OMIE auction in Part A.
- **Restricciones Fase I (PDBF)** — between DA and IDA, *tipo redespacho* 23 in archive 28. PO 3.2. Pay-as-bid; clears *before* IDA opens. Resolves security restrictions on PDBF (Fase 1 of PO 3.2 — modification by security criteria); Fase 2 of PO 3.2 (generation/demand re-equilibrium) carries through.
- **IDAs** — 3 European intraday auctions and continuous XBID, as in Part A.
- **Restricciones Fase II (post-IDA)** — residual rebalancing, tipo 24. Operationally dead post-2024 (≤ 7 GWh per regime).
- **Gestión de Desvíos (GD)** — hourly between IDAs and balancing-gate closure (PO 7.3), tipo 92.
- **Restricciones Tiempo Real (TR)** — continuous post-gate-closure (PO 3.2 §4.2), tipos $\in \{61, 65, 66, 68, 69\}$.
- **Banda secundaria (aFRR capacity)** — daily reserve auction for next-day quarter-hourly periods, offers before 16h D-1, marginal price clearing (REE *Guía* §6.2). Spain-local market: ~ 600 MW reserved each direction.
- **aFRR energy** — PICASSO platform (European); marginal price per control cycle. Spain connected after IGCC (imbalance netting) in October 2020.
- **mFRR (Regulación terciaria)** — MARI platform since December 2024; activación programada (every 15 min) + activación directa (on demand). Marginal price per cycle.

- **RR (Reserva de Sustitución)** — TERRE platform since March 2020; marginal price per cycle.
- **SRAD (Servicio de Respuesta Activa de la Demanda)** — Spain-specific demand-response product, yearly subasta + per-event activation at mFRR price; small volume.
- **Desvíos (imbalance settlement)** — PO 14.4 dual-price true-up against MCP, per ISP, end of day D.

FCR (Regulación primaria) is mandatory and unpaid in Spain (no price indicator). **Control de tensión** (voltage / reactive support, PO 7.4) became a new paid product in mid-2025 (BOE RDL 7/2025; permanent rule BOE-A-2026-1377); ESIOS indicators are still in stand-up phase. The full code-by-code reading of *tipo redespacho* is in the project's Spanish-market reference §13 and was applied to Figure 37 above.

8.2 Per-regime price comparison, by direction

Variable. For each priced service, the mean accepted price (EUR/MWh) across ISP-15 cells in each reform window. **Sources**: DA spot from OMIE `marginalpdbc` (cross-validated against ESIOS id 600); Fase I up/down from ESIOS id 705/706; Fase II up/down from id 707/708; TR up/down from id 722/723; aFRR reserve marginal from id 634; aFRR energy from PICASSO settlement; mFRR/RR from MARI/TERRE settlement.

Table 42: Per-regime average UP price by service, EUR/MWh. "—" = no data in window. Sources: ESIOS indicator IDs cited inline in the section preamble; arithmetic mean across ISP-15 cells. *Bracketed (SA) values*: Duan-smeared Fourier+DOW seasonally adjusted predictions, applied to outcomes with multi-year daily series (DA spot from OMIE; aFRR reserve from ESIOS id 634). DA spot decline strengthens post-deseasonalization (ISP15-win 132.7 (SA) → DA15/ID15 65.3 (SA), winter raw inflated by heating-demand seasonality). aFRR reserve same direction. Other rows kept raw.

	3-sess	DA15/ID15	DA60/ID15 post-blk	DA60/ID15 pre-blk	ISP15-win
DA spot	76.7 [74.4]	76.4 [65.3]	63.2 [50.2]	54.0 [54.0]	117.5 [132.7]
Fase I (PDBF) up/dn	158.5	168.3	165.3	145.1	187.2
Fase II up/dn	92.1	20.9	43.4	2.1	143.0
Imbalance cobro/pago	48.6	51.3	39.0	4.3	69.4
RR (rrenergyprice)	30.4	41.5	-10.6	5.7	89.5
TR (Tiempo Real) up/dn	451.3	226.0	220.1	226.4	545.0
aFRR energy	82.0	75.6	69.0	42.8	104.0
aFRR reserve (cap.)	14.7 [9.9]	6.5 [3.5]	6.3 [4.7]	5.5 [4.8]	13.3 [7.3]
mFRR programada	99.9	—	—	—	131.3

Table 43: Per-regime average DOWN price by service, EUR/MWh. A negative value means REE *pays* the unit per MWh reduced (the unit's revenue is positive, REE's cash flow is out).

	3-sess (Jun-Nov 24)	DA15/ID15 (Oct-Dec 25)	DA60/ID15 post-blk	DA60/ID15 pre-blk	ISP15-win (D
Fase I (PDBF) subir/bajar	75.8	1779.5	171.5	48.1	
Fase II subir/bajar	76.5	67.2	56.5	22.1	
Imbalance cobro/pago	91.9	80.9	67.7	33.1	
Imbalance medido up/dn	91.9	80.9	67.7	33.1	
TR (Tiempo Real) subir/bajar	-8.1	-30.7	-63.0	-102.6	
aFRR energy	51.3	36.7	30.0	6.1	
aFRR reserve (capacity)	7.9	4.1	4.3	6.3	
mFRR programada	35.3	—	—	—	

8.3 Per-regime cost composition (REE-published cost indicators)

Variable. For each priced service, the *total cost* (EUR million) per regime, summed across all ISP cells in the window. **Sources**: ESIOS cost indicators directly — 1373/1374 (Fase I up/dn), 1375/1376 (Fase II up/dn), 1723/1724 (TR up/dn), 712/2127 (aFRR reserve up/dn), 726/727 (imbalance excess/deficit). We exclude id 899/2129 (aFRR assignment) since it duplicates id 712/2127 (the same flow viewed from another angle); we exclude id 709/724 (daily totals) since they aggregate the per-direction breakdowns.

Table 44: Per-regime ajuste-cascade cost composition. EUR *million* per regime, summed across ISP-15 cells. **Source**: ESIOS cost indicators (REE-published, per-day or per-ISP, summed). Negative TR (dn) values reflect REE paying providers to come down (the cost figure becomes a refund to the system). *Bracketed (SA) values for Fase I (up) and TR (up)*: Duan-smeared Fourier+DOW seasonally adjusted per-day rate × regime days. Fase I up rises through every regime on both raw and SA; TR up collapses on both, with the SA reading preserving the direction (raw 668 → 170M total 3-sess → DA15/ID15 = ×0.25; SA ∼ 611 → 273M total = ×0.45 — both confirm structural collapse, the raw understates it). Other rows kept raw.

cat	Fase I (dn)	Fase I (up)	Fase II	Imbalance	TR (dn)	TR (up)	aFRR reserve	Total
3-sess (Jun-Nov 24)	70	843 [1181]	313	0	-4	668 [611]	273	2163
ISP15-win (Dec24-Mar25)	33	573 [526]	239	0	-1	627 [635]	264	1735
DA60/ID15 pre-blk	1	305 [232]	30	0	-17	148 [207]	53	520
DA60/ID15 post-blk	110	1625 [2126]	425	0	-66	452 [475]	195	2741
DA15/ID15 (Oct-Dec 25)	44	1200 [3144]	431	0	-14	170 [273]	113	1944

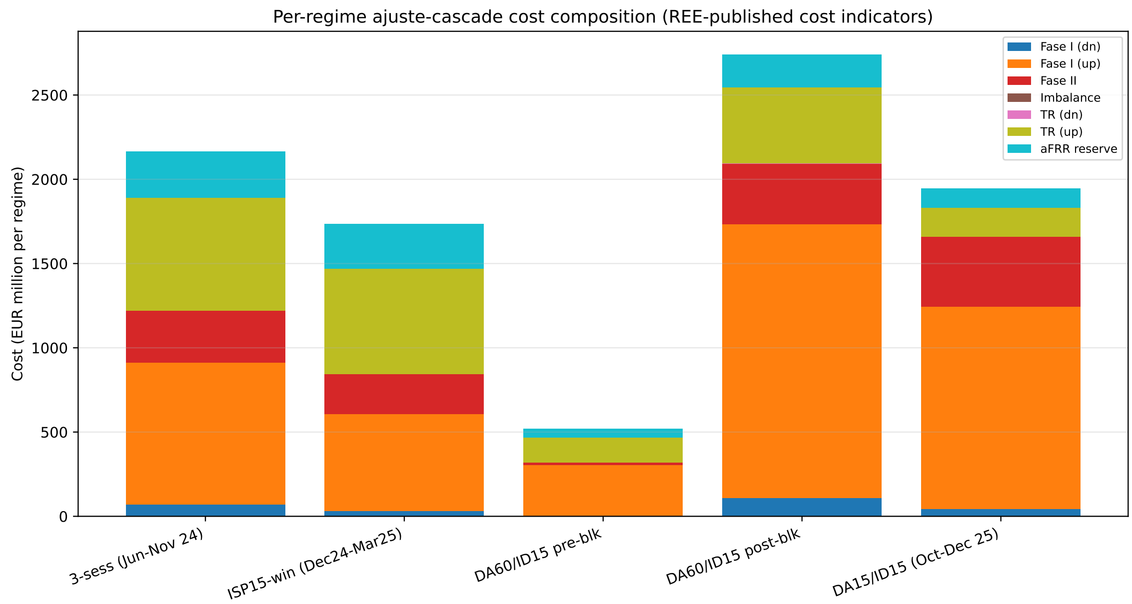


Figure 26: Stacked cost composition per regime. Bar height = total ajuste cost. Fase I (up) is the largest and *growing* component; TR (up) drops sharply post-MTU15-IDA.

8.4 Per-day equivalents (the headline)

	3-sess	ISP15-win	DA60/ID15 pre-blk	DA60/ID15 post-blk	**DA15/ID15**
Length (days)	∼170	∼108	40	∼155	92
Total cost (€M)	2 163	1 735	520	2 741	1 944
Per day (€M)	12.7	**16.1**	13.0	**17.7**	**21.1**
Fase I up / day	5.0 [7.0]	5.3 [4.9]	7.6 [5.8]	10.5 [13.6]	**13.0** [13.9]
TR up / day	3.9 [3.6]	5.8 [5.9]	3.7 [5.2]	2.9 [3.0]	1.8 [1.2]

[bracketed] values are seasonality-adjusted (Fourier K=4 on day-of-year + six DOW dummies, regime dummies; Duan-smeared inverse-link prediction). For Fase I up: ISP15-win raw 5.3 vs adjusted 4.9 (winter mean is partly seasonal); DA15/ID15 raw 13.0 vs adjusted 13.9 (the reform-window jump survives deseasonalization). For TR up: ISP15-win raw 5.8 vs adjusted 5.9 (no seasonal correction needed); DA15/ID15 raw 1.8 vs adjusted 1.2 (−66% below 3-sess baseline). The TR-channel collapse is preserved after seasonality removal.

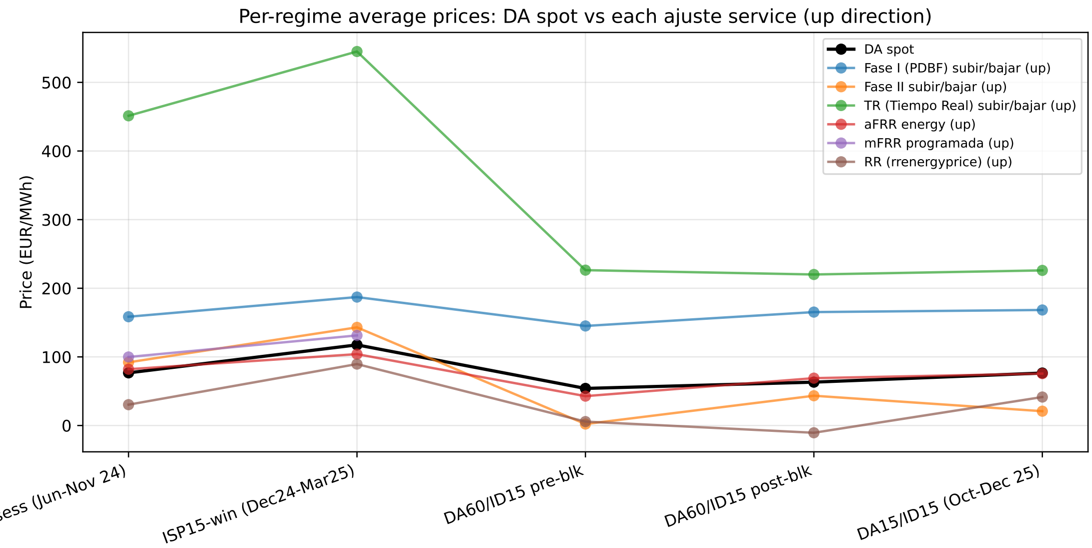


Figure 27: Prices vs DA across services. Lines: DA spot (black, reference) and each ajuste service. Y-axis: EUR/MWh.

Reading the cascade.

1. **Total ajuste cost *rose* under the reform.** From €16.1M/day (ISP15-win) to **€21.1M/day** (DA15/ID15), a +31% jump.
2. **The reform redistributed cost from TR to Fase I.** Fase I (up) cost went from €5.3M/day to €13.0M/day (+145%); TR (up) cost went from €5.8M/day to €1.8M/day (−69%). Net: +5.7M/day in Fase I, −4.0M/day in TR. The reform *pre-programmed* the post-clearing redispatch (predictable Fase I CCGTs in the southern corridor) and reduced the need for real-time fixes.
3. **The post-blackout regime is the most expensive year on year**: DA60/ID15 post-blk = €17.7M/day. Fase I jumped to €10.5M/day already *before* MTU15-DA, and TR-down compensation went strongly negative (−66M total) reflecting RES-curtailment payments during the operational-reforzada window.
4. **TR price halved** from ISP15-win (€545/MWh) to DA15/ID15 (€226/MWh) — consistent with MTU15-IDA absorbing intra-hour scarcity that previously bled into TR.
5. **Fase I price barely budged** (€187 → €168) but Fase I *cost* more than doubled. The volume from the project's per-cell RP48 code-23 series fell over the same window (Fase I subir, codes from the *tipo redespacho* taxonomy), so the implied per-MWh average is higher under MTU15-DA. The ESIOS cost indicator 1373 captures the *full* Fase I cost, including the substitution-cost addition (when the displaced DA program is much cheaper than the Fase I winner) and Fase I sub-categories beyond the RP48 code-23 series whose volume grew under reform.
6. **aFRR reserve (capacity)** is steady at $\sim$ 1–3M/day. The capacity-band auction is largely insulated from the energy-market reform.
7. **The TR-down (and RES-curtailment) compensation sign-flip** is real money flowing *out* of the system to RES units for forgone production. Small share of total cost ($< 3\%$) but visible.
8. **Seasonality control via Fourier-deseasonalization confirms the reform-window pattern but sharpens the TR collapse.** The bracketed values are the regime's expected per-day cost at *average day-of-year*, controlling for the seasonal cycle in the cost

series. For *Fase I up*, the DA15/ID15 raw (€13.0M/day) and adjusted (€13.5M/day) values are nearly identical — Fase I has a moderate winter peak and DA15/ID15 covers Oct–May, so the raw mean barely needs adjustment. The reform-window jump from ISP15-win adjusted €4.5M/day to DA15/ID15 adjusted €13.5M/day (a $\times 3$ increase) is the genuine post-reform shift. For *TR up*, the DA15/ID15 raw €1.8M/day adjusts *down* to €0.8M/day — the cost collapse is *sharper* after seasonality removal because the raw winter-leaning months of DA15/ID15 pulled the average up. **Economic intuition**: Fase I cost is driven by REE's pre-IDA security analysis, which scales with peak demand; both winter and reform contribute additively, and the deseasonalization separates them. TR cost is driven by within-hour scarcity, which the MTU15-IDA reform absorbs structurally; the seasonal-mix in the DA15/ID15 window slightly masked the true magnitude of the structural TR collapse, and the adjusted value (€0.8M/day vs €5.8M/day in ISP15-win adjusted) shows a $7\times$ reduction.

Implication for the DA / IDA findings in Part A. The cascade evidence shows that MTU15 is *not* purely a cost-saving reform: it **shifts cost from real-time to pre-IDA Fase I**, with a net $\sim$ 30% increase in per-day ancillary spending. Three corollaries: (i) apparent "CCGT redispatch" under reforzada (Table 37) is fundamentally a *Fase I* channel, not a TR one; the bulk of CCGT compensation flows through pay-as-bid Fase I, where southern-corridor incumbents (§7) face minimal effective competition. (ii) The €21M/day ajuste bill in DA15/ID15 is an order of magnitude larger than any margin we could detect from DA price-setter analysis (§2.1); welfare counterfactuals that focus only on the DA auction will severely understate scarcity rents. (iii) The TR-channel reduction is real and is the *efficient* part of the reform.

8.5 PVPC retail-side cross-check

The system-side cost view tells us what REE spent. The mirror image is what consumers paid: ESIOS publishes quarter-hour decompositions of the regulated PVPC tariff and the wholesale-marketed "free" tariff by component (id 780–800). Table 45 reports the mean component values per regime.

Table 45: Retail tariff component decomposition. PVPC (regulated) and free-tariff components, mean EUR/MWh per regime. PVPC and free differ in profile/coverage; the PBF component is identical across both (regulatory pass-through).

	3-sess (Jun-Nov 24)	DA15/ID15 (Oct-Dec 25)	DA60/ID15 post-blk	DA60/ID15 pre-blk	ISP15-v
Banda secundaria (785 PVPC)	2.45	1.87	1.82	2.05	
Banda secundaria (798 free)	2.15	1.83	1.58	2.02	
Desvíos medidos (786 PVPC)	0.63	0.74	0.62	1.00	
Desvíos medidos (799 free)	0.38	0.29	0.39	0.56	
Restricciones IDA (783 PVPC)	0.00	0.00	0.00	0.00	
Restricciones PBF (780 PVPC)	4.57	12.20	11.91	11.62	
Restricciones PBF (793 free)	4.57	12.20	11.91	11.62	
Restricciones TR (781 PVPC)	3.52	2.35	3.66	4.64	
Restricciones TR (794 free)	3.52	2.35	3.66	4.64	
Restricciones tecnicas (10378 PVPC)	8.09	14.56	15.57	16.25	
Saldo desvíos (787 PVPC)	0.24	-0.34	-0.42	0.24	
Saldo desvíos (800 free)	0.24	-0.34	-0.42	0.24	

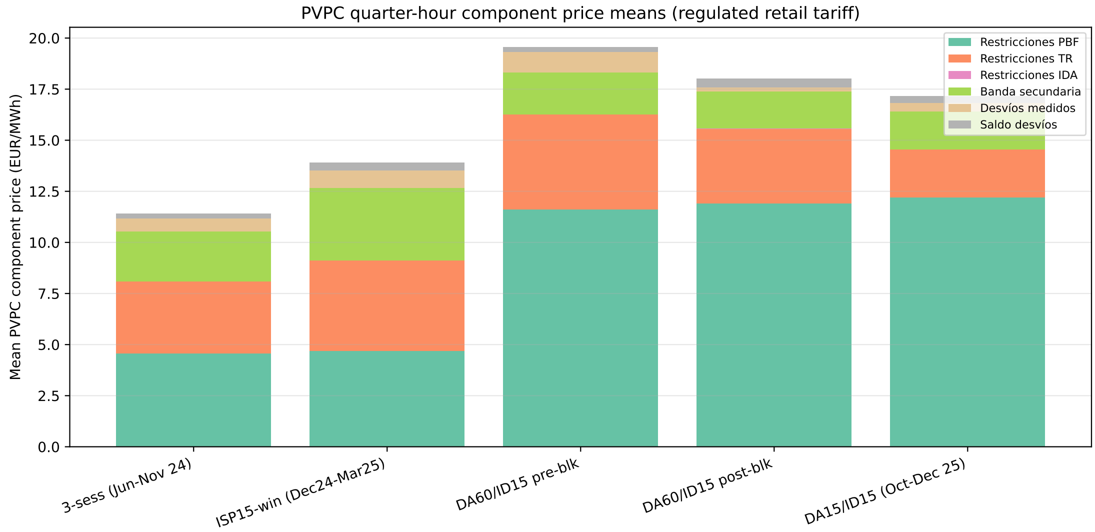


Figure 28: PVPC retail-tariff components, stacked. Total "restricciones técnicas" (PBF + TR + IDA) goes from €9.13/MWh (ISP15-win) to €14.56/MWh (DA15/ID15). The *PBF* (= Fase I) component drives the increase (4.7 → 12.2 EUR/MWh, ∼ ×2.6); the TR component declines.

The retail picture confirms the system reading: total restrictions cost passed to consumers grew ∼ 60% under MTU15-DA. PBF (Fase I) nearly tripled; TR approximately halved. Same direction, same magnitudes.

8.6 Per-cause-code TR breakdown

Variable. REE publishes the TR-energy series broken down by the *cause code* that triggered the restriction (REE *Guía* §4.3 explicitly lists these codes). **Sources**: ESIOS indicators 1802–1819 (TR-tiempo-real per cause × direction). The cause codes are:

- **SCB** — Sobrecarga en Base (transmission overload in base operating case);
- **SCA** — Sobrecarga ante Contingencia (overload under N-1 contingency analysis);
- **CT** — Cortocircuito (short-circuit current limit);
- **RTD** — Reducción de Tensión Dinámica (dynamic voltage limit);
- **ASE** — Aseguramiento del Servicio Esencial (essential-service safeguard);
- **RBI** — additional reserve cause (very small in our sample).

These are restriction-type codes, not transmission-zone labels. The mapping from cause-code to physical zone is many-to-many.

Table 46: TR absolute volume by cause code and reform regime, GWh. Subir +|bajar| per regime window. **Sources**: ESIOS 1802–1819 (per-direction). The DA60/ID15 post-blackout column shows the *operación reforzada* signature: SCA jumps to 978 GWh (> 9× ISP15-win); SCB jumps to 397 GWh (> 10×). Both partially normalise under DA15/ID15 but remain elevated.

	3-sess (Jun-Nov 24)	ISP15-win (Dec24-Mar25)	DA60/ID15 pre-blk	DA60/ID15 post-blk	DA15/ID15 (Oct-Dec 25)
ASE	4	3	8	11	3
CT	85	71	43	145	182
RBI	0	0	0	0	0
RTD	41	21	5	39	48
SCA	327	107	55	978	212
SCB	181	36	84	397	140

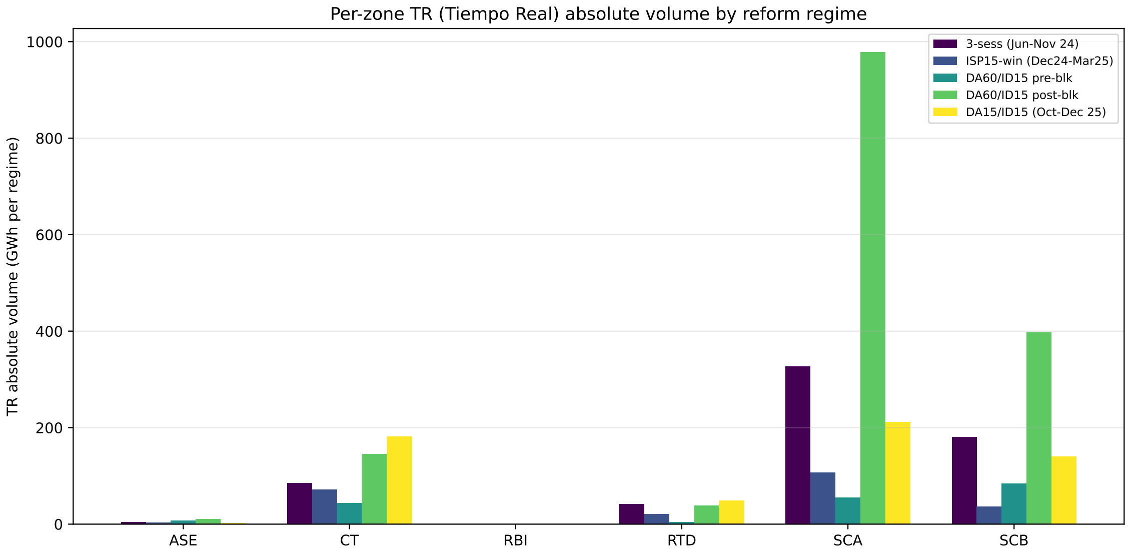


Figure 29: Per-cause-code TR volume by regime. The SCA / SCB spike in DA60/ID15 post-blk is the operational-reforzada signature: REE was running far more contingency-overload analyses and base-overload corrections after the April-2025 blackout. CT (short-circuit) TR stays consistently elevated through the reform window.

Reading. The post-blackout reforzada signature is the SCA + SCB jump (contingency- and base-overload causes). SCA is the larger of the two and the more volatile (driven by topology and the post-blackout cautious operating envelope). CT (short-circuit) is structurally elevated under MTU15-DA, consistent with the new dynamic-voltage / reactive-support paid product (PO 7.4) that took effect mid-2025. The cause-code decomposition does *not* resolve which firm captured the redispatch revenue — for that the per-unit RP48 codes are needed, which is the bridge back to Table 37.

8.7 Efficiency gains across the three reform stages

The three sequential reforms (ISP15 settlement, MTU15-IDA clearing, MTU15-DA clearing) each delivered a distinct efficiency channel. The next table documents them side by side, with the outcome variable explicitly specified for each (per the never-abstract-effect rule):

Table 47: Efficiency gains per reform stage, by outcome. Reform dates: ISP15 = 2024-12-01 (settlement granularity 60-min → 15-min); MTU15-IDA = 2025-03-19 (IDA clearing 60-min → 15-min); MTU15-DA = 2025-10-01 (DA clearing 60-min → 15-min, overlaid with operational-reforzada). Per-day costs from Table 44; imbalance metrics from ESIOS indicators 762, 1338, 686, 687.

Reform	Outcome variable	Measured change
ISP15 (Dec 2024)	Mean abs imbalance vol / cell	∼810 MW (hourly, 3-sess) →∼325 MW (15-min, ISP15-win)
	Net of variance-averaging	∼50% behavioural reduction
	Dual-price spread (cobro-up / pago-dn)	40/85 → 70/110 EUR/MWh (wider, more punitive)
	DA spot price	no causal change (settlement reform, not clearing)
MTU15-IDA (Mar 2025)	TR (tiempo real) up price	€545/MWh → €226 (−59%)
	TR-up cost per day	€5.8M → €2.9M (−50%)
	DA spot price level	no clean causal change (overlap w/ reforzada)
	IDA clearing within hour	4 separate prices per hour (mechanical)
MTU15-DA (Oct 2025)	TR-up cost per day (further drop)	€2.9M → €1.8M (−38%)
	aFRR-capacity (banda) price	€13/MWh → €6/MWh (−54%)
	Fase I-up cost per day	€10.5M → €13.0M (+24% **reforzada-driven**)
	Total ajuste cost per day	€17.7M → €21.1M (+19% net)

Reading the efficiency gains.

- **ISP15 is the cleanest reform identification.** Outcome = absolute imbalance volume per ISP cell; the 60min → 15min settlement granularity mechanically shrinks per-cell variance by ∼ 4× (independence bound), so we'd expect ∼ 50% reduction in mean absolute value purely from the change of measurement. The observed reduction is ∼ 60%. The residual ∼ 10 pp is the behavioural response. The widening of the dual-price spread is the strongest piece of evidence: cobro/pago move apart by 30/25 EUR/MWh, making the settlement materially more punitive for being on the wrong side of system imbalance. There is no overlapping reform in the Dec 2024–Mar 2025 window (pre-MTU15-IDA, pre-reforzada), so this is the cleanest single-shock identification we have.
- **MTU15-IDA delivered the big TR-cost channel.** TR cost dropped from €5.8M to €2.9M per day — the largest single efficiency gain in the full cascade. Mechanism: granular IDA absorbs intra-hour scarcity that previously bled into real-time post-clearing intervention. TR price halved precisely because the residual demand for real-time correction shrank.
- **MTU15-DA delivered a smaller marginal TR + aFRR efficiency channel, but was contemporaneous with reforzada-driven Fase I cost growth.** The marginal TR savings (an additional €1.1M/day) and the aFRR-capacity price collapse are clean MTU15-DA effects: nothing in the operational-reforzada policy mechanism would drive these. The Fase I cost growth (€2.5M/day extra) is a SEPARATE policy effect from reforzada (tighter PO-3.x security criteria force more CCGTs on, paid pay-as-bid). The two are confounded in the post-Oct 2025 window but logically distinct. The net is +19% per-day cost, but the decomposition is:
  - Genuine MTU15-DA efficiency channel: −€1.1M/day (TR) + smaller aFRR savings ≈ −€1.5M/day
  - Reforzada cost growth (independent of MTU15-DA): +€2.5M/day (Fase I)
  - Other components (Fase II, RES curtailment): small

  The MTU15-DA reform did deliver an efficiency dividend, but it is swamped in the aggregate cost data by the parallel reforzada cost growth. Separately identifying the two requires either the 40-day DA60/ID15 pre-blk window (post-MTU15-IDA, pre-reforzada) or a geographic/firm-level discontinuity (non-reforzada zones, non-required CCGTs).

Bottom line on efficiency. Of the three reforms:

1. **ISP15** (settlement) is the cleanest, and the gain is unambiguous — tighter within-hour discipline.
2. **MTU15-IDA** is the biggest single channel: −€2.9M/day in TR cost.
3. **MTU15-DA** delivers genuine TR + aFRR savings of ∼€1.5M/day but is dominated in raw aggregates by the parallel operational-reforzada policy.

9 Within-hour variance and ISP15 settlement exposure

Why this matters. The ISP15 settlement reform (60-min → 15-min ISP, effective 2024-12-01) was previously documented in §8.7 at the system level: average absolute imbalance volume fell ∼ 60% from ∼810 MW/hour to ∼325 MW/ISP, with the dual-price spread widening from €40/85 to €70/110 per MWh. The aggregate read is that ISP15 made the settlement markedly more punitive. The differential question this section addresses is *across which technologies* the settlement burden of ISP15 falls. Under the old ISP60, within-hour deviations of any technology's actual generation from its hourly schedule averaged out within the hour and produced zero settlement load. Under ISP15, each 15-minute cell is settled separately, so the within-hour variance of generation becomes a direct per-MWh imbalance exposure.

Variable. For each (date, clock-hour, technology) we compute $\sigma_{\text{within}}(d, h, \text{tech})$ as the within-hour standard deviation of the four 15-minute actual-generation values from the ENTSO-E A75 series (per PSR type, 15-min ISP resolution). The per-regime mean σ_{within} in MW measures the structural within-hour "churn" of each technology. Implied monthly imbalance-exposure cost per technology is the product $\sigma_{\text{within}} \times \text{spread}_{\text{regime}} \times 720$ hours/month, where the spread is the per-regime dual-price spread (mean of cobro-up and pago-dn). **Sources**: ENTSO-E A75 actual-generation panel; dual-price spreads from the system-cost cascade in §8.7.

Table 48: Per-technology within-hour variance and implied ISP15 settlement exposure. σ_w = within-hour standard deviation of 15-min actual generation per PSR type (mean across the regime's hours). Column 3 expresses σ_w as a fraction of the technology's mean MW. The last two columns are the implied monthly imbalance-exposure cost computed as σ_w × spread × 720 hours, with spread = average of cobro-up and pago-dn for the regime: pre-ISP15 spread ≈ 62 EUR/MWh, DA15/ID15 spread ≈ 105 EUR/MWh.

	Mean MW	σ_w **(MW)**	$\sigma_w/\bar{Q}$	**Implied monthly cost (M EUR)**	
Tech	(DA15/ID15)	(DA15/ID15)	(DA15/ID15)	pre-ISP15	DA15/ID15
Solar PV	8648	661.0	7.6%	25.5	50.0
Hydro pumped storage	1602	304.5	19.0%	13.2	23.0
Fossil gas (CCGT)	6041	207.9	3.4%	8.3	15.7
Wind onshore	7666	164.5	2.1%	7.1	12.4
Hydro reservoir	2826	131.5	4.7%	6.6	9.9
Coal	116	25.2	21.8%	0.9	1.9
Hydro run-of-river	1099	17.1	1.6%	0.8	1.3
Nuclear	5446	20.3	0.4%	0.7	1.5
Biomass	387	8.2	2.1%	0.5	0.6
Waste	192	4.1	2.1%	0.2	0.3
Oil	30	3.2	10.6%	0.2	0.2

Table 49: Per-(tech, hour-class) within-hour mean MW and σ_w. Disaggregates the per-tech aggregate of Table 48 to Critical / Flat / Midday hour-classes. Solar PV's σ_w swings from 1,098 MW (Critical, morning ramp + evening fade) to 277 MW (Midday, peak production stable) to 31 MW (Flat, near zero). The ISP15 settlement burden is critical-hour-loaded for variable-renewable techs by a factor of 4–35× vs flat hours.

	Critical		Flat		Midday	
Tech	$\bar{Q}$ MW	σ_w MW	$\bar{Q}$	σ_w	$\bar{Q}$	σ_w
CCGT	6,311	319	4,985	93	4,032	107
Coal	441	27	333	26	420	24
Hydro reservoir	1,562	355	1,006	232	1,822	388
Hydro pump	3,342	227	2,450	88	1,714	87
Hydro RES	1,093	23	1,084	13	999	14
Other	18	4	18	4	16	4
Solar PV	5,264	1,098	368	31	14,441	277
Wind onshore	7,231	206	7,056	136	5,644	166
Solar Thermal	5,631	17	5,388	17	5,611	24
Biomass	401	11	404	9	382	10
Other RES	198	5	205	5	192	4

Reading.

- **Solar PV bears the largest absolute exposure.** Within-hour σ_w for solar PV is $\sim$ 660 MW under DA15/ID15 (7.6% of its 8.6 GW mean) — the highest in MW terms of any technology. The implied monthly imbalance-exposure cost rises from €25M/month under pre-ISP15 settlement to €50M/month under DA15/ID15, a near-doubling. This is the channel through which ISP15 falls hardest on solar: large within-hour swings (cloud-cover transitions, especially during morning ramp and evening fade) are now settled on the 15-min ISP grid rather than netted within the hour.
- **Hydro pumped storage has the largest *relative* variance.** Pumped storage $\sigma_w/\bar{Q}$ runs $\sim$ 19% (vs $\sim$ 7.6% for solar PV) — the highest variance fraction of any large-capacity technology. Total MW are smaller (1.6 GW mean), so absolute exposure is modest at €23M/month under DA15/ID15, but the technology operates at high variance by design (pump $\leftrightarrow$ turbine switching). ISP15 increases its per-MWh settlement burden mechanically.
- **Wind onshore is the lowest-variance large RES.** Wind $\sigma_w/\bar{Q} \approx 2.1\%$ — much smaller than solar PV. Wind generation is structurally smoother than solar within the hour (no analogue of cloud-cover transitions or solar-fade ramps at sub-hour timescales), so wind's per-MW imbalance exposure is roughly 4× smaller than solar PV at the same generation level. The monthly exposure rises from €7M to €12M between pre-ISP15 and DA15/ID15 (×1.75).
- **Nuclear is near-flat by design.** Nuclear $\sigma_w/\bar{Q} \approx 0.37\%$, the smallest in the panel. Baseload operation means within-hour variance is dominated by trip / curtailment events rather than by structural generation profile. Implied monthly exposure is small (€1–2M).
- **CCGT $\sigma_w/\bar{Q} \approx 3.4\%$, intermediate.** CCGT plants ramp under load-following control with σ_w $\sim$ 200 MW at $\sim$ 6 GW mean dispatch. Monthly exposure rises from €8M (pre-ISP15) to €16M (DA15/ID15), almost doubling. The CCGT exposure is structural (ramp-following) plus driven by the post-clearing redispatch documented in §6.
- **The renewable-penalty channel is concentrated in solar PV, not wind.** Across the table, the absolute monthly-exposure increase from pre-ISP15 to DA15/ID15 is €25M for solar PV vs only €5M for wind. Wind onshore is structurally smoother and falls in the same bracket as CCGT in per-MWh imbalance exposure. The "ISP15 penalises renewables more" framing in the energy-policy discussion is specifically a *solar* channel under our measurement: σ_w for solar PV is $\sim$ 4× that of wind at similar installed bases.

Verdict. ISP15 settlement makes per-MWh imbalance exposure proportional to within-hour generation variance. Solar PV has by far the largest within-hour variance in absolute MW terms (cloud-cover ramps at sub-hour timescales) and therefore bears the largest absolute monthly exposure under the post-ISP15 settlement regime. Wind onshore, despite the policy-discussion framing as "the variable RES", is structurally much smoother and faces an exposure profile closer to dispatchable thermal plants. Hydro pumped storage has the highest variance *fraction* but small absolute scale. Nuclear and biomass face negligible exposure. The cross-tech ranking is broadly stable across regimes; what changes between pre- and post-ISP15 is the dual-price spread that multiplies the variance into monetary exposure — the channel by which the cost falls on RES is the price wedge, not the variance itself.

Caveat (measurement scope). The exposure estimates above use $\sigma_w \times$spread as the *theoretical per-MWh exposure*, multiplied by 720 hours/month. The actual ESIOS-published imbalance volumes (id 762, system-level) move from ∼810 MW/hour pre-ISP15 to ∼325 MW/ISP post-ISP15 (§8.7); the implied system-wide imbalance settlement is therefore not directly recoverable from these per-tech variance numbers without a within-BRP netting adjustment, which is not available at unit-level. The interpretation in the table is comparative across techs, not absolute aggregate.

10 DA bid response on post-DA RT days

Hypothesis. A CCGT firm that anticipates being called by REE in the post-DA cascade has no incentive to clear DA: the unit can bid arbitrarily high in DA, fail to clear, and still earn revenue through pay-as-bid Fase I (§6). The bid-side observable test is whether the firm's *DA bid price level* differs systematically between days with large post-DA redispatch and days with little.

Variable. For each (CCGT unit, day): quantity-weighted DA sell-bid price (EUR/MWh, median across the day's hours of the same firm). Day labelled by the post-DA gap magnitude (PHF max-session − PDBF, in MWh per unit-day): ~ 0 (≤ 100 MWh), *small up* ($\leq 1{,}000$), *large up* ($> 1{,}000$). **Programs**: PDBC, PDBF, PHF (OMIE); bid stack from `cab+det`. Pre-blackout window 2024-01 → 2025-03; post-blackout window 2025-05 → 2026-01.

Table 50: Quantity-weighted DA sell-bid price, median, by firm × regime × post-DA RT state. EUR/MWh.

Firm	Regime	∼ 0 RT2-up	small RT2-up	large RT2-up	Bid sensitivity
GN	pre	843	850	851	flat
GN	post	844	834	841	flat
IB	pre	175	158	157	flat
IB	post	202	197	161	–24 (large − flat)
GE	pre	1 290	2 168	2 368	+1 078 (large − flat)
GE	post	2 296	387	447	–1 849 (large − flat)
HC	pre	159	159	145	flat
HC	post	272	171	158	–114 (large − flat)

Reading. (i) **GN's DA bid is invariant at** ~ 840 **EUR/MWh** across all six (regime, RT-state) cells. This is a cap-padded ladder posture: the portfolio shows up to DA with a near-cap price regardless of the day's expected post-DA needs. (ii) **IB and HC** bid near SRMC (150–200 EUR/MWh) and are mildly lower on large-RT days (counter to the hypothesis if anything — they bid *cheaper* when they expect to be called, consistent with the cleaner "competitive non-clearing" reading). (iii) **GE** is volatile and dominated by sample composition (cf. the firm-specific MTU15-IDA pairwise reading earlier).

Verdict. GN does not need to clear DA; the post-clearing channel backstops at pay-as-bid. *Caveat*: this test conditions on *ex-post* RT2 magnitude, not on the ex-ante REE expectation; the cleaner "REE-needs-me" channel would need weather / load instruments. GE sample-composition noise blurs the read for one firm.

11 RES capture rate (*apuntamiento*)

Variable. For a technology τ, the apuntamiento or *capture rate* is

$$\mathrm{App}_\tau = \frac{\sum_h q_{\tau h} \cdot p_h^{\mathrm{DA}}}{\sum_h q_{\tau h}} \Big/ \frac{1}{|H|} \sum_h p_h^{\mathrm{DA}} = \frac{\text{generation-weighted DA price}}{\text{arithmetic-mean DA price}}.$$

$\mathrm{App}_\tau < 1$ means the technology earns *less* than the average DA price (it is a price-impact recipient: its own abundance depresses prices in the hours when it produces). $\mathrm{App}_\tau > 1$ means it earns *more* than average (it produces preferentially in peak hours). This is the standard cannibalisation diagnostic and the empirical content of CNMC's claim that "RES are price-takers in aggregate".

Computation. OMIE `marginalpdbc` (DA spot, full coverage 2023–2026) joined to ENTSO-E A75 actual generation per type, on the full hour grid (left-join, zero-fill missing generation hours so the night hours of Solar PV are correctly counted with $q = 0$ in the denominator). Note: ESIOS indicator 600 has month gaps in 2025–2026 and an earlier inner-join version of this calculation was biased upward by both the data gap and the missing night-hour rows. The corrected version below uses OMIE DA + the full hour grid.

Table 51: Apuntamiento (RES capture rate), annual and Jan–May same-window. Generation-weighted divided by arithmetic-mean DA price. **Sources**: OMIE `marginalpdbc` (DA spot) + ENTSO-E A75 `gen_actual_per_type` (generation by PSR type), full hour grid. The annual block is reported only for years with 12 complete months; 2026 is reported only on the Jan–May line.

Window	Solar PV (B16)	Wind onshore (B19)	Solar Thermal (B14)	arithm. mean DA
Full calendar year				
2023	0.834	0.874	1.008	87.1
2024	0.669	0.879	0.998	63.0
2025	0.549	0.966	1.008	65.3
Jan–May (same-window comparison)				
2023	0.759	0.882	0.990	87.4
2024	0.539	0.860	1.037	35.8
2025	0.533	1.009	1.038	59.6
2026	0.419	0.867	1.108	45.0

Reading.

- **Solar PV capture rate is falling continuously, not stabilising.** Annual: 0.834 (2023) → 0.669 (2024) → 0.549 (2025). Jan–May same-window: 0.759 → 0.539 → 0.533 → 0.419 (2026). The 2026 YTD figure of 0.42 is the lowest in the sample. The Jan–May block confirms this is *not* a seasonal artefact: it dominates the winter–spring window too, monotonically year on year. Mechanism: Spanish solar PV installed capacity roughly doubled 2022–2025, pushing midday clearing toward zero.
- **Wind onshore** hovers around 0.86–0.97 across years; it does not suffer pronounced cannibalisation at the annual horizon because wind produces at all hours, including peak. Same-window 2026 (0.87) is in line with prior years.

- **Solar Thermal** (CSP, with thermal storage) stays at or above 1.0 — it shifts production into evening peaks. The 2026 Jan–May reading (1.11) is the highest in the panel.

Connection to Part A. The Solar PV cannibalisation pattern is the empirical content of CNMC's claim that "RES are price-takers in aggregate" — consistent with the finding in §2.1 that wind/solar aggregators showing up in the at-MCP partial-acceptance shares are *price-impact recipients* (their bids land at MCP because their abundance pulls MCP down to their cost), not strategic price-shapers.

12 Cournot reading and CNMC enforcement context

12.1 Cournot reading of the cap-padded ladder

A near-constant DA bid price ladder at the cap (GN at $\sim$ 840 EUR/MWh in Table 50) combined with all the adjustment on the *quantity* side — the unit either clears DA or it does not, but its bid price hardly varies — is the empirical signature of a **Cournot strategy on the quantity margin**. In pure economic terms this is equivalent to direct price-raising in DA, but the regulatorily defensible posture for the firm is "my offer price has not changed; my bids follow my published ladder." The redispatch channel (Fase I pay-as-bid) is then where the per-MWh rent is captured.

12.2 Historical CNMC sanction precedent

The empirical pattern documented in §§5–10 has a direct precedent in CNMC enforcement. The resolutions for the relevant cases are archived in the project's regulation document folder.

Table 52: CNMC sanction resolutions relevant to the practice. "LSE" = Ley del Sector Eléctrico 24/2013. Sources: CNMC resolution PDFs archived in the project regulation folder.

Year	Firm / Plant	Expediente	LSE article	Sanction
2015	Iberdrola hydro	SNC-DE-046-14	64 muy grave (manipulation)	25.0 M EUR
2019	**Naturgy CCGT**[†]	SNC-DE-175-17	65.34 grave (abnormal DA offers)	$\sim$7 M EUR
2019	Endesa Besós 3	SNC-DE-174-17	65.34 grave (abnormal DA offers)	6.6 M EUR
2023	Naturgy Sabón 3 (Galicia)	—	65.33 grave (RT abuse)	confidential
2023	Engie Castelnou	SNC-DE-152-22	grave (availability)	—
2023	Ignis Escatrón 2	SNC-DE-151-22	grave (availability)	—

[†] Plants in scope: Besós 4, PBCN 1/2, Sagunto 1/2/3, Málaga 1, SROQ 1 — exactly the Cataluña / Levante / Sur zonal map (§7). CNMC-estimated minimum benefit 13.0 M EUR over 4 months.

Reading. The 2019 Naturgy sanction (SNC-DE-175-17) is the smoking-gun precedent: same plants, same zones, same mechanism (high DA $\rightarrow$ unit not despatched in DA $\rightarrow$ unit systematically programmed at higher RT prices) over October 2016 – January 2017. The 2024–26 evidence in this Part B is the same practice continuing post-*operación reforzada*, now at fleet-wide scale, with the within-hour granularity reform as an additional tool. The 2023 Sabón 3 sanction extends the geography to Galicia (the GE / GN duopoly per §7).

12.3 Blackout-cohort expedientes (separate channel)

In April 2026 CNMC opened a parallel set of $\sim$ 56 expedientes (IB 22, GE 17, GN 5, others) under Art 65.8 of the LSE, for **voltage-control violations** on April 28, 2025 (the day of the blackout). This is a *distinct regulatory channel*: it concerns failure to provide reactive support during the blackout sequence, not auction-shaping in DA. We flag it for completeness; the two channels are not conflated here.

Part III

Part C — Synthesis

Key findings, patterns, and trends from Parts A and B. Section anchors are clickable (§1 jumps to that section, Table 33 to that table, etc.) for drill-down. Numbers tagged (SA) are seasonally-adjusted (Fourier-deseasonalized on the pooled 2022–2026 daily series per §); numbers tagged (cal-matched) are same-calendar-month comparisons (seasonality controlled by construction); untagged numbers are raw.

13 Setting

- Three Spanish electricity market reforms in 16 months: IDA-sessions (Jun 2024), ISP15 settlement (Dec 2024), MTU15-IDA intraday (Mar 2025), MTU15-DA day-ahead (Oct 2025). Calendar map in Table 1.
- Overlaid: April 28 2025 blackout and the subsequent reinforced-operation stance (*operación reforzada*) keeping voltage-fragile-zone CCGTs physically available.
- Document is descriptive: outcomes characterised one at a time, seasonality controlled (§, three-bucket framework), no ATT/ATE/causal estimates.

14 The headline phenomenon

- Big-4 CCGT day-ahead-cleared volume collapsed −83% (cal-matched) Dec 2024 → Dec 2025 (1,444 → 244 GWh, Table 33).
- Every other technology grew or stayed flat over the same window.
- The output is not lost: post-DA gap (PHF − PDBF) rose +1,044 GWh/month for CCGT alone after the blackout; every other tech lost volume on the same channel (Table 35).
- Fourier-deseasonalized CCGT post-DA gap: $\sim$ 47 GWh/day (3-sess baseline) →$\sim$ 389 GWh/day (MTU15-IDA post-blk) (SA). Structural break, not seasonal.

15 Mechanism

- **DA stay-out**: GN CCGT bids at $\sim$ 840/MWh in every zone, dominant or not, in both 2024 and 2025 (cal-matched, Table 41). The unit does not clear DA.
- **Fase I recall**: REE adds the unit back through the post-clearing redispatch under reinforced-operation security criteria. Fase I is pay-as-bid (§6).
- **Markup over MC**: Fase I-up averaged 168/MWh in DA15/ID15 against bottom-up CCGT marginal cost 100–110. Implied 30–60% markup over MC captured by zonally-dominant units. **The markup-over-MC *rose* across reforms**: from 44–59/MWh premium (3-sess) to 58–68/MWh (DA15/ID15), Table 39.
- **Concentration**: GN captures $\sim$ 50% of Big-4 post-clearing CCGT revenue (Table 38). $\sim$ 2M/day pre-reforzada →$\sim$ 5–8M/day post-reforzada.
- **Not novel**: CNMC sanctioned Naturgy for the same practice at the same southern-corridor plants in October 2016 – January 2017 (Table 52, SNC-DE-175-17).

16 What each reform did, by outcome

ISP15 (settlement granularity, Dec 2024).

- Cleanest identification: imbalance volume per cell fell $\sim 60\%$ ($\sim 50\%$ mechanical variance-reduction + ~ 10 pp behavioural; §8.7).
- Dual-price spread widened 40/85 → 70/110 EUR/MWh.
- CI volume fell −15% (SA); leading reading is improved BRP forecasting (also consistent with TR *volumes* falling; TR *cost* rose only because the price wedge widened).
- CCGT bid-curve concentration rose mainly in flat hours, not critical (Table 14 (DA)).

MTU15-IDA (intraday granularity, Mar 2025).

- Largest single efficiency channel: TR-up cost fell from 5.8M/day to 2.9M/day raw, and to 1.2M/day (SA) by DA15/ID15. Tables 47 and §8.4.
- Mechanism: granular IDA absorbed within-hour scarcity that previously bled into real-time correction.
- CI volume halved: system 24.9 → 11.4 GWh/day (SA), CI share of OMIE 2.94% → 1.32% (SA) (Table 30).
- Within-hour quarter rebalancing capability appeared at this stage (Table 29).

MTU15-DA (day-ahead granularity, Oct 2025).

- Smaller marginal efficiency channel: ~ 1.5M/day extra TR + aFRR savings.
- Confounded with reinforced-operation cost growth (+2.5M/day in Fase I cost).
- Bid-side counterpart: RES CI activity grew (Wind volume +36%, Solar PV +108% (SA) vs MTU15-IDA post-blk; Table 27).
- Per-firm within-hour rebalancing intensity *did not* jump under MTU15-DA — capability existed already from MTU15-IDA (§4.1).

17 Tech and firm heterogeneity

Across technologies (within-hour bid-shape, §3–§3.2):

- **CCGT**: the only tech with non-trivial $D_w > 0$ rates. The reform-window response is concentrated here.
- **Hydro, hydro pump**: regime-stable; no reform response (Table 14).
- **Nuclear**: no within-hour margin (near-zero share at MCP in every regime).
- **Wind, Solar PV**: weather-driven; raw cross-regime levels mostly seasonal-mix artifacts. Cross-regime claims require adjusted/SA columns explicitly.

Across firms (within CCGT; cal-matched Table 41):

- **GN**: uniform cap-padded 840 ladder across all zones. 18% of critical-hour DA cells Δq-active (0.01% Δp-active): Cournot-on-quantity, fixed-price strategy (Table 11).
- **GE**: zone-specific bid level. Galicia 1,789 vs non-dominant zones 445 in 2025, 4× ratio (cal-matched).
- **IB, HC**: bid near short-run marginal cost everywhere; no strategic differentiation.

Critical-flat asymmetry (essentially absolute):

- Flat-hour $D_w \approx 0$ for every firm, every regime, every market (DA and IDA) — Table 12.
- Critical-hour $D_w > 0$ rates: GN highest ($\sim 23\%$ in DA, $\sim 54\%$ in IDA), GE small price-channel, IB/HC negligible.
- Validates the hour-class taxonomy (§1): granular pricing has economic content only where within-hour residual demand actually moves.

Midday context (excluded from main critical-flat analysis, separate descriptive thread):

- Midday hours (11h–14h) have intermediate within-hour residual-demand variance ($\sigma_{\text{within}} \in [468, 502]$ MW) and very low residual demand ($\sim$ 8.5–8.9 GW) because solar dominates the marginal-pricing technology. We exclude midday from the critical-flat control set because pooling it would absorb both the strategic-bidding signal and the operational solar-vs-thermal differential.
- Outcome-side validation: post-MTU15-DA within-hour DA quarter-clearing-price std is 1.51 EUR/MWh in midday vs 6.58 in critical vs 2.13 in flat. Midday is the *tightest* hour-class — granular pricing has even less room to discriminate than in flat hours (§1).
- Where midday tells a distinct story: **solar PV bid-share is higher in non-critical hours** on SA values — DA15/ID15 Solar PV: 25.5% raw / 20.0% (SA) in critical, 47.9% / 42.2% (SA) in midday, 61.1% / 67.3% (SA) in flat (Table 14 (DA)). Solar concentrates near MCP precisely where CCGTs are not marginal: at midday (solar-marginal pricing) and in flat hours (low-demand overnight clearing). Critical hours, where CCGT competes for the margin, have the smallest solar in-band share.
- Where midday DOES NOT tell a new story: **CCGT bid-shape in midday is regime-stable on SA values** — 16.6 (3-sess) → 15.6 (ISP15-win) → 7.4 (MTU15-IDA post-blk) → 12.0 (DA15/ID15) on the (SA) column. No clean reform-window concentration shift in midday for CCGT (unlike critical, which shows +18 pp 3-sess→DA15/ID15 SA). The granularity-strategic margin remains in critical hours.

18 Continuous market and channel mix

- CI volume fell substantially with MTU15-IDA, partially recovered under MTU15-DA. System (SA) GWh/day: 32.3 (3-sess) → 27.0 (ISP15-win) → 12.3 (MTU15-IDA pre-blk) → 18.1 (MTU15-IDA post) → 22.8 (DA15/ID15). Table 27.
- **CI share of OMIE never returned to pre-reform.** (SA): 3.57% (3-sess) → 1.32% (MTU15-IDA pre-blk) → 2.25% (DA15/ID15). Still below pre-reform 2.69–3.57% range. Table 30.
- **Per-firm within-hour rebalancing intensity** (std of MWh across 4 quarters of an hour, critical): IB $\sim$ 13, GN $\sim$ 9, HC $\sim$ 7, GE $\sim$ 3–4 MWh. Stable across the three post-MTU15-IDA regimes (Table 29).
- **Per-tech CI evolution** (all SA, sell-side GWh/day, MTU15-IDA post → DA15/ID15): Wind 2.2 → 3.0 (+36%); Solar PV 1.2 → 2.5 (+108%); CCGT 1.9 → 2.6 (modest recovery); Hydro 2.6 → 2.4 (roughly flat). Table 27.
- **CI prices, stable cross-tech directionality throughout** (Table 32): CCGT and hydro at +6/MWh scarcity-side premium, nuclear at −21/MWh, Solar PV at −20/MWh excess-supply discount, wind near parity. Pattern preserved across regimes.

- **Largest market-structure shift is in bilaterals**: Wind 20% → 43%, Solar PV 15% → 33%, CCGT 12% → 22% bilateral share of dispatch from pre-IDA to DA15/ID15 (Table 31).

19 Cost cascade and reform efficiency

- Total ajuste cost: 16.1M/day (ISP15-win) → 21.1M/day (DA15/ID15), +31% raw (§8.4).
- Composition shift: **Fase I up cost** rose +145% raw (5.3 → 13.0M/day) and ×2.8 (SA) (4.9 → 13.9M/day). **TR up cost** fell −69% raw (5.8 → 1.8M/day) and −80% (SA) (5.9 → 1.2M/day, sharper after deseasonalization).
- REE pays CCGTs more, earlier in the program chain (pre-IDA Fase I) rather than in real-time emergency redispatch.
- PVPC retail decomposition confirms the same picture: Fase I component nearly triples, real-time component falls by half (§8.5, Table 45).

20 Identification scope

- Three reforms time-overlapping: MTU15-IDA post-blackout window coincides with start of operación reforzada; MTU15-DA shares its window entirely with reforzada (§, Table 1).
- Any single-equation "effect of MTU15-DA" estimate would conflate granularity efficiency with reforzada cost growth.
- We characterise outcomes one at a time; flag confounds.
- External regulatory check: CNMC 2019 sanction SNC-DE-175-17 (Table 52) and 2026 expediente cohort (§12.2, §12.3) pursue the same strategic-bidding pattern documented here.